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INVESTMENT MANAGEMENT

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ABOUT THIS BOOK

This Course Book **"Investment Management"** has been exclusively written by experts in the discipline to up-date your general knowledge of English Language in order to equip you with the basic tool you will require for your professional work as a teacher and an administrator.

This three-credit course book of thirty-six (36) sessions has been structured to reflect the weekly three-hour lecture for this course in the University. Thus, each session is equivalent to a one-hour lecture on campus. As a distance learner, however, you are expected to spend a minimum of three hours and a maximum of five hours on each session.

To help you do this effectively, a Study Guide has been particularly designed to show you how this book can be used. In this study guide, your weekly schedules are clearly spelt out as well as dates for quizzes, assignments and examinations.

Also included in this book is a list of all symbols and their meanings. They are meant to draw your attention to vital issues of concern and activities you are expected to perform.

Blank sheets have been also inserted for your comments on topics that you may find difficult. Remember to bring these to the attention of your course tutor during your face-to-face meetings.

We wish you a happy and successful study.

Mr. Patrick Kwashie Akorsu
Mr. Siaw Frimpong

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Any limitations in this course book, however, are exclusively mine. But the good comments must be shared among those named above.

Prof. Isaac Galyuon
(Provost)

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UNIT 1: THE SCOPE OF INVESTMENT

Session 1: Introduction to Investment
Session 2: Setting Investment Objectives
Session 3: Investment Instruments
Session 4: Investment Instruments (Continued)
Session 5: Steps in Investment
Session 6: Investment Procedures In Ghana

Unit Overview

A parent spends money in the education of his child so the child can acquire knowledge for future use. Without this, the child may be dependent on the parent forever. A farmer spends money and energy cultivating mango, oil palm, orange, cocoa, etc and is waiting patiently till harvest time. A businessman spends money and time in his business in order to reap benefits later. A young person (usually women) spends huge sums of money on clothes and cosmetics to make them looks more attractive. What is common to all the people mentioned above? They have all spent money on various projects/activities and look forward to reaping the rewards in the future. This is the concept of investment.

In this unit, we shall attempt to define the concept of investment and outline the process of investing. We wish you well as you read through the unit.

Unit Objectives

By the end of this unit, you should be able to:

1. Explain the concept of investment.
2. Explain the issues to consider when setting an investment objective
3. List the steps in the investment process

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 1: INTRODUCTION TO INVESTMENT

Overview

Virtually everyone has some form of interest in investment because investment is an integral part of life. People undertake investments in various forms and reasons. In this session, we would attempt to address some fundamental issues of investment in order to make subsequent sessions reader friendly.

Objectives

By the end of this session, you should be able to

1. Define and explain investment in your own words.
2. Differentiate between investment and savings.
3. Explain the types of assets and investors.

Now read on…

Definition of Investment

Investment involves the commitment of resources into a venture in anticipation of reaping future benefits. The benefits should be enough to compensate the investor because of the concept of time value of money and risk. Stated differently, investment involves sacrificing present consumption with the expectation of receiving more money in the future. For example, when you purchase shares of a company, you would anticipate that future returns from the shares would be enough to justify both the time that the money is tied up as well as the risk of losing your money. As a student, the time you are spending in the study of this course is an investment, because you are sacrificing something of value - money (fees), leisure or income - in expectation that your future career prospect would be greatly enhanced to justify the commitment of money, time and effort.

From the above, we would say that the thrust of investment is to forego something of value now in expectation of benefits in the future. Note, however, that investment should not be confused with idle cash. If you keep thousands of Ghana cedis at under your bed, it does not constitute investment. In fact, idle cash is not an investment because, even though you have postponed the use of the money, you have not put it in a venture that would generate any return. Instead, its value reduces with time due to inflation.

Distinction between Investment and Saving

There are three basic things that people do with their incomes. It is either consumed or saved, or a combination of the two. Saving is depositing or putting money in a bank account, with little risk, if any at all, because such funds are usually protected against

bank failure. Investment also puts money away, but in a risky venture, such as buying shares in a listed or unlisted company. Therefore, the main difference between investment and saving is chances of losing money. With investment, the chances of losing money (risk) is high since one cannot predict with certainty the returns expected, unlike saving where the saver is certain of the return on the saving. In addition to risk, the expected rate of return on investment is higher than savings. Indeed, in risk-return analysis in finance, the higher the risk, the higher the return. For instance while people who put money in their savings accounts have few worries, if any at all, investors have lots of worries due to the uncertainties surrounding their investments and as such, must be compensated for those worries. Hence, the higher rate of return expected. The final distinction is liquidity. It measures the ease with which an asset can be turned into cash without significant loss of value. Unlike savings, it may take time to turn investments into cash, and therefore, we would say that savings are more liquid than investments.

Low and High Risk Investments

One way of differentiating between investments is on the basis of the risk inherent in that investment. If an investment is considered safe with regards to the receipt of a positive return, then that investment is considered as a low-risk investment. Low risk investments include Treasury bills and government bonds. Even though different investment instruments have basic risk characteristic, the exact risk level vary from one instrument to the other. For instance, Treasury bills are considered risk-free investment. That is to say that risk is almost non-existent with Treasury bills.

High-risk investments on the other hand are not very safe with regards to the receipt of positive returns. That is to say that the variability of possible returns or outcomes is wide. Shares are generally thought of as being more risky than bonds, but it is equally possible to find bonds that have higher risk than shares. Investors are therefore confronted with a continuum of low-risk government securities to high-risk shares or commodities. Of course, as has been said over and over again, due to the wide-spread variability (greater risk) of the expected returns associated risky investments the expected returns are usually high. We shall consider the issue of investment risk and return more closely later on this module.

Types of Assets

The definition of investment in the field of finance involves putting money or investing in financial or real assets. An asset is anything that has value that a business or an individual owns such plant, buildings, cash, art works, shares, etc. There are two types of assets, namely, real assets and financial assets.

Real assets tangible assets that can be seen, felt, held or collected. These assets are used to produce goods and services. Examples of real assets are houses, equipment,

inventories, cars etc. This asset generates the net earning of a country and therefore is the backbone of the productive capacity of an economy.

Financial assets on the other hand constitutes financial claim on an asset that is documented by some form of legal representation. These assets contribute indirectly to the productive capacity of an economy. Rather, they are means by which people hold their claim on real assets. Examples of financial assets are shares and bonds.

Types of Investors

There are two types of investors in securities. These are individual investors and institutional investors. **Individual investors** manage their own funds in order to achieve their investment objectives or goals. The objective may be to raise enough money to pay for a child's school fees in future, retirement income or to put up a house. Whatever the objective is, the individual investor commits his funds to investment activities and expects to earn a decent return. Individuals with large sums of money to invest but lack the technical know-how and the time to manage their investments may employ the services of professional investment advisors to manage their investment portfolio.

Institutional investors on the other hand are firms that Invests in securities for their clients. In other words, institutional investors are firms paid to manage other people's funds. They include as investment banks, pension funds, mutual funds and insurance firms. In principle, the techniques used by both institutional and individual investors are similar, but in practice because institutional investors usually have larger funds to manage, they employ sophisticated methods in their operations than individual investors.

The type of reward and the time (period) it takes to get the reward (return) from an investment though depends on the type of investment undertaken. For instance, whereas it may take at least ten years to receive any reward from education, it may take between three to four years for a cocoa or oil palm farmer to start enjoying the fruits of his investment. For investors in stocks, the time period for enjoying the returns can be much shorter.

Direct and Indirect

Investors can invest directly or indirectly. A direct investment is one in which an When an investor personally selects securities (say bonds, shares or even real estates) and registers such securities in his/her name, the investor is said to have engaged in direct investment. An indirect investment on the other hand is an investment made in mutual funds, unit trust, investment trusts, and pension plans, which are diversified portfolios of securities. Mutual funds are a pool of investments managed investment professionals on behalf of a group of investors with little means in order to participate in the investment process.

Financial Markets and Institutions

The concept of investment is basically the process of bringing together suppliers of funds and businesses that needs the through intermediaries such as financial institutions or a financial market. The financial institutions include banks, savings and loan companies. Financial markets on the other hand are markets for buying and selling financial securities, often through intermediaries such as organized securities exchanges. Common examples of financial markets stock markets, bond markets, futures market and options markets. The major role played by financial institutions in financial markets, is the transfer of funds between suppliers and users of funds. It is important to note that financial institutions also supply funds and use funds as well.

Financial markets are classified as money market or capital market. Money markets are short term with a life of up to a year. The long-term market is the capital market, with a life of a year and more, and is dominated by various securities exchanges.

Summary

We have attempted to explain some basic concepts of investment. Granted, these are not exhaustive. We shall fill in as we move on. You should be aware though that investment in the field of finance is about investing in marketable, public traded securities or investing in real assets. You should know that a security is a legal instrument that shows ownership interest, and securitization is the process of converting an asset into a more marketable form.

SESSION 2: SETTING INVESTMENT OBJECTIVES

Overview

It is often said that if you do not know where you are going, every road would take there. This adage can be applied in investment. If one has an objective or a goal when investing, then the person can measure the success or failure of in investing. In this session, we are going to learn about the issues to consider when setting objectives for investment.

Objectives

By the time you finish this session, you should be able to:

1. Explain the key issues to consider when setting an attainable investment objective.
2. Distinguish between risk-averse, risk-lovers and risk-neutral investors.

Now read on…

The setting of investment objectives may be equally important as the selection of the investment because the objectives of an investor would guide him/her as to the type of investment to undertake. A number of key areas should be considered in setting investment objectives. These include:

i. Risk and Safety of Principle

The first factor when considering an investment is the amount of risk you are prepared to assume. In an efficient financial market risk tends to correlate with returns. Therefore the higher the risk, all things being equal, the higher the return of the investment.

The age and economic circumstances of an investor are important variables in determining an appropriate level of risk. Young people are more likely to take on more risk than people nearing their pension. This is because young people have long time on their investment horizon and are therefore prepared to take on risk. Any losses in the short run would be compensated for in the long run. Old people and pensioners have not much time and would normally invest in safe investments such as treasury bills. This is because any loss incurred could be costly to them.

Investors can be categorized by the types of investment they own, and are usually one of the following: Conservative, speculative and balanced.

Conservative investors are also known as risk-averse investors. These investors are normally not willing to take too much risk unless the compensation for taking on risk is

adequate. This type of investor is usually content with earning a lower return if the amount invested can be protected largely against loss.

Speculative investors are referred to as risk lovers. They are willing to assume high risk. They invest in new ideas and companies, hoping that the venture would be successful in the long-run in order to earn very high returns.

Balanced or risk neutral investors are investors who wants the "best of both worlds", and so are willing to take some risk but also want some security. These investors usually own a balance between speculative and conservative investments and is sometimes said to have a balanced portfolio of investments.

ii. Current Income versus Capital Appreciation

There are two types of stocks, growth stocks and high-yielding stocks. Whether an investor will choose a growth stock or high yielding stock depends on their *current financial needs*. Investors in need of current income, e.g. pensioners, may opt for high-yielding mature firms in industries like public utility, food and industries, apparel or clothing industry. Such firms pay high dividend to investors compared to their counterparts in emerging firms. Firms in these industries are not normally adversely affected by the worsening state of the economy and are sure of having consistent flow of income. For instance, whether there is recession or not people will eat and therefore firms in the food industry will not be hit hard in terms of sales reduction compared to firms in the construction industry.

Investors looking for price gains (growth stocks) may look towards investing in smaller, emerging firms, firms in high technology industry, energy or electronic industries. Such firms rarely pay cash dividend, but the value of their stocks continues to grow to give the desired return, example Microsoft.

There is generally a trade-off between growth and high-yielding stocks. It is unlikely to have a stock, which have both characteristics.

iii. Liquidity

Liquidity is measured by the ability of the investor to convert an investment into cash within a relatively short term at its fair market value or within a minimum capital loss on the transactions. Thus, liquidity refers to the ease and speed with which an asset can be converted into cash. Liquidity has two dimension, ease of conversion and loss of value. Any asset can be converted quickly into cash if the price is cut down. Therefore a high liquid asset is the one that can be sold quickly without significant price reduction. An illiquid asset, on the other hand, is the one that cannot be sold quickly without a substantial price reduction. Generally, stocks and bonds have high degree of liquidity compared to real estate.

In selecting investment, investors must assess their own situation to determine their liquidity needs. For instance, a student who will need money to pay for his University fees next academic year would consider immediate liquidity as essential. Therefore it will be unwise investment decision to invest such funds in real estate, instead of buying short-term investment like treasury bills.

Liquidity is also essential to firms. The more liquid a business is, the less likely that the business will experience financial distress (i.e. difficulty in paying debts or buying needed assets). Unfortunately, liquid assets are generally less profitable to hold. For example, cash is the most liquid asset of all investments, but sometimes it earns no returns at all (a cash in a current account). Therefore, there is a trade-off between the advantage of liquidity and foregone potential profits.

iv. Tax Consideration

The choice of investment sometimes depends on the tax liability of the investor. Investors in high tax bracket have different investment objectives compared to those in lower tax bracket or tax-exempt charities, foundations or similar organizations. Investors in high tax bracket may prefer municipal bonds, real estates (with its depreciation and interest written off), treasury bills or investments that provide tax credit or limited tax. In certain countries, capital gains and dividends may be taxed differently. Capital gains are not taxed in some countries whereas others tax capital gains less, compared to dividends. An investor in a high tax bracket will consider all these factors before undertaking an investment. Investors in lower tax bracket would not be affected much by tax and therefore tax consideration may not be a factor in the selection of an investment.

v. Ease of Management

Another factor worth considering in choosing an investment is the ease of management. The investor must consider the amount of time that would be needed in managing the investment portfolio. If too much time is spent on the investment, its opportunity cost may be too high to warrant undertaking such an investment personally. This explains why high-income earners such as sportsmen, entertainers etc normally use the devices of professional investors. The time used in managing an investment could be used to make more money than the return from that investment.

The investor must also consider whether he has enough knowledge in the field of investment to be undertaken. For example investors who do not know how swap or options work would not reap a full return in investing in these investments. They may end up making losses. Decisions such as these may also depend on your trade-off between work and leisure. An investor may truly find it satisfying and intellectually stimulating to manage a portfolio and may receive psychic income from mastering the

art of investment. However, if you rather play soccer, tennis or enjoy some leisure activities then using professional management may be the best choice.

vi. **Retirements and Estate Planning Consideration**

Every investor must consider the effect of their investment decisions on their retirement and the estate they will like to leave for their generation. Planning for retirement should be done at very early years when you start employment and not when you are about to retire. Therefore it will be of importance to start putting aside some amount out of your monthly salary in a mutual fund or similar investments for your retirement. The investor can also plan for the family and future generations by setting aside some money to invest in assets, which will be of benefit to their children in the future. For example, a fixed deposit account can be opened for your child and will be accessible at his eighteenth birthday. Other areas where investment can be undertaken for the family include investing in real estate and land for your children.

Summary

Investment objectives are very important to a successful investment programme. In fact, the list of objectives shown above is not exhaustive. We believe that you are in a position to establish your own investment objective after reading this session. We wish you success as you move on to the other sessions of this unit and believed that when you are called upon to answer questions on this session, you will pass with distinction.

SESSION 3: INVESTMENT INSTRUMENTS

Overview

Investors and prospective investors have many investment instruments at their disposal through which they can fulfill their investment objectives. Investments may also be classifies as short-term security, common stock and fixed income security with unique characteristics. These are the subjects for consideration in this session.

Objectives

By the time you finish this session, you should be able to:

1. Explain what constitute short term investment instruments.
2. Explain specific short term investment instruments.

Now read on…

Short–Term Investment Instruments

i. Certificate of Deposit

A certificate of deposit is an instrument issued by a bank or other depository institutions representing funds placed on deposit at the bank for a certain period of time. It is considered money market instrument because they can be traded among investors after their initial issue. Because of this characteristic, they are sometimes called negotiable certificate of deposit. They are not redeemable before maturity but an investor who purchases it may sell to another investor before the maturity date.

ii. Commercial Paper

It is a term for short-term promissory note issue by large corporations with high credit ratings. Funds raised from commercial paper are normally used for current transactions-to purchase inventory, pay taxes, meet payrolls and cover short-term obligations-rather than for capital transactions. It is traded mainly in the primary financial market.

iii. Money Market Fund

It represents an instrument to buy short-term fixed income security through a mutual fund arrangement. An individual with small amount to invest may pool funds with others to buy high-yielding large certificate of deposit and other similar instruments indirectly through the fund.

iv. Treasury Bills

Treasury bills are short-term IOUs issued by the government. It is safe and popular investment medium for individuals, financial institutions and companies of all sizes. It is one of the best-patronized short-term investments in Ghana. Most individual, corporations and financial institutions invest large sums of money in the treasury bills issued by the Ghana government.

Ordinary Shares

Ordinary shares are equity investment representing fractional ownership in a company. It is the most important form of corporate shares. It represents a residual claim against the assets of the issuing company entitling the owner (holder) to share in the net earnings of the firm when it is profitable or share in the net market value of the firm's assets if it is liquidated. Investors of ordinary shares take full risk of ownership. Ordinary shareholders are the risk-takers because they own the portion of the firm, which is not guaranteed. In return for taking this risk, they share in the growth of the firm. Their shares entitle them to a voting right that allows them to vote on important issues and election of board of directors.

Preference Shares

Preference shares have priority over ordinary shares in times of bankruptcy. Preference shares are technically equity interest in a company but its characteristic is more like those of a bond. It carries a stated annual dividend expressed as a percentage of the share's par value. This characteristic makes it a fixed income security. The dividend on preference shares are usually cumulative in the sense that dividends that are not paid cumulates until they are paid in full before any dividend can be paid to ordinary shareholders. Preference shares occupy the middle ground between debt and equity including advantages and disadvantages of both forms of raising long-term funds. However, it is more flexible financial arrangement for the issuing firm because dividend may be passed if earnings are inadequate.

Fixed Income Securities

They are investments that offer a fixed periodic return to the investor. Some have contractual guaranteed return, for example bonds, while others have specified but not guaranteed return, for example preference shares. Fixed income securities include bonds, debentures and preference shares.

i. Bonds

Bond is the basic fixed income security. It is a debt obligation issued by a business firm or a unit of government that covers several years usually over five years. Normally,

bonds represent a long-term contractual obligation of a firm or government unit issuing it to pay interest to the bondholder (investor) as well as the face value of the bond at maturity. The interest rate (corporate rate) on the bond is either paid semi-annually, or annually. The interest rate is call coupon, and the amount that is repaid at the end of the loan is called the bond's par value or face value or maturity value. The annual coupon divided by the face value is called the coupon rate. Some bonds do not pay coupons, only the principal at maturity. Such bonds are called zero coupon bonds. The number of years until the face value of a bond is paid is called the bond's time to maturity.

ii. **Debentures**

A debenture is a type of bond that has no specific charge on the assets of the company issuing it. The holder of this bond depends mainly on the earning power of the issuing company for the safety of the bond. In the event that the company defaults in payment or goes bankrupt, investors in debenture become general creditors and are treated as such. Since debentures are unsecured, they offer attractive returns to investors. Large companies with excellent credit standing such as Unilever, Ghana Breweries, Anglogold Ashanti are those most likely to issue debentures instead of secured bonds.

Summary

Short-term investments instruments are good for investors who are very much concern with liquidity. It is popular among investors who wish to earn something on idle funds in the short-run while considering long-term investment. It is normally associated with conservative investors because of the low risk associated with such investments. Remember, the instruments considered were mutual funds, certificate of deposit, treasury bills and commercial paper.

This is a blank sheet for your short notes on

- Difficult topics if any; and
- Issues that are not clear

SESSION 4: INVESTMENT INSTRUMENTS (CONTINUED)

Overview

In the preceding session, we discussed some instruments that investors can use to achieve their investment objectives. We are introducing you to additional instruments in this session and wish you well in all your investment endeavours as you read on.

Objectives

By the time you finish this session, you should be able to:

1. Explain the various types of option.
2. Explain how an option works out as an investment instrument.
3. Explain real assets as an investment instrument.

Read on now…

Convertible Securities

A convertible security is a fixed income security that can be converted into equity at the option of the holder. Thus, the holder has a fixed-income security that can be transferred to ordinary shares if and when the performance indicates that such a conversion will be profitable. The most common of convertible securities are bonds, with preference shares and warrants following in that order. The characteristics of convertible securities are that, while they are sold initially as fixed-income securities receiving an appropriate rate of interest, the holder is given the option of converting this security within a given time period into ordinary equity shares at a specified price. The advantage to the company of this form of security is that, initially if it is a loan (bond) it helps the company's gearing and the company also enjoys tax shield, which reduces cost of capital in the years when it remains a debt. The holder of the convertible security has the advantage of waiting to see how the firm's shares perform before deciding whether to invest in equity. In general, the best time to buy a convertible is when interest rates are high and stock prices are relatively low.

Derivative (Speculative) Investment Instruments

Derivatives are securities whose value is derived from the value of some underlying assets such as stocks and bonds, or the relationship between several other assets. The speculative investment vehicles include options, futures and swaps.

i. Options

These are rights (but not obligation) to buy or sell an asset at a fixed price within a specified period of time. This definition is general and applies to puts, calls, warrants

and real estate options. An option is a contract giving the holder the right to buy or sell a stated security at a specified price, called the strike or exercise price on or before a specified date. Options can be a call or put option. A call option gives the holder the right to purchase securities at a fixed price during a set period of time. A put option on the hand gives the holder the right to sell a given number of securities at a specified exercise price during a set period of time. Note however, that these are just rights, and that the holder of the right may choose not to exercise the right since he/she is not obligated to exercise the right. Options are usually purchased with the motive of taking advantage of a possible decrease or increase in share prices.

But how do investors benefit from options? Suppose an investor buys an option to purchase SIC Company shares for 12 Ghana pesewas today (June 1, 2008). The expiration date is November 1, 2008, while the exercised price is 60 Ghana pesewas. Until this contract expires on November 1, 2008, the investor who purchased this call option is entitled to buy SIC shares for 60GP. Now, let us assume that on the day the option was purchased (June 1, 2008), SIC shares sold for 55GP If the share price stays below the 60GP, the call would expire without it being exercised and the investor would lose the 30GP he paid for the call. However, if on expiration of the option, SIC sells for 80GP, it would be profitable for the investor to exercise the option since he/she would pay 60Gp for shares worth 80GP. The mathematics of the above is shown below:

Proceeds from investment in options = share price – exercise price
= 80GP – 60GP = 20GP
Profit from investment = proceeds from investment-original investment
= 20GP – 12GP = 8GP
This represents a 66.7% holding period return in six months.

Notice that a put option works in a reverse manner. That is, a put option is exercised only if the share/asset price is less than the exercise price. When an option results in a positive payoff for the investor, it is described as *in the money*. On the other hand, if an option would produce a negative payoff to an investor when exercised, it is referred to as *out of the money*. Finally, when the exercise price and the asset price are equal, it is termed *at the money*. Note also that there are organized exchanges for options, but some options trade on over-the-counter markets.

A right is a type option that allows the holder to buy a percentage of newly issued shares of a company below the current market price over a short period of time, usually up to two months. A warrant is also a type of option similar to a right. Warrants give the holder the opportunity to buy shares of companies above the initial market price over a period, typically 2 to 10 years or even more.

ii. **Futures**

A futures contract is an agreement between two parties to exchange at some future date an asset for cash at a price that is fixed today. Futures are like options in the sense that they specify the price at which an asset could be bought or sold at some future date. For example, if you agree on January 31 to buy 100 ounces of gold at a price of GH¢400 on July 31 from your local goal dealer, you have created a six-month futures contact. Futures are more liquid in that they are traded on a regulated exchange. The exchange guarantees the integrity of the contact because if one of the parties goes into bankruptcy it will not affect the other party. With futures contact the broker will require an initial deposit in a margin account. The account is debited or credited with losses or gains as the futures price changes. What this means is that even though technically the futures contract mandates the delivery of an asset at a future date, delivery rarely happens. This is because the parties involved usually close out their positions before the expiration of the contract, resulting in profits or losses.

Investors make profits and losses by a process called *marking to market*. This involves the establishing of a margin account by the two parties at the beginning of the contract. The margin act as a guarantee or security account to ensure that parties to the futures contract discharge their obligations. The margin account may consist of cash and/or near cash securities. The futures exchange does not wait till maturity date before investors realize gains and losses. Rather, it is done daily. This means that funds are taken from the margin account to offset changes in prices. Put differently, the futures contract price is adjusted each day as the futures price for the contract changes. Therefore, actual daily cash settlements occur between the investors (buyers and sellers of contract) in response to this marking to market process.

It is worth noting that with futures contract, the person who commits to buying a commodity is said to have taken a *long position*. The person committing to selling at the delivery date, on the other hand is said to have taken a *short position*.

iii. **A Swap**

According to Hirscey & Nofsinger (2008), swaps are arrangements whereby two companies agree to lend to each other on different terms, such as in different currencies or at different interest rates. In other words, a swap is an agreement between two parties to exchange cash flows at a stated future times according to the terms of the agreement. Swaps are usually entered into for currency and interest rates, but there are commodity and equity swaps too

Real Assets

The last investment instruments we want to talk about are real assets. Real assets include gold, art works, real estate and land. These are assets that can be touched and

seen. They normally have high unit values. Most of the time, they appreciate in value over time and therefore good for long term investment. However, they may have low liquidity compared to assets such as securities. Therefore, investors with immediate liquidity needs may not find this type of asset attractive. Investment in real estate such as residential houses and warehouses would give the investor continuous flow of return in the form of rent. Real estate business is a major investment in most developed countries where condominiums put up large building made up of flats and rent them out to individuals and organizations. Generally, increases in asset values and favourable tax treatments enable real estate investments to offer decent returns in the form of rental income, depreciation write–offs, and capital gains that are mostly not available with other investment instruments. Some investors use gold as a hedge against the dollar.

Summary

Once again, we want to emphasize that the selection of the right investment instruments are key to reaching your investment goals. In fact one may have the most comprehensive investment objectives in place but if the right instruments are not selected, it will come to naught. We hope you have mastered the issues and would respond appropriately when called to do so. We wish you all the best

SESSION 5: STEPS IN INVESTMENT

Overview

Undertaking an investment is a very important decision in life. Therefore care must be exercised to ensure the success of such investments. After going through how to formulate investment objectives and also going through the various instruments at your disposal, we are ready to take you though the steps in actualizing your investment agenda. In fact, a careful consideration of this session could be the difference between a successful investment and a failed investment venture. We have attempted to explain some of the steps to follow when undertaking an investment.

Objectives

By the time you finish this session, you should be able to:

1. List and explain the various steps in the investment process.
2. Distinguish between constructing a diversified portfolio and managing an investment portfolio.

Read on now…

The following issues must be carefully considered in order to achieve investment objectives.

i. Meeting Investment Prerequisite

Before you undertake any investment, the investor must make sure the basic necessities of life are met. You cannot invest your entire fund and starve to death. The investor must ensure that the basic needs of food, clothing and shelter are provided for the family before thinking of investing other than that he/she will have to sell the investment to meet these necessities. The investor can establish a savings account from which contingencies can be met. Such account should be readily accessible to avoid financial problems when the need arises. The investor can also ensure against losses such as death through accident, illness or negligence on the part of a third party, disability, loss of property through fire and theft by taking insurance policies

ii. Establishing Investment Goals

For an investor to be successful, he needs to establish specific investment goals and work towards achieving those goals. Without a target, the investor will not have something to work towards achieving. For example, the investor may have the following goals; accumulate five hundred thousand Ghana cedis within the next ten years (2008-2017) to supplement retirement income, or accumulate one million Ghana cedis by

2020 to build a house. In setting such targets, the investor must be sure that the target can be met considering his financial status, investments instruments available and the return of such investments. For instance, a worker whose annual income is two thousand Ghana cedis cannot achieve an investment target of twenty thousand Ghana cedis in 5 years when the highest return of all available investments is say 15%. This is because even if he invest all his income he is not likely to get that amount. Therefore when setting investment goals the investor should be realistic as much as possible looking at his financial standing and the returns of available investments

iii. **Evaluating Investment Vehicles**

Once the investor has set a target, he/she must evaluate the available investment vehicles with his financial goals and select which investment vehicles that can help to achieve the set goals. The investor should therefore determine assets/investments that will be eligible. The investor then should assess the expected return over a holding period and risk associated with each investment.

iv. **Selecting Suitable Investment**

The selection process is very important because it determines a course of action. Whether the investor will be successful in achieving his/her investment goals will depend on the type of investment vehicles that were selected. When selecting an investment, the investor may have to consider not only the return and risk of a particular investment but also other factors such as tax liability and liquidity. For example a student investing to meet next semester's school fees should consider liquidity and risk in terms of loss of principal as very important factors.

v. **Constructing a Diversified Portfolio**

A portfolio of investment is a collection of investments to meet a set investment goal. The investor will have to combine assets in certain proportion to form a desired investment portfolio. One factor that should be considered in portfolio construction is diversification. Diversification leads to a higher return at a possible minimum risk. The portfolio that gives the possible minimum risk is known as minimum variance portfolio. The minimum variance portfolio is normally good for risk-averse investors who want to get a return higher than Treasury bill's (risk-free) rate. The investor can also use optimization techniques to determine investments that will be suitable for a portfolio mix. This is because certain assets on their own may be risky but when combined with other assets reduce the risk drastically. In constructing a portfolio, the investor will have to consider the covariance of the assets that form the portfolio because covariance determines the level of risk of a portfolio.

vi. **Managing the Portfolio**

Once a portfolio of investment has been formed, the investor must monitor the portfolio to make sure the targeted goals are met. The investor has to measure the actual returns of the portfolio in relation to expected performance. If there is deviation from the set target, the investor will need to take corrective measures. This may include selling certain investments and using the proceeds to acquire another investment or changing the portfolio mix. In certain situations, it may mean selling the entire portfolio and investing in a new set of assets altogether.

Summary

We have attempted to walk you through the actual process of investing. In other words, we have explained the key steps in the investment process. These steps are Meeting Investment Prerequisite, Establishing Investment Goals, Evaluating Investment Vehicles, Selecting Suitable Investment, Constructing a Diversified Portfolio, and Managing the Portfolio. Can you explain each of the above in your own words?

This is a blank sheet for your short notes on:

- issues that are not clear; and
- difficult topics, if any.

SESSION 6: INVESTMENT PROCEDURES IN GHANA

Overview

The first five sessions of this unit has been devoted to investing in an already established business. Having taken you through the fundamentals of such investments, we conclude the unit with a look at the procedures to follow if one wants to invest in Ghana by setting up a firm or going into a joint venture. Notice that the procedures are for both Ghanaians and foreigners alike, but where there is a difference, it shall be highlighted.

Objectives

By the time you finish this session, you should be able to:

1. Explain procedures involve for a Ghanaian investor who intends to set up a firm as an investment venture.
2. Explain the steps in Foreign Direct Investment in Ghana.

Read on now …

The source of date for this session is *http://www.gipc.org.gh*. Under the Ghana Investment Promotion Centre Act, 1994 (Act 478), investment in all sectors of the economy, other than mining, petroleum, free-zones and portfolio investments, can be established without prior approval by Ghana Investment Promotion Centre (GIPC). Mining and petroleum sector projects have to be approved or licensed by the Minerals Commission and the Ministry of Mines and Energy, respectively. Free-Zone operations are administered by Ghana Free Zones Board while portfolio investments are handled by the Ghana Stock Exchange.

Foreign investors intending to invest in Ghana are by law to register with the Ghana Investment Promotion Centre under the GIPC ACT 478. The registration will be concluded only when the legal minimum equity contribution has been met.

To satisfy the minimum equity requirement, the investor must follow the following steps:

STEP 1 Incorporate a company at the Registrar General's Department.
The department has five (5) working days to complete formalities if all documents are in order.

STEP 2 Foreign investors should comply with the GIPC Act 478 regarding minimum equity requirements either in cash or in kind as follows:

a. Open a bank account in the name of the company.

b. Effect a Bank to Bank transfer of minimum equity requirement. This transaction should be confirmed to the Bank of Ghana by the investor's local authorized dealer bank. Bank of Ghana in turn, confirms this transaction to GIPC for the company's registration purposes.

c. Physical cash carried into Ghana by individuals for investment purposes should be declared on Bank of Ghana Form T5 on arrival and subsequently deposited in a bank account within the shortest possible time. This transaction should be confirmed by Dealer Bank and the Bank of Ghana as in (b) above.

d. Enterprises are free to implement their projects by importing the relevant plant, machinery and equipment. Zero rated and concessionary duty items should be cleared automatically and directly through CEPS. Essential plant, machinery and equipment which fall under Section 24 of the GIPC ACT 478 should be cleared with the GIPC

e. In the case of equity in kind in the form of imported machinery, equipment and goods, all documents covering such imports should be in the name of the registered company and evidenced by the following which should be submitted to GIPC for registration purposes:

- Bill of lading/Airway bill (original)
- Destination (Ghana) Inspection Certificate.
- Customs Bill of Entry (original)
- Import Declaration Form (IDF)
- Certified/Final Invoices.
- Evidence of Capitalization - Form 6 from the Registrar General's Department.

STEP 3 The investor then registers with the GIPC which has five (5) statutory working days to complete the registration process, provided the registration forms are in order.

STEP 4 Wholly Ghanaian-owned enterprises do not need to register with GIPC since the minimum foreign capital requirement does not apply to such enterprises.

STEP 5 All wholly Ghanaian-owned enterprises and enterprises with foreign participation seeking immigrant quota facilities in respect of expatriate personnel (experts) for their businesses should satisfy the relevant minimum capital requirements specified under Section 30 of Act 478.

STEP 6 All enterprises must register directly with the Internal Revenue Service and the Value Added Tax (VAT) Secretariat for purposes of statutory tax e.g. taxes, rebates, and exemptions thereof.

STEP 7 Enterprises must register and obtain an environmental permit from the Environmental Protection Agency (EPA).

For all of the above the GIPC may assist as the need arises, either directly or through referral to accredited management, accounting or legal firms.

Summary

We have successfully taken you through the procedures involved in setting up a firm as a means of investment. You should master the procedures since you would be answering questions on it. As a student of investment, you may be called upon to advice prospective investors, both local and foreign, and this should be within your reach.

This is a blank sheet for your short notes on:

- **issues that are not clear; and**
- **difficult topics, if any**

UNIT 2: SECURITIES MARKET

UNIT OUTLINE

Session 1: Types of Securities market
Session 2: Functions of the Securities Market
Session 3: The Ghana Stock Exchange
Session 4: Brokerage Business
Session 5: Functions of Brokerage Business and Section of Brokerage Business
Session 6: Types of Orders

Overview

You are welcome to unit 2. We hope you enjoyed reading the first unit and will find this unit enjoyable. Some of us might have investments in the form of stocks and bonds. How do we acquire these investments? The acquisition is made possible through the securities market which brings people with excess funds together with firms/organizations in need of funds. In this unit, we are going to discuss the operations of the securities market.

Objectives

By the end of this unit, you should be able to:

(a) Distinguish between the various types of securities markets.
(b) Outline the functions of the securities markets.
(c) Explain brokerage business and its functions.
(d) Explain the different types of orders in the brokerage business.

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 1: TYPES OF SECURITIES MARKET

Introduction

You are warmly welcome to the first session of unit 2. In this session, we are going to discuss the securities market and the types of the securities market. I hope you will understand the concept of the securities market and the various types of securities that we have.

Objectives

By the end of this session, you should be able to:

(a) Explain what is securities market
(b) Distinguish between the various types of securities market.

1.1: Meaning of Securities Market

The term 'securities' according to the dictionary of international insurance and financial terms, has two meanings.

(1) It is anything (usually property) pledged as collateral against a loan, or the documents that sets out the terms of such collateral.
(2) It is any financial instrument that is traded on the stock exchange and that yields an income. Securities represent a loan that would be repaid at some time in the future. In this sense the word is most often used in the plural and securities include bills of exchange, bonds, debentures, gilt-edged securities, options, shares and stocks.

The securities market is a market where the above mentioned financial instruments are traded. In other words, it is a market for bills of exchange, bonds, debenture, gilt-edged securities, options, shares and stocks. The securities market can be divided into two main categories. These are the ***capital market***; which can be divided into two sub categories namely the primary and the secondary market and the ***financial market*** which can also be sub-grouped into money market and capital market.

1.2: Types of Securities Market

The securities market can be divided into different segments based on the characteristics of the securities being traded or the type of transaction that takes place in the market or the needs of investors. For example we have Money market and Capital market. The securities market can also be viewed as Primary or Secondary market.

1.2.1: Money Markets

The money market is designed for the provision of short-term funds. It is an institution through which corporations and individuals with funds meet the needs of borrowers with temporal shortage of funds. A security or loan maturing within one year or less is considered as a money market instrument. One major function of the money market is to finance the working capital needs of corporations and provide the government with short-term fund. The money market also supplies funds for speculative buying of securities and commodities.

1.2.2: Capital Market

Capital markets are designed to finance long-term investment by government, corporations and households. Securities traded in the capital market take more than one year to mature and range in size from small loans to multimillion credits. For instance, if the government wants to undertake road construction, it will have to go for a long-term loan. A company that wants to embark on an expansion of its factory will also need to finance such an activity through a long-term source of fund such as issue of stocks or contracting long-term loan through the capital market.

1.2.3: Primary Markets

Companies can raise external funds through borrowing or issue of shares**.** If securities are issued they are sold in the Primary Market.

The Primary Market is purposely for trading of new securities (IPOs) never before issued. That is to say, securities available for the first time are offered through the Primary Market. The principal function of the Primary market is to raise financial capital to support new investments in building, equipment and inventory.

Though companies can sell directly to the general public, they usually use underwriters (investment bankers) who handle the details of the new issue and sale to the investor. Underwriters normally buy the new issue at an agreed price from the issuer and sell it at a higher price to the public. Sale through the investment bankers (underwriters) can take the form of “best-efforts” or agency agreement. Under such agreement, the investment banker does not buy the new issue but uses his best effort to sell it. Any unsold securities are returned to the issuer.

1.2.4: Secondary Markets

Once an investor purchases new issues, they change hands in the Secondary Market. Secondary markets are for existing securities that are currently traded between investors. These markets create price and allow for liquidity. If secondary markets did not exist, investors will have no place to sell their investment. Without liquidity many investors will not invest al all. Thus, the chief function of the secondary markets is to provide an avenue for converting financial instruments into ready cash (liquid asset). Volumes of trading in securities take place in the secondary market than the primary market. However, the secondary market does not support new investment. Broadly,

there are two types of secondary markets namely organized exchanges (Stock exchange) and over the counter (OTC) markets.

Summary

In this session we have discussed the securities market. We looked at the meaning of securities market. We also considered the different types of securities market and the role that each type plays.

Self Assessment Questions

Exercise 2.1

1. Explain the term ‘securities market’.
2. Distinguish between the following
   a. Money market and Capital market
   b. Primary market and Secondary Market

This is blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 2: FUNCTIONS OF THE SECURITIES MARKET

Introduction

You are once again welcome to session 2 of unit 2. In our previous session, we discussed securities market and the types of securities market. In this session, we are going to discuss the functions of the securities market.

Objectives

By the end of this session, you should be able to:

(1) Explain the benefits of the securities market to the economy.
(2) Outline some benefits of the securities market to the investor.
(3) Explain the benefits of the securities market to business organizations.

Enjoy your reading

2.1: Functions of the Securities Markets

The securities market performs very important functions that benefit the economy, the corporations and the individual investors. It also performs other general functions relating to investment. Below are the functions/benefits of the securities market to the economy, investors,

2.1.1: Gains of the Securities Markets to the Economy

a) It improves utilization of existing domestic savings and encouraging new savings. The existing of the securities market for example the Stock Exchange makes it possible for people to invest their savings.

b) It helps to channel these savings to where they are needed most. The stock market makes it possible for firms performing well to attract more investors. The prices of stock of such firms go up. Investors sell the stock of firms that are not doing well and buy the shares of companies doing well. By so doing, funds are transferred to where they are most needed, that is to firms that need the funds to make good use of these funds.

c) It also attracts foreign investors through portfolio investment. The Ghana Stock Exchange is open to foreign investors. Therefore, foreigner with surplus funds can invest in Ghana through the exchange. Without it such investment would not be possible.

d) It assists in the privatization process. The Ghana Stock Exchange made the divestiture process that the government of Ghana embarked upon in the 1990s possible. The government was able to sell its stocks in the divested companies through the exchange. This made the divestiture process easy and made shares more accessible to the public.

2.1.2: The Benefits of the securities market to the Investor

a) It enables investors to spread their risk through diversification. The existing of a stock market makes it possible for an investor to spread and diversify his/her risk by holding a well-diversified portfolio.

b) It provides a liquid investment opportunity. Liquidity is very important to every investor. This is because every investor wants to convert his/her investment quickly into cash when the need arises. The existing of stock market makes it possible because any investor with a security that is traded on the security market can easily dispose off such asset when the need arises without having to lose large part of the principal investment or having to incur high commission.

2.1.3: The Benefits to the Corporation

a) Securities market (stock exchange) makes access to investment capital possible. Firms in need of investment capital can raise it through the exchange.

b) It provides opportunity for fast growing and young companies to obtain finance. Such companies can raise capital through the issue of shares (initial public offer).

c) It creates an opportunity for corporations to become known to national and international investors. Former Ashanti Goldfields Company (AGC, now Anglogold Ashanti, is listed on the New York Stock Exchange. This has made it an international firm making it possible for people who do not know Ghana to know and invest in AGC.

2.1.4: Other Functions of the Stock Market

a) It provides continuous market for the purchase and sale of securities.

b) It provides a mechanism for determining a fair market price for securities traded on the exchange.

c) It imposes some standardization regarding the release of financial information by companies whose securities are traded on the exchange. For example, all listed companies are supposed to publish their audited final accounts each year.

d) It attempts to protect investors who maintain account at member brokerage firms. It does this by regulating trading practices and imposing financial standards upon member firms.

SUMMARY

Summary

In this session, we have looked at the functions of the securities market. We considered the significance of the securities market to the investor, business organizations and the economy as a whole.

Self Assessment Questions
Exercise 2.2

1. State three functions of the of the securities market to the economy.
2. Outline two benefits that business organizations are likely to get from the securities market.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics, if any

SESSION 3 THE GHANA STOCK EXCHANGE

Introduction

This is session 3 of unit 2. In session 1 we discussed different types of the securities market. We said that one of the secondary markets is the Stock Exchange. The other one is over-the-counter market. A stock exchange is an organized market for trading securities. In this session we are going to discuss the Ghana Stock Exchange.

Objectives

By the end of the session you should be able to:

1. Get the historical background of the Ghana Stock Exchange
2. Outline the listing rules of the Ghana Stock Exchange
3. Comprehend the membership of the Exchange
4. Have an insight into the characteristics of listed companies

3.1: The Ghana Stock Exchange

3.1.1: Historical Background

The Ghana Stock Exchange was incorporated in July 1989 as a private company listed by guarantee under the Ghana Companies Code of 1963 (Act 179). The following year, in October 1990, the GSE was given recognition as an authorized stock exchange under the Stock Exchange Act of 1971 (Act 384). On 9th January 1991 the Council of the GSE with approval of the Secretary for Finance, brought into being regulations governing the Exchange.

3.1.2: Performance of the Ghana Stock Exchange

The Ghana Stock Exchange has performed relatively well as an emerging market since it started operating in 1990. In 1993, the GSE was the sixth best performing emerging stock market index with capital appreciation of 113.74 percent. In 1994, it was the best performing market among all emerging markets with an appreciation of 124.3 percent in its index level. The international financial community described the GSE as the world's best performing stock exchange in 1998. According to London-based financial analysts, the GSE All-Share Index, which is used as a measure for the performance of the market, increased by more than 60 percent in dollar terms. This was repeated in 2003 when the market was judged the best performing stock market with capital appreciation over 140% in dollar terms. In 1994, the GSE was one of the biggest exchanges in sub-Saharan Africa with market capitalization of over USD2 billion.

Steady improvements have been made in trade volumes and values. From 1990 to 1992, only four million shares at about 350 million cedis were traded on the Exchange. However, in 1994 alone, 93 million shares worth 73 billion cedis were traded. Trading activities however declined in 1995 and only 55 million shares worth 27 billion cedis were traded. Primary market activities on the Exchange have also been increasing. For

example, over 124 billion cedis were raised through the Exchange in 1996. Between January and September of 1998, total shares of 55 million were traded valued at 93.4 billion cedis.

3.1.3: Characteristics of Companies Listed on the Ghana Stock Exchange

The shares of 37 listed companies are currently traded on the GSE with a market capitalization of GH¢15,030.27m (May 2008). The composition of the firms listed on the Exchange is as follows; finance/insurance 11 companies, paper converters companies, manufacturing 6 companies, agriculture/agro processing 4 companies, metals/oil 5 companies and pharmaceutical/beverages 6 companies. Ownership of the companies listed on the stock exchange varies from company to company. The company with the highest foreign ownership is Paterson Zochonis Ghana Limited with 94.45 per cent of foreign ownership and 5.55 per cent Ghanaian ownership. Mechanical Lloyd Company Limited has the highest Ghanaian ownership of 99.99 per cent with only 0.01 percent foreign ownership

3.1.4: Membership

Membership of the GSE consists of licensed dealing members (LDMs) and associate members. Licensed dealing membership is open to companies incorporated under the Companies Code 1963 (Act 179) and partnerships incorporated under the Incorporated Private Partnerships Act 1962 (Act 152). There are 16 stock broking firms (LDMs), which are licensed to deal in securities listed on the Exchange. These LDMs serve both local and non-resident investors. Associate membership is open to individuals, bodies of persons or professionals who are law abiding in the opinion of the Council of the Exchange. There are 41 associate members. The licensed dealing members have the right (and license) to deal in securities but associate members do not.

For a company to qualify as a member of the Exchange:

1. Its regulations should be approved by the Council of the Ghana Stock Exchange;
2. Each of its directors enters into an agreement with the Exchange to be jointly and severally liable for the debts and obligations of the company;
3. All its directors have recognized professional or academic qualification or experience in banking, law, accountancy, economics, business administration, secretarial practice, dealing in securities or any other relevant qualification or experience acceptable to the Council;
4. Its sole business is dealing in securities and business activities relevant to such dealings;
5. It is solvent;
6. It has the required financial resources to meet its obligations as a member of the Exchange.

For a partnership firm to qualify as a member;

1. The firm should be a registered firm under the Incorporated Partnership Act 1962 (Act 152); each partner must be at least 21 years of age;
2. Each partner has the required professional and academic qualifications or experience;
3. No partner has been convicted of an offence involving fraud;
4. Partners are resident in Ghana and the firm has the required financial resources to meet its obligations as a member of the Exchange.

3.1.5: Governing Council

A Council governs the Exchange. The Council is made up of representatives from the body of LDMs, listed companies, banks and the public. The managing director of the Exchange is an ex-officio member of the Council. The Council sets the policies of the Exchange and its functions include preventing fraud and malpractice, maintaining good order among members, granting listings and regulating stock market business.

3.1.6: Listing Rules

A GSE-listed company is a company whose securities are traded on the Ghana Stock Exchange. An application for listing needs to be sponsored by a licensed dealing member of the Exchange. The sponsoring member shall:

1. Ensure that the Council is made aware of all the information which should be brought to its attention;
2. Be responsible for filing all the needed documents with the Exchange;
3. Satisfy itself that the applicant is suitable for listing;
4. Examine the composition of the Board of Directors of the applicant to ascertain whether they have the requisite qualification, experience and skill for efficient functioning of the board.

The sponsoring member may act as a market maker in the securities of the company it sponsors. As a condition for listing, an applicant must comply with the listing rules and regulations of the Exchange and accepts continuing obligations that will apply after the admission of its securities on the Exchange.

A company applying for listing of any class of its shares is expected to have a stated capital at list of one million Ghana cedis (GH¢1,000,000) in the case of application relating to the first list and five hundred thousand Ghana cedis (GH¢500,000) in the case of an application to the second list. For a company's securities to be eligible for admission to the first list, the company must have published or filed accounts for at least five years preceding the application in accordance with the Companies Code 1963 (Act 179). For a company's securities to be eligible for the second list, the company must have published its accounts for at least three years preceding the application. In each case, the company must have made a reasonable profit throughout the period specified for each category.

3.1.7: Trading procedure

a) Trading shall be conducted under the Call-over system, or such other system as the council may decide from time to time.
b) The Authorized Dealing Officers shall assemble at fixed times on the floor and a designated official of the Exchange shall preside over the transactions on the floor and direct the conduct of business.
c) As the Presiding Officer calls the list of quoted securities, ADOs may shout their offer or bid prices and the number of securities involved as the case may be.
d) The highest bidder is satisfied with the maximum number of his bid at his price before the next bidders are awarded allotment at their respective prices.
e) When offers are invited during trading, ADOs would be given the chance to adjust their prices downwards up to the lowest offer price before bids are invited.
f) When bids are invited ADOs will be given the chance to adjust their prices upwards to the level of the highest bid before allocation is made.
g) When a buyer and a seller agree on a price, a bargain is completed and the price is marked on the Price Board. Continuous markings shall be made of bargains done during such times, as the floor remains open for trading. Only prices marked shall be considered to be official prices for the day.

When a bargain is completed, the facts relating to the transaction, that is, the date, the name of the security, the amount and the price, shall be duly recorded on the Bargain Slip approved by the Council and both parties to the bargain shall sign respectively as ADOs for buyer and seller.

A copy of the Bargain Slip shall, immediately after the transaction, be handed over to the designated presiding official at the Call-over.

Authorized Dealing Officers must enter all bargains in the appropriate Market Books. Market Books of dealers may be examined by the Managing Director in the event of the need to investigate discrepancies in the records of two dealers with regards to trade.

Every licensed dealing member should, within 48 hours after completing a transaction on behalf of a client, forward to the client a contract note setting out the date of the transaction, the number of units of the securities bought or sold, the price, the amount of commission and the amount of statutory duty, if any, to be paid by the client in respect to the transaction.

All delivery is made or tendered at the office of the Licensed Dealing Member effecting the purchase on a business day before bank closing time that day. It is the responsibility of the seller to ensure that all documents in connection with the security are genuine.

3.1.8: External investors

Any foreign investor who wishes to invest through the Exchange by purchasing shares of listed companies on the Exchange is regulated by the Exchange Control Act 1961, (Act 71). According to the Act, an external resident (individual and institutional) investor cannot hold more than 10 per cent in one security listed on the Exchange. The total holding of all foreign investors in one security should not exceed 74 per cent of that security. External investors are allowed full and free foreign exchange remittability on the original capital or principal amount invested, any capital gains, dividends and interest payments and related earnings and refunds. Capital gains are tax-exempt (up to November 2005) but a withholding tax of 10 per cent is charged on all dividend income.

Summary

In this section we discussed the Ghana Stock Exchange. We discussed the historical background of the Exchange, the performance of the Exchange, governing council of the exchange, listing rules among others.

Self-Assessment Questions

(1) State five (5) conditions for companies to fulfill in order to qualify as members of the Ghana Stock Exchange.

(2) Discuss the characteristics of the companies listed on the stock exchange.

(3) What are the trading procedures on the Ghana Stock Exchange?

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any

SESSION 4: BROKERAGE BUSINESS

You are welcome to session 4 of unit 2. In this session we are going to discuss brokerage business. Security holders cannot sell their securities to investors who are prepared to purchase such securities. In the same vein individuals and corporate investors cannot go to the Stock Exchange to transact business. They have to do so through the broker. Brokerage business is, therefore, very important because it acts as an intermediary between buyers and sellers of securities.

Objectives

By the end of the session you should be able to:

1. Identify the various types of brokerage firms.

4.1: Types of Brokerage Firms

Brokerage firms and dealers in securities can be classified according to the three broad functions they perform. They function as:

4.1.1: Investment Bankers

Brokerage firms act as investment bankers because they sell securities to the public without using the facilities of an exchange. Investment bankers normally buy new issue from the issuer at an agreed price and hope to resell it at a higher price to the general public. In this capacity, investment bankers are said to underwrite an issue. Sales through an investment banker can take the form of *best-efforts* or agency agreement. In this case, the investment banker does not underwrite the issue but use his best effort to sell it. Any unsold securities will be returned to the issuer. This best-efforts activity will normally be used in the sale of new and small companies' shares that are thought to be highly risky.

4.1.2: As Buyers and Sellers of Securities on Behalf of Customers

A brokerage firm functions as a buyer and seller of securities for customers. It is this function that most brokerage firms are known by the investing public and the function with which the individual investor comes in contact with the broker. A broker trading in listed securities is acting as an agent on behalf of a client. Compensation for this service is in the form of commission. It is important that the brokerage firm exercises care and demonstrate a reasonable amount of skill in filling customers' orders. The brokerage firm may be liable for any losses that result from its mistake. The care with which brokerage firm issues orders is determined by what is reasonable practice in the brokerage business. The exercise of broker's skill requires that instructions are followed and the order placed in the market where the securities are traded fastest possible time. The broker is required to refrain from making secret profits or crossing orders in its office by acting as both dealers and brokers in the same transaction. The brokerage firm cannot act as dealer and broker in the same transaction because there could be a conflict

of interest and this could result in the client paying double commission. All Securities listed on an exchange must be traded on the floor of that exchange. They cannot be executed off the floor by the broker except in certain circumstances.

4.1.3: As Principal Making Markets in Securities

The broker functions as a principal and makes markets in securities. The broker's main principal activity is to bring sellers and buyers together. Thus, the brokerage firm is a middleman. As a true intermediary, brokers bring sellers and buyers together, thus creating a market. This is generally done through over-the-counter issue, but can also take place in listed securities. If the securities to be traded are not listed on an exchange but traded in over-the-counter market, the broker might own the shares himself. He/she will be acting as a principal or dealer in the transaction. Many brokerage firms specialize in making a market in a certain securities. In this case the brokerage firm will sell the security to the customer at the asking price and will not charge a commission for handling the transaction. The brokerage firm makes its fee from the difference between the price it pays for the securities for its own account and the price it sells them to the investor. The difference between the asked and the bid price is called the spread and is the compensation for making a market in that security.

Summary

In this session we discussed the types of brokerage business. The brokerage firms act as investment bankers, as buyers and sellers of securities on behalf of customers and as principals making markets in securities.

Self Assessment Questions

Exercise 2.4

1. Discuss the function of brokerage firm as an investment banker.
2. How does the broker act as an intermediary between the seller and buyer of securities?

SESSION 5: FUNCTIONS OF BROKERAGE BUSINESS AND SECTION OF BROKERAGE BUSINESS

Introduction

This is the fifth session of unit 2. In this session we are going to discuss the functions of the brokerage business. We will also look at the factors we have to consider when it comes to the selection of a brokerage business.

5.1: Functions of a Brokerage Firm

A traditional function of a brokerage firm is providing customers with information to help them purchase suitable securities. Most brokerage firms inform their customers about individual companies and industries. Many brokerage firms believe that they cannot serve their client adequately, unless they provide them with information. These are some of the services that are provided by brokerage firms to their clients.

1) Most brokerage firms continue to provide research report on individual companies and industries to their clients. Analysts in the research department of the firm normally carry out the research. The information is made available to retail clients through published reports available in the broker's office.
2) Many large brokerage firms provide economic forecast. It is very difficult to invest successfully without knowledge about how the economy will be doing. It is therefore essential for investors to know the overall economic conditions within which companies operate. Knowing the economic condition will help the investor to make a right choice of investment. Some companies do well when the economy is doing well, for example the construction industry. Therefore if an investor have knowledge that there is going be economic boom, investing in a constructing firm will be an advantage.
3) Many brokerage firms analyse portfolios and recommend specific changes that need to be made to meet the investor's objectives. For example, an investor who wanted recommendation on securities he owns could ask a broker to appraise his portfolio. The analysis will enable the broker to make any recommendations on the portfolio as to whether it needs reconstruction or not.
4) Few brokerage firms also provide indirect investment management services. For example an investor who feels that he cannot manage an investment properly may ask a brokerage firm to recommend an investment advisor. This is particularly true for people who have little knowledge about investment management or people who are highly paid and therefore find it financially wise to manage their own portfolio of investments. Most brokerage firms that provide this service continually employ the services of outside investment advisory firms.

5.2: Selecting a Brokerage Firm

There are a number of factors to consider when it comes to the selection of a brokerage firm. These factors include:

1. The investor's first step in establishing a satisfactory relationship with a broker is to choose a firm that is suitable for his investment needs and to select a representative of the firm with whom he can work. Both the brokerage firm and the representative (broker) must be suitable for the investor because they cannot work independently of each other. The broker chosen should be prepared to work closely with the investor at each stage of the investment process. If the investor is new in the investment business, it is important he selects a broker who will be willing to spend time to educate him in the fundamentals of investment. In selecting a broker, an investor can seek advice from friends or his bank.
2. The brokerage firm selected must be well-known and long-established institution. A brokerage firm must have an unquestionable credit rating and it must be of sufficient size and experience to provide the necessary investment service.
3. Brokerage firms differ enormously in the type of clientele they attempt to attract. Some develop a large business with investors of modest means who are interested in buying and selling odd lots while others seek wealthier investors and institutional investors. In short while all brokers welcome all types of clients some are skewed towards particular class of investors. In selecting a brokerage firm that will best serve the needs of the investor, he should try to judge the type of clientele sought by the firm in relation to the type of investor he is.
4. Selecting a brokerage firm that provides research facilities and adequate information is paramount to a successful investment programme. A brokerage firm with full-time highly qualified and competent analysts is likely to be successful in any investment compared to a firm with part-time persons who process the analytical work of their clients and run out second-hand analyses. The information provided by the firm should contain data on the general economic conditions, industry analysis, company analysis and securities appropriate for the individual investor. The information should include a historical and economic and business forecasts for the industry selected.
5. Many brokerage firms combine their regular brokerage business wit a volume of activities such as underwriting of new securities. Such a firm will be ideal for an investor who is interested in purchasing new issues. Since order execution is of paramount importance to any investor, the brokerage firm selected should be a registered dealer who can deal in shares listed on an exchange and also knowledgeable in over-the-counter market.
6. Once the individual investor has selected a reliable firm, a qualified broker (registered representative) must be chosen. An investor can evaluate a

representative by inquiring into his business and educational background together with his investment philosophy and goals. The investor should also determine whether the broker would be available at all times during working hours and is not overburdened with other accounts that will prevent him from having adequate time for the investor. The broker should also have adequate knowledge about the securities held by an investor and should inform him of any news that is relevant to these holdings. It is important for brokers to give prompt and efficient service. That is, the broker should be able to confirm a purchase or sales within minutes and provide price quotations quickly. This calls for selecting a broker who is experienced in the brokerage business and who is working for a well-established firm with good records.

Summary

In this session we discussed two issues about the brokerage business. We looked at the functions that brokerage firms perform. These include provision of research report on firms and industries to clients, economic forecast, and investment management services among others. We also considered the factors to take into consideration when selecting a brokerage firm. The firm selected must be well- known and long established, provide research facilities and adequate information and have qualified brokers as employees.

Self-Assessment Questions

Exercise 2.5

1. You are an investor who is considering selecting a brokerage firm. Discuss five (5) factors you will take into consideration in the selection of the ideal brokerage firm.Outline four (4) functions of a brokerage business

This is blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 6: TYPES OF ORDERS

Introduction

You are welcome to the last session of unit 2. In this session you will be discussing the various types of orders that clients issue to brokers when it comes of execution of transactions on the stock exchange.

Objectives

At the end of this session you should be able to:

(1) Enumerate the various types of orders when it comes to trading of securities on the stock market.
(2) Distinguish between different types of orders.

Enjoy your reading

6.1: Types of Orders

There are different types of orders that an investor may give to his broker to execute in buying or selling of securities. For the broker to satisfy his clients these orders must be followed strictly. Some of the orders are discussed below.

6.1.1: Market Orders

Market orders are executed as fast as possible and at the prevailing market price on the exchange. It is simply buy and sell orders, which are to be executed immediately at the current market price. For example, an investor might call his broker and ask for the price of ABC Ltd. If the broker quotes the price as 500 cedis and 600 cedis meaning the bid price is 500 cedis and the asked price is 600 cedis. The investor might direct his broker to buy 100 shares “at market” meaning he is willing to pay 600 cedis per share for immediate transaction. Similarly, an order to sell “at market” will result in stock sale at 500 cedis. The advantage of a market order is the speed at which it is executed. The problem with market orders is that the investor may pay different prices for the same order. For example, if 500 cedis and 600 cedis represent commitment up to 1000 shares, an investor who wishes to buy more may have to pay a higher price for the rest of the order.

6.1.2: Limit Orders

In limit orders investors specify prices at which they are willing to buy or sell a security. For example, an investor may instruct his broker to buy ABC Ltd shares if and when the share price falls below 570 cedis. Correspondingly, a limit to sell order instructs the broker to sell as soon as the share price goes above the specified limit. A limit order with a price distant from the prevailing market price is said to be “away from the market”. The main advantage of limit orders is that the investor has advance knowledge of the potential transaction. It sets the boundaries of risk the investor is

willing to assume. The major disadvantage of a limit order is that it may never be executed, because limit orders are normally away from the market.

6.1.3: Time Limit Orders

Orders can also be limited by time. At the time the investor place an order, the period for which the order is to remain in force is specified. There are different types of time orders. Day orders, for example, expire at the close of trading day. Open or good-till-cancelled (OTC) orders remain in force until either executed or cancelled by the customer. However, it normally remains in force up to six months unless cancelled by the investor. As a matter of convenience to customers, a broker may accept an order for a week, month or through a specified period.

6.1.4: Stop-Loss Orders

Stop-loss order is similar to limit orders because the trade is not to be executed unless the stock price hit a price limit. Stop-loss order to sell will be executed if the price of the stock falls below a stipulated level. Similarly, stop-loss orders specify that a stock should be bought when its price rises above a limit. As the name suggests, the order lets the stock be sold to stop further losses from accumulating. These orders normally accompany short sales and are used to limit potential losses from the short position. However, stop orders can be used to protect profits.

6.1.5: Stop Limit Orders

Stop limit order is a device to overcome some of the uncertainty of the stop-loss order. In a typical stop-loss order, the investor does not know what the execution price will be after the order becomes a market order. To overcome this problem, an investor may go further to specify the limit price. An investor may fix a minimum price at which he is willing to sell but go further and specify a price below which trading cannot be executed. For example, an investor may instruct his broker to sell his stocks if the price falls below $50 but enter a stop limit to sell at $48. Nothing below $48 will be acceptable to the investor. The only danger with a stop limit order is that the order may never be executed if the market moves downwards. This may result in a huge loss.

6.1.6: Odd-Lot Orders

Orders in an exchange are normally done in established unit of trading called a round lot, which are usually 100 shares. However, it is possible for orders to be made which are less than 100 shares. This is known as odd-lots. For example if an investor wants to purchase 150 shares on the exchange, this will be a combination of a round lot and an odd lot. Odd-lot trading permits an investor who has limited financial resources to purchase modest number of shares in many companies and thus obtain some diversification for his investment. The disadvantage is the odd-lot price differential and the minimum commission charged.

Summary

In session six (6) we discussed the various types of orders when it comes to execution of transaction on the stock market. The orders discussed include Market orders, time orders, limit orders, time limits, stop-loss orders, stop limit orders and odd-lot orders.

Self-Assessment Questions

Exercise 2.6

1. Distinguish between stop loss orders and stop limit orders.
2. Explain the following:
   a. Market orders
   b. Time orders
   c. Time limit orders

This is blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

UNIT 3: FINANCIAL STATEMENT ANALYSIS

UNIT OUTLINE

Session 1: Overview of Financial Statement Analysis
Session 2: Profitability Ratios
Session 3: Liquidity Ratios
Session 4: Activity ratios
Session 5: Leverage/Gearing Ratios
Session 6: Investment Ratios

Overview

You are welcome to unit 4. We hope you have enjoyed reading the first two units and will continue to enjoy your reading of this unit. In this unit we are going to discuss financial statement analysis using ratios. When people are going to invest, they want to find out whether it will be profitable to undertake that investment. One way to find out about the firm and whether is worth investing in it is to undertake fundamental analysis of the firm. This analysis can be done through various means including financial statement analysis using ratios.

Objectives

By the end of this unit, you should be able to:

1. Explain what are financial statements
2. Outline the various users of financial statements and their information needs
3. Outline the means by which financial ratios are compared (bases of comparison)
4. Explain the limitations of financial ratios as a means of analyzing a firm.
5. Calculate and interpret the major ratios under financial statement analysis.

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 1: OVERVIEW OF FINANCIAL STATEMENT ANALYSIS

Overview

You are welcome to session 1 of unit 3. In this session we are going to discuss the overview of financial statement analysis. Issues to discuss will include what a financial statement is, users of financial statements, types of ratios and the limitations of financial statement analysis.

Objectives

After reading this session you should be able to:

1. Explain what a financial statement is
2. Outline the various users of financial statement and the use of the financial information
3. Bases of comparison
4. List limitations of financial statement analysis.

1.1: Financial Statements

The basis for financial planning, analysis and decision-making is the financial information. Financial information is needed to predict, compare and evaluate the firm's earnings ability. It is also required to help in economic decision-making such as investment and financing.

A firm communicates financial information to the users through financial statements and reports. The financial statements contain summarized information of the firm's financial affairs organized systematically. Financial Statements (Profit and Loss Accounts, Balance Sheet, Cash Flow Statements, etc.) are products, not merely for their own sake but to communicate economic measurement of, and useful information about, the resources and performances of the reporting company to those who have reasonable rights to such information.

1.2: Users of Financial Statements

These identified groups are the users of financial statements:

1. **Owner/Investor Group:** Existing and potential shareholders and stockholders.
2. **Employee Group:** Existing and potential employees including management, trade unions and past employees.
3. **Loan Creditor Group:** Existing and potential debenture holders, and loan stockholders and providers of short term secured and unsecured loans.
4. **Business Contact Group:** customers, suppliers, clients, bankers, potential mergers and take over bidders etc.
5. **Advisor/Analyst Group:** economists, statisticians, researchers, investment advisors, management consultants, credit rating agencies, etc.
6. **Government:** Central government, agencies like IRS, CEPS, Price and Incomes Board (PIB) and local government agencies.

7. **General Public:** Community, political parties, pressure groups etc. tax payers, environmentalists.

These interested groups must be able to interpret the financial statements in order to:

- Evaluate the performance of the entity to assess its viability.
- Assess the effectiveness of the entity in achieving the objectives established by its management, its owners or the general public at large.
- Evaluate managerial performance, efficiency and objectives.
- Assess the economic stability and vulnerability of the reporting entity.
- Assess the liquidity of the entity.
- Estimate the future prospects of the entity, including its capacity to pay dividends and other cash outflows.
- Attest to compliance with company law and other legal obligations.
- Ascertain the ownership and control of the entity.

1.3: Types of Financial Ratios

To evaluate the financial condition and performance of a firm, the financial analyst needs certain yardsticks. The yardstick generally used is a ratio or index, relating two pieces of financial data to each other.

The relevant ratios can be grouped under the following headings:

1. Profitability Ratios
2. Liquidity /Solvency Ratios
3. Activity Ratios/Efficiency Ratios
4. Leverage/Gearing Ratios(Long-term solvency) Ratios
5. Investment Ratios

1.4: Basis of Comparison for Analysis

Evaluating the performance of a company by the use of the various ratios would be less meaningful and beneficial unless a reliable and appropriate basis of comparison is defined.

1. Comparisons Over Time

One of the most helpful ways in which accounting ratios can be used is to compare them with previous periods' ratios for the same organization. For example, if 1994 ratios are compared with a trend of ratios from say 1989 – 1993 or a selected year's ratios, it would clearly indicate whether there has been an improvement or otherwise. This is intra-firm comparison.

2. Comparison with Other Firms

Since firms operate in a competitive environment, it is always necessary to have some basis of comparison with other organizations particularly those in the same type of business, operating in the same economic and political environment and using similar

accounting policies. This is inter-firm comparison, for example, comparing the performance ratios of Standard Chartered Bank with those of Barclays Bank.

3. Comparison with Industrial Average

The performance ratios of one firm in a particular period may be compared with what are regarded as standard ratios for the industry in which the firm finds itself.

4. Comparison with budgeted Ratios

The actual performance ratios of one firm in a particular period may be compared with the budgeted ratios to assess the extent of achievement or expected performance targets.

1.5: Limitations of Ratio Analysis

1. The selection of the appropriate basis for the comparison is as difficult as it may be misleading.
2. Different conditions exist in one business in different years; different companies may be operating under different conditions, using different policies and bases; this may make comparison difficult or may negate the usefulness for the comparisons as a means of assessing operational performance.
3. Instability of price levels may negate the potency of performance ratios as a means of assessing operational performance and make interpretation of ratios unreliable, unless adjustments are made for price level change.
4. Many accounting ratios use end of period figures which might not be a true representation of the actual situation existing over the entire accounting period. Unless weighted mean values for the period are used the correct picture may not come out.
5. If ratios are computed on the basis of historic cost accounts figures, the results obtained and their interpretation may be a variance with the current value and may not portray current existing situation.
6. Many ratios are computed from past financial statements and can therefore not be regarded as a reliable tool for future prediction, unless a number of periods' ratios are arranged for a trend to be discerned.
7. The interpretation of the ratios must take into consideration the existing circumstance and the nature of the business; otherwise the interpretation may be misleading. For example a stock turnover rate of five (5) times may be low for one business but not acceptable to another.

Summary

In this session, we discussed some aspects of Financial Statement Analysis. Among the issues discussed were users of Financial Statements and the use of the Financial Statements, types of ratios used to analyze financial statements (to be discussed into details later), bases of comparison and limitations of financial ratios as a means of analyzing financial statements.

Self-Assessment Questions

Exercise 3.1

1) Outline the bases of comparing financial statements.
2) State five (5) limitations of ratio analysis.
3) Mention five (5) major groups of users of financial information and their information needs.

SESSION 2: PROFITABILITY RATIOS

Overview

In the previous session we mentioned the five main types of ratios. In this session we are going to discuss one of the ratios namely profitability ratio. We hope you will find reading this session enjoyable and meaningful.

Objectives

By the end of the session, you will able to:

1) Calculate the various ratios under profitability ratio.
2) Interpret the implication of such ratios

2.1: Profitability Ratios

These ratios measure the degree of profit maximization in terms of certain values that contribute towards profit earning. Profitability ratios are of two types, those showing profitability in relation to sales and those showing profitability in relation to investment. Together, these ratios indicate a firm's efficiency of operation. These ratios are of great significance to company's owner and its creditors. Profitability ratios include the following:

2.1.1: Return on Capital Employed (ROCE)

This ratio is described as primary ratio and it indicates the overall profitability of the business. It is a measure of a firm's operating performance and indicates the earning power of the firm. It has different meanings and bases of calculation depending on how one defines capital employed.

i. Where Capital Employed is defined as Total Capital, i.e. Shareholders Funds and Total Liabilities, ROCE is computed as:

$$\frac{\textbf{Net Profit before Interest and Tax}}{\text{Total Capital}}$$

ii. Where Capital Employed is defined as long-term capital then ROCE is:

$$\frac{\textbf{\textit{Net Profit before Interest and Tax}}}{\text{Shareholders Fund plus Long Term Liability}}$$

iii. Where Capital Employed is defined as shareholders fund, then ROCE is:

$$\frac{\textbf{\textit{Net Profit after Interest and Tax}}}{\text{Stated Capital + Surpluses}}$$

2.1.2: Return on Assets (ROA) Ratio

The return on asset ratio (ROA) is a guide to the overall profitability of a firm and measures after-tax returns without regard to the manner in which the assets were financed. It is computed as:

$$\frac{\textit{Net Profit after Interest \& Tax}}{\text{Total Asset}}$$

2.1.3: Return on Equity (ROE)

Unlike the ROA that measures profitability without regard to the manner in which the assets are financed, the ROE recognizes both, the profitability of the underlying assets and the manner in which the assets are financed. It is measured as:

$$\frac{\textit{Net Income (PAT)}}{\text{Shareholders Equity}}$$

2.1.4: Gross Profit Ratio/Gross Profit Margin

It is one of the trading account ratios and it indicates the average gross margin on sale of goods. It is computed as:

$$\frac{\textbf{Gross Profit}}{\text{Sales}}$$

A high gross profit ratio is better in terms of profit earning potential but it does not necessarily result in a large absolute figure of gross profit unless a large volume of sales accompanied it.

2.1.5: Net Profit to Sales Ratios/Net Profit Margin

It indicates the relative efficiency of the business after taking into account all revenues arising from operations and expenses. Net profit is obtained when operating expenses, interest and taxes are subtracted from the gross profit. Where extraordinary items/figures are substantial it may be determined from the profit. It is computed as :

$$\frac{\text{Net Profit (after Interest \& tax)}}{\text{Sales}}$$

2.1.6: Expenses to Sales Ratio

It indicates the proportion of Sales Value spent on administrative and selling overheads. The lower the ratio, the more efficient the cost control system. It is computed as:

$$\frac{\text{P \& L Expenses}}{\text{Sales}}$$

Summary

In this session, we discussed the various ratios under profitability ratio. Some of the ratios discussed were return on capital employed, return on equity, return on assets, gross profit margin, net profit margin and expenses to sales ratios.

Self-Assessment Questions

Exercise 3.2

1. Distinguish between return on asset ratio and return on equity ratio.
2. Explain return on capital employed ratio and how it is measured.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics, if any

SESSION 3: LIQUIDITY RATIOS

Overview

You are once again welcome to session 3. In this session we are going to discuss the liquidity ratios. Liquidity talks about the firm's ability to raise funds to meet short-term obligations. In view of this, suppliers of short-term funds to the firm will be particularly interested in liquidity ratios. This is because it will give them an idea as to whether the firm will be able to meet its short-term obligations or not.

Objectives

By the end of this session, you will be able to:

1) Explain the various types of liquidity ratios and their significance.
2) Calculate the various types of liquidity ratios.

3.1: Liquidity /Short-Term Solvency Ratios

It is extremely essential for a firm to be able to meet its current obligations. This is because the survivability or otherwise of the firm depends on this. Liquidity ratios measures the short-term financial stability of the company and reveals its ability to meet current liabilities or short term obligations when the fall due.

In fact, analysis of liquidity needs the preparation of cash budgets and cash flow statement, but liquidity ratios, by establishing a relationship between cash and other current assets and current liabilities, provide a quick measure of liquidity. A firm should ensure that it does not suffer from lack of liquidity s this can lead to poor creditworthiness, loss of creditors' confidence or even bankruptcy. A very high degree of liquidity is also bad as the firm ties up funds unnecessarily. The most important liquidity ratios are:

3.1.1: Current Ratio

Known as working capital ratio, it measures in general terms, the ability of a business to meet its short-term liabilities as they fall due out of its current assets. It is computed as:

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Generally, Current Ratio around 2:1 is preferred but what is an ideal current ratio differs from industry to industry. The higher the ratio, the more solvent (in the short-run) the business is. However, if the ratio is too high it indicates under-trading or holding of excessive liquid funds.

3.1.2: Liquidity or Quick Ratio

It indicates the relative amount of assets in cash, or current assets which can quickly be converted into cash, available to meet current liabilities Liquid or quick asset consists of cash and bank balances, debtors and marketable securities. It is computed as:

$$\frac{\text{Current Assets} - \text{Stocks}}{\text{Current Liabilities}}$$

Ratio of about 1:1 is preferred but what is an ideal liquid ratio depends on the industry.

3.1.3: Acid Test Ratio

It indicates the amount of cash and cash equivalents available to meet current liabilities. It is calculated as:

$$\frac{\text{Cash + Cash Equivalents}}{\text{Current Liabilities}}$$

In computing of cash plus cash equivalents stocks and debts are deducted from current assets.

3.1.4: Vulnerability Ratio

It answers the question: "from which Current Assets shall we have the last cedi or pesewa to settle our current liabilities, beginning from cash to stock-in trade and what percentage of this current asset do we require?"

In assessing a company's financial strength on vulnerability ratio, the magnitude of the ratio will not matter much but the type of current asset from which the last cedi will come is the most important. A firm with a vulnerability of 60% of Debtors is better than a firm with a vulnerability ratio of 20% of stock-in-trade.

Summary

In this session we discussed the various types of liquidity ratios. These ratios are indicators of the firm's ability to meet its short term financial obligations when they fall due. Among the ratios are current ratio, quick ratio, acid test ratio and vulnerability ratio.

Self-Assessment Questions

Exercise 3.3

1) Distinguish between current ratio and quick ratio and show how they are calculated.
2) Explain vulnerability ratio and its implications to business organizations

SESSION 4: ACTIVITY RATIOS

Overview

You are warmly welcome to session 4. In this session we are going to discuss activity ratios. These ratios are used to measure the efficiencies of management. Among the ratios to be discussed are stock turnover ratio, stock turnover period, debt collection period, creditor payment period, sales to total assets ratios and sales to capital employed ratio.

Objectives

By the end of the session you will be able to explain the various ratios under activity ratios and how to calculate them.

4.1: Activity Ratios

Activity ratios sometimes called Efficiency or Asset Utilization ratios indicate how efficient management utilizes and manages its assets in generating revenues by comparing sales to different types of assets. These ratios measure the degree of efficiency in the use of company's resources to generate revenue and profit. Thus, activity ratios involve a relationship between sales assets. The intent is to obtain the speed at which assets generate revenue. The relevant ratios are:

4.1.1: Stock Turnover Ratio

It indicates the velocity, in number of times per period, at which the average figure of trading stock is being sold and replaced. It is calculated as

$$\frac{\text{Cost of Sales}}{\text{Average Stock}}$$

The denominator, average stock is often taken as a simple average of opening and closing stock. Where stock fluctuates widely, especially in businesses where trade is seasonal, a weighted average figure is used.

4.1.2: Stock Turnover Period

It reveals the average number of days, weeks or months taken to turnover average stock. It is computed as:

$$\frac{\text{Average Stock}}{\text{Cost of Sales}} \times 365 \text{ days}/52 \text{ weeks}/12 \text{ months}$$

Or

$$\frac{365\text{days}/52\text{weeks}/12\text{months}}{\text{Stock Turnover Ratio}}$$

4.1.3: Debtor Collection Period

It indicates the average period measured in months, weeks or days for which debtors remain uncollected. It shows the average credit period granted to credit customers and calculated it is calculated as:

Average Debtors x 356days/52weeks/12months

Credit Sale

Debtors reflect only credit sales. The denominators should contain only credit sales often total sales is given in the financial statement where the credit sales cannot be found, use total sales. In other words, this ratio is a measure of the efficiency of a firm's credit policy.

4.1.4: Creditor Payment Period

It indicates the average period measured in months, weeks or days for which creditors remain unpaid. It shows the average credit period granted by credit suppliers and is calculated as:

Average Creditors x 356days/52weeks/12months

Credit Sales

Use the purchases figure where you cannot find credit purchases. Purchases figure can be determined if you have your closing and opening stock and the cost of sales.

4.1.5: Sales to Total Assets Ratio

This ratio indicates efficiency or utilization of assets in generating revenue. It is calculated as:

$$\frac{\text{Sales}}{\text{Total Assets}}$$

4.1.6: Sales to Capital Employed Ratio

This ratio indicates the efficiency or utilization of capital employed in generating revenue. It is calculated as:

$$\frac{\text{Sales}}{\text{Capital Employed}}$$

Summary

In this session, we discussed the various ratios under activity ratio. We looked at stock turnover ratio, stock turnover period, debtor collection period. The various ratios were explained and how they are measured was also discussed.

Self-Assessment Questions

Exercise 3.4

1). Explain the following ratios and how they are measured:
   a). Sales to Capital Employed Ratio
   b). Debtor collection period
   c). Creditor payment period
   d) Stock turnover ratio

2) What is activity ratio? What are the uses of such ratios?

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 5: LEVERAGE/GEARING RATIOS

Overview

You are welcome to session 5 of unit 3. In this session we are going to continue with our discussion on ratio financial statement analysis. In session 3, we discussed liquidity ratios and said that they are indicators of the firm's ability to meet its short term obligations. In this session we will discuss leverage ratios. They are ratios which indicate the strength of the firm in terms of long term financing.

Objectives

By the end of this session you should be able to:

1. Explain the various ratios under leverage
2. Calculate and interpret the various ratios.

Enjoy your reading

5.1: Leverage/Gearing/Long-Term Solvency Ratios

The liquidity ratios measure the firm's ability to meet its short-term obligations when they fall due. Therefore, short-term creditors like suppliers of goods and bankers are more concerned with these ratios. On the other hand, long-term creditors like debenture or bond holders and financial institutions are more concerned with the firm's long-term financial strength. To judge the long-term financial position of he firm, financial leverage of capital structure (gearing) ratios are calculated. These ratios indicate the mix of funds provided by owners and lenders. They include the following:

5.1.1: Debt Ratio

This ratio measures a firm's degree of indebtedness, that is, the proportion of the firm's assets financed by debt relative to the proportion financed by equity. Debt ratio is measured as:

$$\frac{\text{Total Debt}}{\text{Total Asset}}$$

Where, total debt comprises current liabilities and long-term liabilities. Total assets here represent fixed assets and current assets.

5.1.2: Debt to Equity Ratio s' contribution. It is measured as:

This ratio shows the relationship between lenders' contribution to owner

$$\frac{\text{Total Debt}}{\text{Equity (Net Worth)}}$$

5.13: Capital Employed to Net Worth Ratio

Another alternative way of expressing the relationship between debt and equity is to determine funds that are contributed together by lenders and owners for each cedi of

owners' contribution. Such measurement is the capital employed to net worth ratio. It is calculated as:

$$\frac{\text{Capital Employed}}{\text{Net Worth}} \quad \text{or} \quad \frac{\text{Net Assets}}{\text{Equity}}$$

5.1.4: Coverage Ratio

The coverage ratio measures the degree to which fixed payments are covered by operating profit. The emphasis here is on assessing a firm's ability to service its financial expenses. It is computed as:

$$\frac{\text{Net Profit before Interest and Tax}}{\text{Interest}}$$

Summary

In this session we discussed the various ratios under leverage ratio. We discussed the four main ratios under leverage namely debt ratio, debt to equity ratio, capital employed ratio and coverage ratio.

Self-Assessment Questions

Exercise 3.5

1. What is the difference between debt ratio and debt to equity ratio? Explain coverage ratio and how it is calculated

SESSION 6: INVESTMENT RATIOS

Overview

This is the last session of unit 3. In this session we shall discuss investment ratios. Whenever people want to invest, they want to do so with the assurance that they will get a positive return on their investment. They therefore need some indicators as to whether the firm they are going to invest in will give them the needed return. How do investors find out about this? One way to do this is to calculate the leverage ratios of the firm.

Objectives

By the end of this session you should be able to:

1. Identify the various leverage ratios
2. Explain the various leverage ratios and how to calculate them.

6.1: Investment Ratios

These ratios indicate the potential and actual growth rate of a business. These ratios help equity shareholders and other investors to assess the value and quality of an investment in the ordinary shares of the company. Therefore, it is of great importance to potential investors who are thinking of investing in a company. Investment ratios include:

6.1.1: Earnings Per Share (EPS)

This ratio is the subject matter of SSAP 3. It indicates the amount of Net Profit after tax and preference dividend (but before taking account of extra-ordinary items) attributable to each ordinary share in issue and ranking for dividend during the period. This is calculated as:

$$\frac{\text{Earnings available to Ordinary Shareholders}}{\text{Number of Ordinary Shares}}$$

6.1.2: Dividend per Share

This indicates the amount of dividend payable on each share in a particular accounting year. When related to Earnings per Share, it shows the dividend and retention policy of the company. It is calculated as:

$$\frac{\text{Total Dividend}}{\text{Number of Ordinary Share}}$$

Where total dividend refers to net dividend proposed or actually paid plus associated tax credit.

6.1.3: Dividend Cover

This indicates the number of times earnings available to equity holders could cover ordinary dividends payable. This is calculated as:

$$\frac{\text{Total Equity Earnings}}{\text{Total Equity Dividend}}$$

It can also be calculated as:

$$\frac{\text{Earnings per Share}}{\text{Dividend per Share}}$$

6.1.4: Payout Ratio

This ratio shows the ratio between Equity Dividend and Equity Earnings and is the reciprocal of Dividend Cover. It can be calculated as:

$$\frac{\text{Equity Dividend}}{\text{Equity Earnings}}$$

6.1.5: Price/Earning Ratio

It indicates the number of years' purchase of the earnings that make up the current market value of an ordinary share and is regarded as an indicator of future performance. It is calculated as:

$$\frac{\text{Market Price per Ordinary Share}}{\text{Earnings per Share}}$$

6.1.6: Earnings Yield

It indicates potential return on investment. Like the P/E ratio, it highlights the amount earned on the ordinary shares relative to their market price. It is calculated as:

$$\frac{\text{Earnings per Share}}{\text{Market Price per Share}}$$

6.1.7: Dividend Yield

It indicates current return actually received on investment by ordinary shareholders. It is calculated as:

$$\frac{\text{Dividend per Share}}{\text{Market Price per Share}}$$

6.1.8: Net Book Value per Share

This shows the valuation per ordinary share based on book value. Net book value is defined in this context as: Total Assets – (Total Liabilities + Preference Shares).

An alternative valuation is Shareholders fund minus Preference Shares or Ordinary Shares plus Surplus. It is calculated as:

$$\frac{\text{Net Book Value of Ordinary Shares}}{\text{Number of Ordinary Shares}}$$

Summary

In this session we discussed the various leverage ratios and how they are calculated. Among the ratios discussed were earnings per share, dividend per share, dividend yield, dividend cover, price/earnings ratios among others.

Self-Assessment Questions

Exercise 3.6

1. Discuss the following leverage ratios and indicate how they are calculated.
    a. Earnings Per Share
    b. Dividend Cover
    c. Price /Earning ratio
2. Explain the significance of leverage ratios to investors.

This is blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

FULLY WORKED EXAMPLE

This is a supplementary exercise to enable you get a feel for how the various ratios are calculated given a financial statement. Note that companies will often include the financial statement of the previous year in the current year's statement to aid computation and comparison.

Question

Wholesale Merchant Limited is a distributor of second clothing in Ghana. The annual financial statement of the company for 2000 is summarized as follows:

WHOLESALE MERCHANTS LTD

Trading profit and loss account for the year ended 31st December 2000

		1999		2000	
		¢m	¢m	¢m	¢m
Sales:	Cash	42,000		44,000	
	Credit	378,000		478,000	
			420,000		523,600
Cost of sales			330,000		417,200
Gross profit			89,000		106,400
Expenses:					
	Warehouse		18,200	19,600	
	Transport		8,400		14,000
	Administration		26,600		26,600
	Selling		15,400		19,600
	Debenture interest		-		2,800
			68,600		82,600
Net Profit			**21,000**		**23,800**

Balance sheet as at 31st December 2000

			1999			2000
Fixed Assets (less depreciation)			42,000			56,000
Current Assets:						
	Stock	84,000		131,600		
	Debtors	70,000		114,800		
	Cash	14,000	168,000	9,800	256,200	
Less current liabilities			70,000		106,400	
Net current assets			98,000			149,800
Net Assets			**140,000**			**205,800**
Financed by:						

Share capital	105,000	105,000
Reserves and undistributed profit	35,000	58,800
Debenture loan	-	42,000
Capital employed	**140,000**	**205,800**

Additional information

1. All sales were from the stocks in the company's warehouse.
2. The range of merchandise was not changed and buying prices remained steady throughout the two years.
3. The debenture loan was received on 1st January 2000, and additional fixed assets were purchased on that date.

You are required to calculate the relevant major ratios.

Solution

1. **PROFITABILITY RATIOS:**

a. Return on Capital Employed (ROCE) = $\frac{\text{Net profit before interest and tax}}{\text{Total Capital}}$

1999	**2000**
$\frac{21,000}{210,000} \times 100$	$\frac{26,600}{312,200} \times 100$
= 10%	= 8.52%

Note: Total capital = shareholders' fund + total liability

b. Return on Equity (ROE) = $\frac{\text{Net profit after interest and tax}}{\text{Shareholders' fund}}$

	1999	**2000**
ROE	$\frac{21,000}{140,000} \times 100$	$\frac{23,800}{163,800} \times 100$
	= 15%	= 14.53%

Note: Shareholders' fund for 2000 = 205,800 – 42,000

c. Gross Profit ratio = $\frac{\text{Gross profit}}{\text{Sales}} \times 100$

1999	**2000**
$\frac{89,600}{420,000} \times 100$	$\frac{106,400}{523,000} \times 100$
= 21.33%	= 20.32%

d. Net Profit Ratio = $\frac{\text{Net profit after interest and tax}}{\text{Sales}}$

	1999	**2000**
	21,000 x 100	23,800 x 100
	420,000	523,600
	= 5%	= 4.55%

e. Return on Assets = Net profit after interest and tax / Total Assets

	1999	**2000**
	21,000 x 100	23,000 x 100
	210,000	312,200
	= 10%	= 7.62%

2. LIQUIDITY RATIOS

a. Current Ratio = Current Assets / Current Liabilities

	1999	**2000**
	168,000	256,200
	70,000	106,400
	= 2.4 times	= 2.4 times

b. Quick Ratio = Current Assets - Stock / Current Liabilities

	1999	**2000**
	168,000 – 84,000	256,200 – 131,600
	70,000	106,400
	= 1.20 times	= 1.17 times

c. Acid Test = Cash + Cash equivalents / Current Liabilities

	1999	**2000**
	14,000	9,800
	70,000	106,400
	= 0.2 times	= 0.092 times

3. ACTIVITY RATIOS

a. Stock turnover ratio = $\frac{\text{Cost of goods sold}}{\text{Average stock}}$

1999	**2000**
330,400	417,200
84,000	107,800
= 3.93 times	= 3.87 times

b. Stock turnover period = $\frac{\text{Average stock}}{\text{Cost of goods sold}}$ x 365 days

1999	**2000**
84,400 x 365	107,800 x 365
330,400	417,200
= 92.80 days	= 94. 31 days

Note: We assume average stock to be the closing stocks for the respective years.

c. Debtors collection period = $\frac{\text{Average Debtors}}{\text{Credit Sales}}$ x 365

1999	**2000**
70,000 x 365	92,400 x 365
378,000	478,000
= 67.59 days	= 70.56 days

d. Sales to Capital Employed Ratio = $\frac{\text{Sales}}{\text{Capital employed (long term capital)}}$

1999	**2000**
420,000	523,600
140,000	205,800
= 3.0 times	2.54 times

4. LEVERAGE/GEARING RATIOS

a. Debt Ratio = $\frac{\text{Total Debts}}{\text{Total Assets}}$ x 100

1999	**2000**
70,000 x 100	148,400 x 100
210,000	312,000
= 33.33%	= 47.53%

b. Debt to Equity Ratio = $\frac{\text{Total Debts}}{\text{Equity}}$ x 100

1999	**2000**
70,000 x 100	148,400 x 100
140,000	163800
=50%	= 90.6%

Exercise

1. Comment on the profitability ratios of the firm and indicate whether the firm is doing well or not.
2. Mr. John Arhinful, a stakeholder of the company made a comment during the company's General Meeting that the increase in the Capital Employed in 2000, has not been utilized profitably. By analyzing the activity ratios above, do you agree with Mr. Arhinful? Support your answer.
3. Calculate the various investment ratios (if any) from the above financial statement and comment on them.

UNIT 4: INVESTMENT RETURN AND RISK

Session 1: The Concept of Return
Session 2: Expected Return
Session 3: Calculating Risk
Session 4: Expected Return and Variance of a Portfolio
Session 5: Risk and Diversification
Session 6: Concepts Related to Risk-Return Trade-Off

Unit Overview

Whenever people invest, they anticipate making a return in the future. The anticipated return may or may not be realized. It is this uncertainty that leads to risk in investment in particular and business activities in general. Investors should be compensated for taking on risk, but before this can be done, the risk must be measured in order to reward investors appropriately for the risk taken. This unit is devoted to the calculation of investment returns and risk, and other related matters.

Unit Objectives

By the time you complete this session, you should be able to:

1. Explain the concept of return and estimate returns for individual securities and portfolios.
2. Explain the various measures of risk.
3. Calculate risk

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 1: THE CONCEPT OF RETURN

Overview

Return and risk are key characteristics of all investments activities. An understanding of these dimensions and their measurement is a vital prerequisite to making wise investment decisions. In this session, we are going to take a closer look at the concept of return on single investments.

Objectives

By the end of this session, you should be able to:

1. Define return and explain how to measure it.
2. Explain risk premium and maturity premium.

Read on now …

Return can be seen as the reward for investing. The return on an investment is measured as the total gain or loss experienced by the investor over a given period of time. Although the return is not necessarily guaranteed, it motivates people to invest in a given investment instrument, hoping that the reward would be good in the end. Every investment instrument guarantees a return, but the quantum of the return is a function of several factors including risk.

When you buy shares or bonds, your return comes in two forms:

a. Interest payment or dividend
b. Capital gain or capital loss.

Suppose you bought shares in Cal Bank at 20GP a share in 2004 when the company went public for the first time. By the end of the year the share price of the stock had appreciated to 31GP. Your return on this investment is 11GP. This return is also referred to as capital gain. If Cal Bank paid a dividend of 3GP a share, then your total return on a share of Cal Bank is 14GP.

Note however that the returns are computed in percentages. Hence, we convert the above into percentages to see how the investment performed. The general formula for percentage return is given as:

$$\text{The percentage return} = \frac{\text{Capital gain + dividend}}{\text{Initial share price}}$$

The percentage return is also referred to as Holding Period Rate of Return.

Substituting the figures from above, we have:

$$\text{The percentage return} = \frac{11GP + 3GP}{20GP} = 0.7 = 70\%$$

This means that the investor made 70% on each share held.

We shall work more examples so that you become conversant with the process. Assume that a company's share price rises from 35GP to 39GP over a one-year period, and the share has also paid a 2GP as dividend per share (towards the end of the year). What is the percentage return of the share?

Solution

$$\text{Percentage Return} = \frac{\text{Capital gain + dividend}}{\text{Initial share price}}$$

$$= \frac{4GP + 2GP}{35GP} = 0.1714 = 17.14\%$$

Tax Consideration

Dividends and capital gains are taxed in real world situations. But the policy (as to whether to tax or not, as well as the rate) may differ from country to country. In some countries only dividends or capital gains are taxed, whereas in others both dividends and capital gains are taxed. The formula above needs to be adjusted when tax is introduced.

$$\text{Percentage Return} = \frac{\text{Capital gain (1-T) + dividend}}{\text{Initial share price}}$$

or

$$\text{Percentage Return} = \frac{\text{Capital gain + dividend (1-T)}}{\text{Initial share price}}$$

or

$$\text{Percentage Return} = \frac{\text{(Capital gain + dividend) (1-T)}}{\text{Initial share price}}$$

Where T is the tax rate.

The tax component (1-T) is multiplied by the component that is taxed. For example if capital gain is taxed in a question, we multiply the (1-T) by the capital gain. The same applied to dividend. If both capital gain and dividend are taxed, then both are multiplied by the tax component (1-T) as can be seen above.

Example

Using the example above, assuming capital gains are taxed at the rate of 20%, what is the Percentage Return?

Solution

$$\text{Percentage Return} = \frac{\text{Capital gain (1-T) + dividend}}{\text{Initial share price}}$$

$$= \frac{4GP\ (1-0.20) + 2GP}{35GP} = 0.14857 = 14.86\%$$

Using the above example, assume that dividend is taxed instead of capital gain. What is the Percentage Return?

$$\text{Percentage Return} = \frac{\text{Capital gain + dividend (1-T)}}{\text{Initial share price}}$$

$$= \frac{4GP + 2GP\ (1-0.20)}{35GP} = 0.16 = 16\%$$

Now let us assume that both dividend and capital gains are taxed. Using the above example what is the Percentage Return?

$$\text{Percentage Return} = \frac{\text{(Capital gain + dividend)(1-T)}}{\text{Initial share price}}$$

$$= \frac{(4GP + 2GP)(1-0.20)}{35GP} = 0.1371 = 13.71\%$$

Real Rate of Return

The Percentage Return as calculated above is the nominal rate of return on an investment. The nominal return measures how much more money you will have at the end of the year if you invest today. It does not tell you the quantum of goods and services you can buy with it. This shortcoming is addressed by the real rate of return. The real rate of return tells you how much more you will be able to buy with your return on an investment at the end of the year if you invest today.

To convert from nominal to real rate of return, we use the following relationship:

$$1 + \text{real rate of return} = \frac{1 + \text{nominal rate of return}}{1 + \text{Inflation}}$$

$$\text{Real rate of return} = \frac{1 + \text{nominal rate of return}}{1 + \text{Inflation}} - 1$$

Assume inflation was 11% in the year under consideration. Using the above example where both dividend and capital gains were taxed, calculate the real rate of return.

$$\text{Real rate of return} = \frac{1 + 0.1371}{1 + 0.11} - 1$$

$$= 1.0244 - 1 = 0.0244 = 2.44\%$$

Question
Suppose you buy a bond for GH¢1020 with a 15 year maturity paying an annual coupon of GH¢80. A year later interest rates have dropped and the bond's price has increased to GH¢1050. What are your nominal and real rates of return? Assume inflation rate is 4% and a capital gain tax of 25%.

Solution:
The bond's price at the end of the year is GH¢1050.
Therefore capital gain on each bond is GH¢1050 - GH¢1020 = GH¢30. Your dollar return is the sum of the income from the bond, GH¢80, plus the capital gain, GH¢30, giving you a total return of GH¢110.

The Nominal Rate of Return is:

$$\frac{\text{Capital gain (1-T) + Income}}{\text{Original price}} = \frac{30\,(1 - 0.25) + 80}{1020} = 0.10 = 10\%$$

$$\text{Real Rate of Return} = \frac{1 + \text{nominal return}}{1 + \text{Inflation rate}} - 1 = \frac{1.10}{1.04} - 1 = 0.058 = 5.8\%$$

Historical Records
When you invest in a stock, you cannot guarantee what your return is going to be. However, by looking at the history of the security's returns, you can get some idea of the return that investors might reasonably expect from investments in different types of securities. Financial analysts cannot track every stock, so they rely on market indexes to summarize the return on different classes of securities.

Market index measures the investment performance of the overall market. The best-known stock market index in the United States is the Dow Jones Industrial Average, generally known as the Dow. The Dow Jones Industrial Average is the index of investment performance of a portfolio of 30 "blue-chip" stocks. ***Blue-Chip*** stocks are industrial shares that are thought to be a safe investment.

The Standard and Poor's Company Index, better known as the S&P 500, includes stocks of 500 major Companies and is therefore a more comprehensive index than the Dow. There are also stock market indexes for other countries such as the Nikkei index for Tokyo, Financial Times Index for London. Morgan Stanley Capital International (MSCI) computes a world stock market index. Ghana has three stock indexes and these are Ghana Stock Exchange All Share Index, Data Bank Stock Index and Gold Coast Security Index.

A popular source of such information on stock performance has been a study by University of Chicago Centre for Research in Security Prices. The Centre has a file of share prices and dividends since 1926 for every stock listed on the New York Stock Exchange as shown below. The portfolios offer different degrees of risk and return, with Treasury bills being as the safest investment one can have. Risk of default is virtually none existent and because they are short term instruments, their prices are relatively stable and therefore guarantee a certain pay off. What cannot be guaranteed is a real rate of return by the Treasury bill. Any investor who moves from Government securities assume an additional risk called *Default Risk.* As can be seen from the table, common stocks are the most risky of the four groups of securities in the table, but they also offer the highest returns.

Average rates of return on Treasury bills, government bonds, corporate bonds, and common stocks. (1926-1992).

Portfolio	Average Annual rate of return (nominal)	Average Annual rate of return (real)	Average risk Premium (extra return versus T bill
T. bills	3.8	0.6	-
T. bonds	5.2	2.1	1.4
Corp. bonds	5.8	2.7	2.0
Common stocks	12.4	9.0	8.6

Source: Stocks, Bonds and Inflation 1993 Year Book.

Treasury bills, the safest investment, had the lowest rate of return of 3.8% a year in nominal terms, and 0.6% in real terms. This implies the average inflation rate over the period was about 3.2% a year.

Treasury bonds, which are long-term government bonds paid a little higher than Treasury bills. The difference between the two returns is called ***Maturity Premium.*** This is defined as extra average return from investing in long versus short term Treasury securities.
Investors in common stocks received an extra return of 8.6% a year on average over the return on Treasury bills. The 8.6% is compensation for assuming risk of common stock

ownership. This compensation is known as ***risk premium,*** and it is defined as the expected return in excess of risk-free return as compensation for risk.

You may be wondering why the long time period to measure average rates of return? The reason is very simple: Averaging the returns over a long period gives a fair idea of the typical return that investors might justifiably expect.

Question
The average rates of return reported by Ibbotson Associates for the period 1947-1992 is as shown below.

Stocks	-	13.4%
Treasury Bills	-	5.0%
Treasury Bonds	-	5.4%

What were the risk premium on stocks and the maturity premium on Treasury bonds for this period?

Solution:
The ***risk premium*** on stocks is the average return in excess of Treasury bills. Risk premium = 13.4% – 5.0%
= 8.4%

The ***maturity premium*** is the average return on Treasury bonds minus the return on Treasure bills
Maturity premium = 5.4% – 5.0%
= 0.4%

Summary

This session has taken you through some basics in estimating returns. We believed that you followed carefully on how the tax element was treated, as well as the difference between nominal and real rate of return. You must be familiar with them since we are going to build upon them in subsequence sessions. Best wishes

SESSION 2: EXPECTED RETURN

Overview

In the preceding session, we discussed the concept of return and worked out some examples. You would have noticed that the discussion centred on post investment returns. In other words the assumption was that the returns on the investment was already known and out duty was to add all the returns together, and also to estimate the proportion of the return in terms of the investment made. In this session however, we do not know with certainty what the returns would be, and yet, we are called upon to estimate it. That is why we are calling this type of return ***expected***.

Objectives

By the end of this session, you should be able to:

1. Differentiate between rate of return and expected return.
2. Calculate expected return on investments.

Read on now …

Expected return is the weighted average of the outcomes anticipated when we invest. The weights are the probabilities of those outcomes. As we mentioned in the overview, because we are not certain of the outcome of an investment, we use various market conditions to estimate the outcome expected. The general formula for estimating expected return is given as:

Expected Return $E(r) = \sum_{i=1}^{n} P_i r_i$

Where P_i is the probability of possible outcomes and r_i is the possible outcomes.

The formula above is what we refer to as probability-weighted average of possible outcomes. The probabilities may be based on past experience, industry ratios and trends, interviews with company executives, and sophisticated simulation techniques.

Let us use this example to explain the formula above. Assume we are examining an investment with the possible outcomes and probability of outcomes shown below:

Outcome	Prob. Of Outcome
¢450m	0.2
¢650m	0.6
¢850m	0.2

The expected return $E(r) = \sum_{i=1}^{n} P_i r_i$ = (450 x 0.2) + (650 x 0.6) + (850 x 0.2)= $650

We hope you followed the example above carefully, and that you are in a position to solve similar problems now. If that is so, then calculate the expected return of the following investments.

Market conditions	Probability	Return (%) Investment A	Return (%) Investment B
Recession	1/3	- 8	+20
Normal	1/3	+ 5	+ 3
Boom	1/3	+18	-20

Investment A:

Expected Return E(r) = $\sum_{i=1}^{n} P_i r_i$ = 1/3(-8 + 5 + 18) = 5%

Investment B:

Expected Return E(r) = 1/3(+20 + 3 - 20) = 1%

Are you saying that the application of the formula is different in this example? No, it is not. We have rather simplified the solution because the probability is the same for all the market conditions indicated above. Applying the formula as we did in the first example would give you the same answer. If you are comfortable with the first application, you are free to use it.

Try your hands on this example before you refer to the solution. You have been provided with the following information for two investments A and B.

State of the economy	Probability	Returns of A (%)	Returns of B (%)
Boom	0.4	20	9
Normal	0.3	15	12
Recession	0.3	10	18

Calculate the expected return of projects A and B.

Solution

Expected Return E(r) = $\sum_{i=1}^{n} P_i r_i$

Expected Return E (r_A) = (0.4 x 20%) + (0.3 x 15%) + (0.3 x 10%) = 15.5%
Expected Return E (r_B) = (0.4 x 9%) + (0.3 x 12%) + (0.3 x 18%) = 12.6%

Again, we urge you to try your hands on this example before you refer to the solution. Suppose you are offered the chance to play the following game. You have been given GH¢20,000 to the investment game. You have been given two coins to flip, and for each head that comes up your starting balance would increase by 20% and for each tail that shows up your starting balance would be reduced by 10%. What is the expected return or outcome on this investment game?

As you can see, there are four equally likely outcomes as shown below:

When 2 heads appear (head + head), you would make 20% + 20% = 40%
When a heads and tail appear (head + tail), you would make 20% – 10% = 10%
When a tail and a head appear (tail + head), you would make -10% + 20% = 10%
When 2 tails appear (tail + tail), you would make -10% -10% = - 20%

There is a chance of 1 in 4, or 0.25, that you will make a 40% gain.
There is a chance of 2 in 4 or 0.5, that you will make a 10% gain,
There is a chance of 1 in 4, or 0.25, that you will lose 20%.

Expected return E(r) = $\sum_{i=1}^{n} P_i r_i$ = (0.25 x 40%) + (0.5 x 10%) + (0.25 x (– 20%)) = + 10%

Summary

Before you commit resources to a venture, you should have some idea as to what you should expect in the end. We have taken you through how to estimate the expected return investments. You should memorize the formula since you may use it extensively in this course.

SESSION 4: EXPECTED RETURN AND VARIANCE OF A PORTFOLIO

Overview

The combination of two or more investments is referred to as portfolio of investment. Investors are free to choose any combination of available investments when constructing a portfolio. Estimating the expected return on a portfolio is easy once we know the expected return on the individual securities in the portfolio. The variance, however, is no simple, especially the variance of a three or more security portfolio. For this reason, we would limit examination questions to a two security portfolio at this level.

Objectives

By the time you are through with this session, you should be able to:

1. Explain how to calculate portfolio return.
2. Explain covariance and correlation coefficient
3. Calculate portfolio risk.

Read on now …

Portfolio Return

Portfolio construction should be carefully planned after taking into consideration the expected return and the standard deviation of the proposed portfolio. The expected return on a portfolio of two (or more) securities is the weighted average of the expected returns of the individual securities. Suppose two investments, A and B have expected returns of 10%, and 20% respectively, i.e. $E(r_A) = 10\%$, and $E(r_B) = 20\%$. You decided to invest 45% of your financial resources in investment A and the remaining 55% in investment B. The expected return of this portfolio $E(r_{Pf}) = 0.45 \times 10\% + 0.55 \times 20\% = 15.5\%$.

Generally**,** the expected return of a portfolio is given as:

$E(r_{Pf}) = X_A E(r_A) + X_B E(r_B) + \ldots + X_Z E(r_Z)$.

Where X_A, $X_B \ldots X_Z$ are proportions of the total amount invested in securities A, B …Z. We would work out more examples later on in the session.

EXPECTED RETURN AND VARIANCE OF A PORTFOLIO

Portfolio Risk

Calculating the expected return on a portfolio is straight forward and easy. The difficulty lies in calculating the risk of a portfolio. When calculating portfolio variance, we need to calculate the individual stock variances, the covariance and correlation coefficient of the investments first since they serve as an input.

Covariance

It measures the variability of the returns on investments in a portfolio, i.e. the extent to which the returns on the investments tend to vary with each other or co-vary. In other words the covariance measure the extent to which the uncertainties of the returns on the two assets tend to reinforce or offset each other. Assuming you have securities A and B in your portfolio, then the covariance of investments A and B is:

$$\text{Cov}\,(_{A,\,B}) = \sum_{i=1}^{n} P_i\,[(r - E(r_A))(r - E(r_B)]$$

Let us use this example to illustrate how to calculate the covariance of two securities. You have been given information on two investments, A and B as shown below. Calculate the covariance of the two securities.

Market conditions	Probability	Return (%) Investment A	Return (%) Investment B
Recession	1/3	- 8	+20
Normal	1/3	+ 5	+ 3
Boom	1/3	+18	-20

Expected Return $E(r_A) = 1/3(-8 + 5 + 18) = 5\%$

Expected Return $E(r_B) = 1/3(+20 + 3 - 20) = 1\%$

$$\text{Cov}\,(_{A,\,B}) = \sum_{i=1}^{n} P_i\,[(r - E(r_A))(r - E(r_B)]$$

$$= 1/3[(-8-5)(20-1)] + 1/3[(5-5)(3-1)] + 1/3[(18-5)(-20-1)]$$

$$= -173.3$$

The covariance of two investments A and B can also be written as $\rho_{AB}\ \sigma_A\sigma_B$, where ρ_{AB} is the correlation coefficient and σ_A and σ_B are the variance of A and B respectively. Note that this can only be used when the correlation coefficient and variances of the securities are known.

Correlation Coefficient

It is difficult to interpret the magnitude of a covariance, hence we employ correlation coefficient to explain this. Correlation coefficient is a statistical measure of relationship and measures the extent to which the returns of investments move with one another. The correlation coefficient is bounded between –1 and +1, and so we can say large or small, strong or weak relative to +1 and -1. The General formula for estimating correlation coefficient is:

$$\text{Correlation coefficient } \rho_{AB} = \frac{\text{Cov}(A, B)}{\sigma_A\sigma_B}$$

Where σ_A and σ_B are the standard deviations of securities A and B respectively?

A correlation coefficient of + 1 indicates that the returns of the investments move in the same direction. This means that a fall in the return of one investment would mean a fall of the same magnitude in the return of the other investment. Such investments are said to be perfectly positively correlated.

When the correlation coefficient is -1, then the return on investments in the portfolio moves in the opposite direction. A fall in the return of one would mean an increase in the return of the other. The investments are said to be perfectly negatively correlated if the magnitude of fall is equal.

If the value of the coefficient is zero, it means there is no relation between the returns of the investments in the portfolio. That is to say a fall or a rise in one investment would not have any effect on the other.

Variance

Having gotten the inputs needed to compute the variance of a portfolio, we are more than ready to work out some examples. The general formula for calculating the variance of a portfolio of two investments is given as:

$\sigma^2_{Pf} = (\sigma_A X_A)^2 + (\sigma_B X_B)^2 + 2(X_A X_B)Cov(_{A, B})$

where X_A and X_B are proportions of wealth invested in two securities respectively.

For a portfolio of three investments the variance of the portfolio is giving as:

$\sigma^2_{Pf} = \sigma^2_A X^2_A + \sigma^2_B X^2_B + \sigma^2_C X^2_C + 2X_A X_B Cov(_{A, B}) + 2X_A X_C Cov(_{A, C}) + 2X_B X_C Cov(_{B,C})$

You need not worry about the variance of a three or more asset portfolio since it is beyond this level.

Let us work out some comprehensive examples now. You have been provided with the following information for two investments A and B.

State of the economy	Probability	Returns of A (%)	Returns of B (%)
Boom	0.4	20	9
Normal	0.3	15	12
Recession	0.3	10	18

a) Calculate the expected return and standard deviation of projects A and B.
b) If projects A and B are combined in the ratio 6:4, what is the expected return and standard deviation of the portfolio?

Solution

$E(r_A) = (0.4 \times 20\%) + (0.3 \times 15\%) + (0.3 \times 10\%)$
$= 15.5\%$

$E(r_B) = (0.4 \times 9\%) + (0.3 \times 12\%) + (0.3 \times 18\%)$
$= 12.6\%$

Variance $(\sigma^2) = \sum_{i=1}^{n} P_i [r_i - E(r)]^2$

$\sigma^2_A = 0.4(20 - 15.5)^2 + 0.3(15 - 15.5)^2 + 0.3(10 - 15.5)^2$
$\sigma^2_A = 17.25$
$\therefore \sigma_A = \sqrt{17.25}$
$\sigma_A = 4.15\%$

$\sigma^2_B = 0.4(9 - 12.6)^2 + 0.3(12 - 12.6)^2 + 0.3(18 - 12.6)^2$
$\sigma^2_B = 14.04$
$\therefore \sigma_B = \sqrt{14.04}$
$\sigma_B = 3.75\%$

b) $E(r_{Pf}) = X_A E(r_A) + X_B E(r_B)$
$= 0.6 \times 15.5 + 0.4 \times 12.6$
$= 14.34\%$

$\sigma^2_{Pf} = (\sigma_A X_A)^2 + (\sigma_B X_B)^2 + 2X_A X_B \text{Cov}(_{A,B})$

To get the covariance of A and B, we use the formula

$$\text{Cov}(_{A,B}) = \sum_{i=1}^{n} P_i[(r_A - E(r_A))(r_B - E(r_B)]$$
$= 0.4(20-15.5)(9-12.6)+0.3(15-15.5)(12-12.6)+0.3(10-15.5)(18-12.6)$
$= -15.3$

$\sigma^2_{Pf} = (0.6 \times 4.15)^2 + (0.4 \times 3.75)^2 + 2(0.6)(0.4)(-15.3)$
$= 1.106$
$\therefore \sigma_{Pf} = \sqrt{1.106}$
$\sigma_{Pf} = 1.05\%$

Now, try your hands on this example before you look at the solution. A company is considering one of two possible portfolios. Details are provided as follows:

Portfolio one

Security	Proportion of Portfolio	Expected Return	Standard Deviation
A	50%	12%	18%
B	50%	24%	24%

The correlation coefficient for the two securities is 0.35

Portfolio Two:

Security	Proportion of Portfolio	Expected Returns	Standard Deviation
A	50%	15%	21%
B	50%	30%	27%

The correlation coefficient for the two securities return is zero.
Required:
a) Calculate the expected return and standard deviation of each portfolio.
b) Advise management as to which company to invest in.

Solution

a) $E(r_{Pf}) = X_1\ E(r_A) + X_2\ E(r_B)$

$E(r_{Pf1}) = 0.5(12\%) + 0.5(24\%)$
$= 18\%$

$E(r_{Pf2}) = 0.5(15\%) + 0.5(30\%)$
$= 22.5\%$

$\sigma_{Pf} = \sqrt{\sigma^2_{Pf}}$

But we need to calculate the variance (σ^2) first before the standard deviation (σ) can be estimated. Recall that we said $Cov(_{A,B}) = \rho_{AB}\sigma_A\sigma_B$, where ρ_{AB} is the correlation coefficient. Since the correlation coefficient has been given in this example, it would be a waste of time and effort to calculate the covariance. Pay particular attention to how the technique works as you go through the example.

$\sigma^2_{Pf} = X_1^2\ \sigma^2_A + X_2^2\ \sigma^2_B + 2X_1X_2\ \rho_{AB}\sigma_A\sigma_B$

$\sigma^2_{Pf1} = (0.5^2 \text{ x } 18^2) + (0.5^2 \text{ x } 24^2) + 2(0.5)(0.5)(18)(24)(0.35)$

$\sigma^2_{Pf1} = 300.6$

$\therefore\ \sigma_{Pf1} = \sqrt{300.6}$
$= 17.34\%$

$\sigma^2_{Pf2} = (0.5^2 \text{ x } 21^2) + (0.5^2 \text{ x } 27^2) + 2(.5)(.5)(21)(27)(0)$

$\sigma^2_{Pf2} = 292.5$

$\therefore \sigma_{P2} = \sqrt{292.5}$

$\sigma_{Pf2} = 17.10\%$

Summary of expected returns and standard deviations of the two portfolios are:

Portfolios	$E(r_{Pf})$	σ_{Pf}
One (1)	18%	17.34%
Two (2)	22.5%	17.1%

c) Based on expected returns and standard deviation (risk), the company should choose portfolio Two (2) because it has a higher expected return and a lower risk.

Summary

Calculating portfolio risk calls for extra care, especially where you are dealing with three more security portfolio. We wish to reaffirm our earlier decision not to ask questions on a three or more security portfolio. Note carefully how to solve questions when you have information on correlation coefficient. You would be wasting precious time by calculating covariance when you have been given information on correlation coefficient

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 5: RISK AND DIVERSIFICATION

Overview

We have been primarily concerned with the risk inherent in individual securities so far in this unit. Even though we estimated the return and risk of portfolios in the preceding session, we also need to consider in details the impact of a given investment on the total risk of the firm. As a prelude to that, think of this: If the market portfolio is made up of individual stocks, why isn't its variability equal to the average variability of the components that make it? The answer is simple - diversification reduces variability (risk).

Objectives

By the time you finish this session, you should be able to:

1. Explain the concept of diversification.
2. Explain why the risk of a portfolio is not simply the weighted average of the individual security risks.
3. Explain the difference between systematic and unsystematic risk.

Now read on …

Usually, portfolios contain two or more securities would ensure proper diversification. The thrust of diversification is risk reduction and it is achieved by the inclusion of a variety of non-complementary securities for this purpose. Put differently, portfolios are diversified to reduce the risk of loss without compromising extensively on the investor's return objective. Let us illustrate the concept of diversification. But first, remember the saying - *do not put all your fresh eggs in one basket.*

Suppose you sell umbrellas. You may make good returns when it rains but you are likely to lose during the dry season. You may conclude that selling umbrellas is a risky business. Selling ice cream is no safer since you would do well in the dry but poorly in the wet season. Now, suppose you invest in both ice cream and umbrella businesses, you would have diversified your investments across the two businesses, and you would make an average level of profit come rain or shine. The diversification has even out the cyclical fluctuations associated with the two businesses and has accordingly reduced your overall risk exposure.

Diversification works because prices of different stocks hardly move exactly together. In fact diversification works best when returns on stocks are negatively correlated, i.e. when one business does well, the other does not.

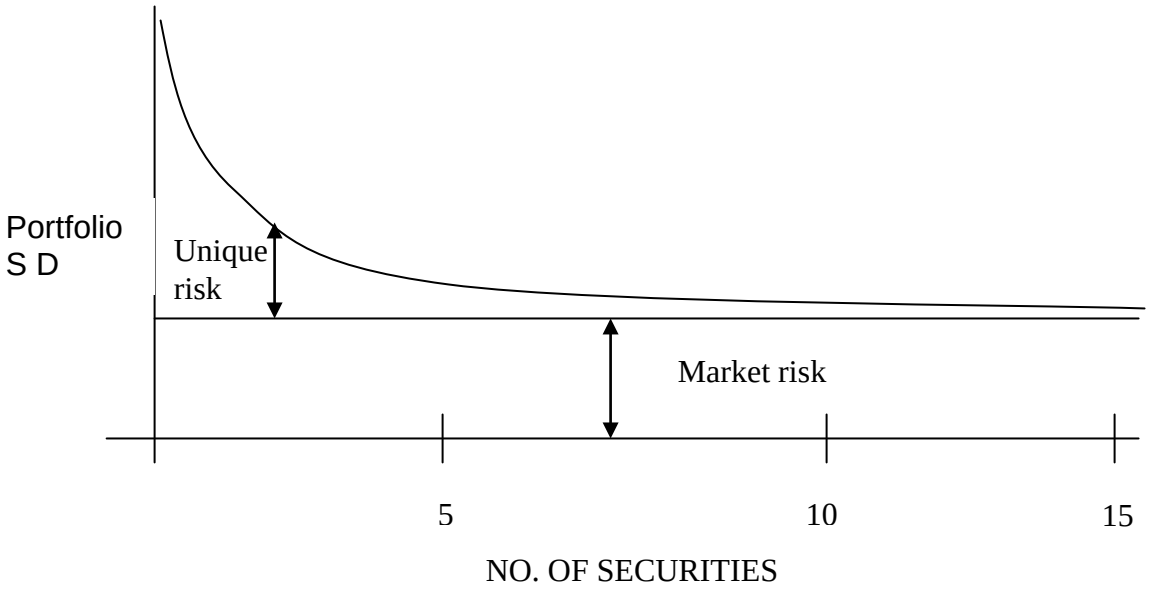
A greater part of the benefit of diversification achieved with relatively few stocks. In fact, the benefit of diversification is infinitesimal or negligible when the number of stocks in a portfolio is increased beyond 15. The point is that no matter how many securities you hold, you cannot eliminate your total risk or exposure.

Types of Risk

There are two basic types of risk namely, unique risk and market risk. ***Unique risk*** is risk that can be eliminated by diversification. In fact, a well diversified portfolio can completely remove this type of risk. Unique risk comes about because many of the perils or challenges that surround an individual company are peculiar to that company and possibly its direct competitors. Put differently, this risk is unique to a particular firm and hence the name unique risk. Unique risk is also called ***diversifiable risk*** because diversification can eliminate it. It is also referred to as ***unsystematic risk***.

Market risk on the other hand is the risk that cannot be eliminated regardless of how well or how much one diversifies. This risk is also called ***systematic risk*** or ***undiversifiable risk***. Market risk results from the existence of economy-wide perils that threaten all businesses. The perils are environmental forces that firms have absolutely no control. For example changes in macroeconomic variables such as interest rates, inflation, exchange rates and taxation etc. affect all companies in the economy and hence cannot be eliminated by diversification. Market risk is therefore a macro risk since it affects the economy as a whole.

Figure 1 Unique and Market risk



As you can see in the figure (figure 1) above, if you have only a single stock in your portfolio, unique risk is very important, but once you have a portfolio of about 15 or more stocks, diversification has done most of what it can to eliminate unique risk. The point we are making is that as more and more securities are added to a portfolio, unique risk reduces. The investor is able to diversify away the unique risk of the individual assets. Notice that when stocks are increased beyond 15, the benefit from diversification is negligible. That is to say that if an investor adds more securities to his/her portfolio, little risk could be diversified away.

Market risk, however, cannot be diversified away by holding more assets. In fact, for a reasonably well-diversified portfolio, only market risk matters. For this reason, an investor with a diversified portfolio will be concerned basically with market risk.

Reasons for Gains from Diversification

Research has found that diversification can remove over half of the risk of a stock market portfolio. Fama (1976) found that 10 randomly selected stocks are needed to obtain most benefits of diversification. The main reason why diversification reduces risk is that it eliminates (or much reduces) unique/diversifiable risk.

Unique events often affect one company's profit and not another's. These events include:

- The company's product loses market
- Production problems occur at the company, for example breakdown of plant, strike action by workers, shortage of raw materials etc…
- The company's competitors come up with lower cost of production alternatives.
- The company's production cost rises.

Unique events often affect one industry and not another, for example,

- The industry's product is no longer in demand.
- Industry input costs rise.
- Industry is disrupted by industry-wide strikes

These types of risks can be minimized by diversifying into:

- Other companies in the industry
- Other industry sectors.

Summary

Diversification reduces risk, and that is why the variability of a portfolio is not equal to the average variability of the components that make it. The two basic types of risk that affect a portfolio, systematic and non-systematic, have other terms that used to refer them. These terms are used interchangeably, so spend time and master them since they can confuse you.

SESSION 6: CONCEPTS RELATED TO RISK-RETURN TRADE-OFF

Overview

Having considered return and risk and how to estimate these for either individual securities or a portfolio, it is time to look at other concepts in the trade-off between risk and return. Unfortunately, time is of essence here, so we may not be able to cover all the issues. We are limiting this to a consideration of Capital Asset Pricing Model and Beta. You need to relax as you go through the material so as to grasp the concepts. We wish you our very best.

Objectives

By the time you complete this session, you should be able to:

1. Explain the Capital Asset Pricing Model and use it to calculate expected returns on either individual stocks or portfolios.
2. Explain beta in relation to its risk effects on stocks on portfolios.

Read on now …

Capital Asset Pricing Model

Although portfolio theory shows investors how to assess or measure risk, it falls short of specifying the exact relationship between risk and expected return. As a result, William Sharpe (1963) John Lintner (1965) and others developed the Capital Asset Pricing Model to describe the relationship between risk and expected return of an investment. The model is used in the pricing of risky securities. The model specifies that the required rate of return of a risky asset is a function of three factors namely:

- the risk-free rate
- the required rate of return on an average risk asset
- the volatility or variability of the asset return relative to the return on the (the beta (β) of the asset).

Using the Capital Asset Pricing Model, the expected return of a portfolio (r_p) is:
$r_p = r_f + \beta(r_m - r_f)$.

Where:
r_f = risk-free rate
β = beta of the security
r_m = expected return on the market

Capital Asset Pricing Model (CAPM) is premised on the notion that investors need to be rewarded in two ways: time value of money and risk. In the model above, the time value of money is represented by r_f, (risk-free rate).This is a compensation or reward to investors for putting money in any investment over a given period of time. The risk, $\beta(r_m - r_f)$, measures the amount of reward or compensation that the investor requires for assuming an additional risk. It is computed by comparing β (which is a risk measure) with the returns of the asset to the market over a period of time and to the market premium ($R_m - r_f$).

From the above, we can explain the CAPM as the expected return of an investment (or a portfolio) is equivalent to the rate on a risk-free investment plus a risk premium. If the expected return is not equal to or more than the required return, then the investment should not be rejected.

Let us solve some problems using CAPM. ABC Company Ltd. is an all equity company with a cost of capital of 15% per annum. It wishes to invest in a new project with an estimated β of 1.5. If the risk-free rate is 7% and the expected market return over the period is 13%, what is the stock's expected rate of return?

Solution

$E(r_p) = r_f + \beta (r_m + r_f)$

$E(r_P) = 7\% + 1.5(13\% - 7\%)$

$= 16\%$

Follow closely on this example as you may be called upon to solve similar one later. Deep Sea Fisheries is considering investing in the canning industry and has estimated that the average β of firms in the industry is 1.2. If it undertakes this investment, it will incur an initial cost of GH¢2million but expect to obtain annual profits of $200,000 indefinitely. If the Treasury bill rate is 4% and the expected return on the market is 12%, should Deep Sea undertake the project?

Solution

Computation of Expected Return (Required Rate of Return)

$E(R_p) = R_f + \beta (R_m - R_f)$

$E(R_p) = 4\% + 1.2 (12\% - 4\%)$

$= 13.6\%$

NPV of the project

$$NPV = (\$2m) + \frac{\$200,000}{0.136}$$

NPV = ($529411.76)

The project must be rejected since it has a negative NPV.

Advantages of CAPM

- It provides a market-based relationship between risk and return.
- It shows why only systematic risk is important and relevant in this relationship.
- It is one of the best methods for estimating a company's cost of capital.
- It provides a basis for establishing risk adjusted discount rate for capital investment projects.
- It helps us to make educative guess as to the expected return on assets that have not yet been traded on the stock market.
- Another usefulness of CAPM is in utility rate-making cases. In this case the issue is the rate of return that a utility should be allowed to earn on its investment in plant and equipment.

Disadvantages of CAPM

- It ignores unsystematic risk, which may be important to investors who do not hold a well-diversified portfolio.
- The model considers only the level of return and not the way that return is received. It views capital gains and dividends as equally important.
- It is strictly one-period model and should be used with caution appraising multi-period projects.
- The most serious drawback is the assumptions under which the CAPM is derived. The assumptions may not be applicable in real life situations.

Measuring Beta

Beta of an asset is the contribution of that asset to the overall risk of a well-diversified portfolio. Beta measures nondiversifiable risk. Beta shows how the price of a security responds to market forces. Even though market fluctuations affect all stocks, the degree of the effect varies. Stocks that are not very sensitive to market fluctuations are referred to as defensive stocks, while the direct opposite is aggressive stocks. The more responsive a security price to changes in the market, the greater its beta. Aggressive stocks have high betas, i.e. betas greater than 1.0. Defensive stocks on the other hand have betas less than 1.0. Therefore it is better to be in aggressive stocks when the market goes up, and when the market is down, defensive stocks are preferred. Beta of the overall market is equal to 1.0 and other betas will be viewed in relation to the market beta. A security with a beta less than 1.0 is said to be less risky than the market. A security with a beta greater than 1.0 is viewed as more risky. The beta of an asset may depend on the covariance with other assets in the portfolio.

The beta of a portfolio can be estimated by the following methods:

a. Regression analysis: Beta can be measured by using a single index model framework by regressing market returns on security returns. Points to consider in this process include:

b. $\beta = \frac{Cov\ (i, m)}{\sigma^2_m}$

Where Cov (i, m) is the covariance between return on a security and the return on the market portfolio and σ^2_m is the variance of return on the market portfolio.

c. $\beta = \frac{E\ (r_i) - r_f}{r_m - r_f}$

This beta formula is derived from the CAPM. We know that the CAPM is given as: $E\ (ri) = r_f + \beta\ (r_m - r_f)$, therefore we can make β the subject of the formula.

Hence, $\beta = \frac{E\ (r_i) - r_f}{r_m - r_f}$

d. $\beta = \Sigma\ W_i\ \beta_i$, Using this method, beta of a portfolio is the weighted average of the betas of the individual assets/securities that form the portfolio.

e. Another formula for computing beta is $\beta = \frac{\rho_{im}\ \sigma_i}{\sigma_m}$

Example

You have been provided with the following information regarding a new investment that a company intends to undertake:

State of the economy	Probability	Investment Return	Market Return
Recession	0.3	10%	12%
Stable	0.4	17%	14%
Expansion	0.3	19%	17%

The risk-free rate is 10% per annum. Is the investment worth undertaken?

Solution

Expected return of the new investment:

$$E(r_i) = \sum_{i=1}^{n} P_i r_i$$

$$E(r_i) = (0.3 \times 10\%) + (0.4 \times 17\%) + (0.3 \times 19\%)$$
$$= 15.5\%$$

Expected return of the market:

$$E(r_m) = (0.3 \times 12\%) + (0.4 \times 14\%) + (0.3 \times 17\%)$$
$$= 14.3\%$$

Variance of the market:

$$\sigma^2_m = \sum_{i=1}^{n} P_i [r_m - E(r_m)]^2$$
$$= 0.3(12 - 14.3)^2 + 0.4(14 - 14.3)^2 + 0.3(17 - 14.3)^2$$
$$= 3.81$$

Covariance between the market and investment return

$$Cov(i, m) = \sum_{i=1}^{n} P_i [r_i - E(r_i)][r_m - E(r_m)]$$
$$= [0.3(10 - 15.5)(12 - 14.3)] + [0.4(17 - 15.5)(14 - 14.3)] + [0.3(19 - 15.5)(17 - 14.3)]$$
$$= 6.45$$

$$\beta = \frac{Cov(i, m)}{\sigma^2_m}$$
$$= \frac{6.45}{3.81}$$

$$= 1.69 \text{ or } 1.7$$

Required Rate of Return $[E(r_i)] = r_f + \beta(r_m - r_f)$

$$E(r_i) = 10\% + 1.7(14.3\% - 10\%)$$
$$= 17.31\%.$$

The project should not be undertaken since the required rate of return (the hurdle rate) using the CAPM is 17.3% but the expected return of the investment is 15.5%.

Example

Praise Jah Ltd. is all equity-financed company with a cost of capital of 19%.
The company is considering the following investments projects with the following information:

Projects	Initial Outlay ($m)	Cash inflow in one year ($m)	Beta
A	1	1.095	0.3
B	1	1.130	0.5
C	1.5	2.385	1.0

If the risk-free rate of interest is 8% and the expected return on the market portfolio is 15%,

a) Calculate the company's beta.
b) Calculate the CAPM required rate of return for each project.
c) If the company decided to invest 40% in project A, 30% in project B and 30% in project C, what will be the beta for the new investment?

Solution

a) $\beta = \frac{E(r_i) - r_f}{R_m - R_f}$

$= \frac{19\% - 8\%}{15\% - 8\%}$

$= 1.57$ or 1.6

b) $E(r_p) = r_f + \beta (r_m - r_f)$

Project A = 8% + 0.3(15% - 8%) = 10.1%
Project B = 8% + 0.5(15% - 8%) = 11.5%
Project C = 8% + 1(15% - 8%) = 15%

c) $\beta = \Sigma W_i \beta_i$

$= (0.4 \times 0.3) + (0.3 \times 0.5) + (0.3 \times 1)$

$= 0.5$

Example

Suppose the CAPM accurately describes the security prices. You forecast that the correlation of returns on your portfolio of risky securities and the return on the market portfolio is 0.5 that the standard deviation of returns on your risky portfolio is 0.4 and the standard deviation of returns on the market portfolio is 0.25.

a) How would you combine the risky portfolio with a risk-free asset to obtain a portfolio that has an overall beta of 1.2?
b) How would your answer differ if the correlation coefficient were 0.8?

Solution

a) $\beta = \dfrac{\rho_{im}\ \sigma_i}{\sigma_m}$

$= \dfrac{0.5 \times 0.4}{0.25}$

$= 0.8$

Let a fraction (X) be invested in risky portfolio

$\beta = \Sigma W_i \beta_i$

Therefore $1.2 = X(0.8) + (1-X)\,0$

$1.2 = 0.8X + 0$

So, $X = 1.2 / 0.8$

$X = 1.5$

This means that the investor will borrow 50% of net wealth at r_f rate and invest 150% in risky portfolio.

b) $\beta = \dfrac{0.8 \times 0.4}{0.25}$

$= 1.28$

$\beta = \Sigma W_i \beta_i$

$1.2 = X(1.25) + (1 - X)\,0$

$1.2 = 1.25X$

$X = 1.2 / 1.25$

$X = 0.96$

Invest 96% in risky asset and 4% in the risk-free asset.

Summary

We have used the Capital Asset Pricing Model to explain the relationship between risk and expected return. This was done because portfolio theory showed investors how to measure risk, but failed to show the relationship. We also explained almost all the basic concepts about beta. We urge you to solve more problems as you read along. We take the opportunity to wish you well in your pursuit to enrich yourself knowledge wise.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 3: CALCULATING RISK

Overview

When we invest funds in treasury bills, the outcome is certain. Because of the certainty of the outcome, we say there is no variability with respect to the outcomes and hence no risk. On the other hand, if we invest the same funds in a petroleum exploration off the coast of Ghana, the variability of possible outcomes is great and hence, we say the project is risky. We would take you through the process of calculating the risk inherent in a particular security in this session.

Objective

By the time you complete this session, you should be able to:

1. Explain the concept of risk.
2. Calculate the various forms of risk.

Now, read on …

Risk is a hazard or a peril or exposure to an injury or loss. It is the chance that an unfavorable event will occur or expectations may not be realized. In finance, risk is defined in terns of variability of possible outcomes from an investment activity. Note that risk is measured not only in terms of losses but also in terms of uncertainty. We say gold mining carries a high degree of risk not just because you may lose your money but also because there is a wide range of possible outcomes.

It is generally accepted that a basic assumption in financial theory is that a good number of investors and managers are risk-averse. That is to say that when called upon to choose between relative certainty and uncertainty, they would prefer relative certainty. What about you? Would you do same? If yes, then you are risk-averse too.

In fact, this is not to say investors or business people are afraid of risk and there are unwilling to take risk. Instead, they require a higher expected return for taking on risky investments. Risk or variability in expected return is measured by three statistics, namely;

i. variance of the return
ii. Standard Deviation of the return
iii. Coefficient of variation of the return.

Variance, Standard Deviation & Coefficient of Variation

Variance and dispersion standard deviation (SD) are a measure of the dispersion of returns. These statistics are indicators of deviations or volatility of possible returns from the expected return.

The general formula for calculating Variance (σ^2) = $\sum_{i=1}^{n} P_i [r_i - E(r)]^2$

Where P_i is the probability of possible outcomes, r_i is the possible outcomes, and E(r) is the expected return.

Standard deviation (SD) is the square root of variance = $\sqrt{\sigma^2}$

Generally, the larger the standard deviation, the greater is the risk. For example assume you have 3 projects/investments; A, B & C. Each has an expected return of GH¢8,000, but with standard deviations of GH¢2,000, GH¢8,000 and GH¢12,500 respectively. Which one would you invest in? As a rational investor, you should invest in project A.

Now, assume that the returns on the investments were different, (say GH¢30,000, GH¢60,000, and GH¢90,000) just as the standard deviation. In this instance, a direct comparison of the standard deviations for each distribution would not be helpful in measuring risk.

The *size* difficulty is eliminated by using the ***coefficient of Variation***. This is done by dividing the standard deviation of an investment by the expected return, [SD/E(r)]. Generally, the larger the coefficient of variation, the greater is the risk.

Assume investment A has expected return of GH¢6000 and SD of GH¢600 and investment B has expected return of GH¢600 and SD of GH¢190. Which of these two investments would you choose and why?

Coefficient of variation of investment A = $SD/E(r_A) = \frac{600}{6000} = 0.1$

Coefficient of variation of investment B = $SD/E(r_B) = \frac{190}{600} = 0.317$

We have correctly identified the second investment as carrying the greater risk since it has a higher coefficient of variation.

Having explained risk and the various statistics for estimating it, let us spend the rest of the time available to this session to solve some examples. Note however that in most examinations, you would be required to calculate the expected return before you proceed to calculate the variance, standard deviation & coefficient of variation.

Question

Calculate the expected return, variance, standard deviation, and coefficient of variation of each of the following investments. As a rational investor, which investment would you prefer?

Market conditions	Probability	Return (%) Investment A	Return (%) Investment B
Recession	1/3	- 8	+20
Normal	1/3	+ 5	+ 3
Boom	1/3	+18	-20

Solution

Investment A:

Expected Return E(r) = 1/3(-8 + 5 + 18) = 5%

$$\text{Variance } (\sigma^2) = \sum_{i=1}^{n} P_i\,[r_i - E(r)]^2$$

$$= 1/3[(-8-5)^2 + (5-5)^2 + (18-5)^2]$$

$$= 1/3(169 + 0 + 169)$$

$$= 112.7$$

$$\text{Standard deviation} = \sqrt{112.7} = 10.6\%$$

$$\text{Coefficient of variation} = \frac{0.106}{0.05} = 2.12$$

Investment B:

Expected Return E(r) = 1/3(+20 + 3 - 20) = 1%

$$\text{Variance} = 1/3[(20-1)^2 + (3-1)^2 + (-20-1)^2]$$

$$= 1/3(361 + 4 + 441)$$

$$= 268.7$$

$$\text{Standard deviation} = \sqrt{268.7} = 16.4\%$$

$$\text{Coefficient of variation} = \frac{0.164}{0.01} = 16.4$$

As a rational investor, you should select investment A since it is less risky.

Try your hands on this question before you look at the solution provided. You have been giving the following information on two investments, A and B.

State of the economy	Probability	Returns of A (%)	Returns of B (%)
Boom	0.4	20	9
Normal	0.3	15	12
Recession	0.3	10	18

Calculate the expected return and standard deviation of projects A and B.

Solution

$E(r_A) = (0.4 \times 20\%) + (0.3 \times 15\%) + (0.3 \times 10\%)$
$= 15.5\%$

$E(r_B) = (0.4 \times 9\%) + (0.3 \times 12\%) + (0.3 \times 18\%)$
$= 12.6\%$

$$\text{Variance } (\sigma^2) = \sum_{i=1}^{n} P_i\,[r_i - E(r)]^2$$

$\sigma^2_A = 0.4(20 - 15.5)^2 + 0.3(15 - 15.5)^2 + 0.3(10 - 15.5)^2$
$\sigma^2_A = 17.25$
$\therefore \sigma_A = \sqrt{17.25}$
$\sigma_A = 4.15\%$

$\sigma^2_B = 0.4(9 - 12.6)^2 + (0.3(12 - 12.6)^2 + 0.3(18 - 12.6)^2$
$\sigma^2_B = 14.04$
$\therefore \sigma_B = \sqrt{14.04}$
$\sigma_B = 3.75\%$

Summary

We explained risk as the probability that actual returns from an investment would differ from the expected return. We explained however, that this is not to say business people are afraid of risk and there are unwilling to take risk. Rather, higher compensation must be ensured for taking on risky investments. And remember that risk or variability in expected return is measured by three statistics, namely; variance of the return, standard deviation of the return, and coefficient of variation of the return.

This is a blank sheet for your short notes on

- Difficult topics if any; and
- Issues that are not clear

UNIT 5: SECURITY ANALYSIS

Session 1: Technical Analysis
Session 2: Fundamental Analysis including global economy
Session 3: Fundamental analysis: Domestic Economy
Session 4: Fundamental analysis: Industry analysis
Session 5: Fundamental analysis: Company analysis
Session 6: Efficient Market Hypothesis

Overview

Security analysis involves the process of finding a value for a stock. There are basically two approaches to security analysis. These are the technical analysis and fundamental analysis. This unit takes a look at these two approaches of finding a price of a stock. Also considered in this unit is the efficient market hypothesis.

Objectives

By the end of this unit, you should be able to:

1. Explain Technical Analysis
2. Explain fundamental Analysis
3. Outline the components in Domestic economy
4. Explain industry analysis and its components
5. Discuss Efficient Market Hypothesis and the three main types.

This is a blank sheet for your short notes on;

- Issues that are not clear; and
- Difficult topics, if any

SESSION 1: TECHNICAL ANALYSIS

Overview

You are welcome to session 1 of unit 5. In this session we are going to discuss technical analysis. This is the first leg of security analysis. In this session we will discuss the meaning of technical analysis, the tool for undertaking technical analysis, the advantages and the criticisms of technical analysis.

Objectives

By the end of the session, you should be able to:

a) Explain what technical analysis is
b) Describe how technical analysis is conducted
c) Outline the advantages and the criticisms of technical analysis.

1.1: Technical Analysis

Technical analysis is essentially the search for recurrent and predictable patterns in the stock prices. Analysts and market technicians examine prior price and volume data as well as other market-related indicators to determine part trends in the belief that they will help forecast future ones. One broad definition of technical analysis includes the use of stock prices, trading volume and other market data to formulate rules telling the technician when to buy or sell stocks.

Technical analysts place much emphasis on charts and graphs of internal market data than on such fundamental factors as earnings reports, management capability or new product development. They believe that even when important fundamental information is uncovered; it may not lead to profitable trading because of timing consideration and market imperfections.

Although technicians recognise the value of information regarding future prospects of the firm, they believe that such information is not necessary for a successful trading strategy. This is because whatever the fundamental reason for a change in stock price, if the change is slow enough, the technical analyst will be able to identify a trend that could be explicated during the adjustment period.

The technical analysts are sometimes called chartists because they study records and charts of past stock prices hoping to find patterns they can exploit to make profit. Many of these technicians do not even consider the influence of other market or company

information to make their investment decision. They believe that stock prices and volume data are all an investor needs, and extra information only clouds their ability to make relevant buy or sell decision. Many technicians can apply their techniques to forecasting the direction of both the stock market and individual stocks. They typically use market indicators to measure the strength of the markets or the economy.

1.2: Assumption of Technical Analysis

Technical analysis is based on a number of basic assumptions:

1. Market value is determined by the interaction of demand and supply.
2. It is assumed that though there are minor fluctuations in the market, stock prices tend to move in trends that persist for long periods.
3. Reversals of trends are caused by shifts in demand and supply
4. Shifts in demand can be detected sooner or later in charts.
5. Many chart patterns tend to repeat themselves

1.3: Tool of Technical Analysis

One major tool use by technical analysts is charting. The chartist may draw lines connecting the high and low prices for the day to examine any trends in the prices. This is called a search for memorandum. Charting is often linked to Dow Theory. The Dow Theory named after its Greater Charles Dow, the founder of Wall Street Journal, is the grandfather of most technical analysis. The aim of the Dow Theory is to identity long-term trends in stock prices. The theory maintains that there are three major movements in the market, daily fluctuations, secondary movements and primary trends.

The primary trend is the long-term movement of prices lasting for several months to several years. The secondary (intermediate) trends are caused by short-term deviations of prices from the underlying trend line. These deviations are eliminated via corrections when prices revert back to trend values Tertiary (minor) trends are daily fluctuations of little importance. The important facet of the secondary movement is that each low is higher than the previous low and each high is higher than the previous high.

This pattern of upward-moving ‘tops’ and ‘bottoms’ is one of the key ways to identify the underling primary trend (bullish). Despite the upward primary trend, immediate trends still can lead to short periods of declining prices. Under the Dow Theory it is assumed that the primary trend will continue for a long period and the analyst should not be confused by secondary movements. However, an upward movement must ultimately end. This is indicated by a new pattern in which a recovery fails to exceed the previous high and a new low penetrates a previous low. A change from a bear to a bull market will require similar pattern of confirmation.

1.4: Advantages of Technical Analysis

Many investors use technical analysis. Some investors are technical analysis exclusively while others combine it with fundamental analysis. Technical analysis offers the following advantages.

1. It is objective: Once the technician has determined the particular role to use in the technical strategy, making the actual investment decision is easy. If technical criterion for a buy recommendation exists, then technician buys. The decision requires no thought, forecasts or other subjective judgments.
2. Technical analysis is easy to learn and requires no specialized financial knowledge. Constructing a chart of a stock's price is easy task. Computing market indicators usually requires only a simple calculation. On the other hand, conducting fundamental analysis requires knowledge in accounting, present value, finance and other topics that take long period to study.
3. Technicians have access to all the information they require in their analysis. Stock prices and volume of activities are easily accessible through the press or the Internet. Fundamental analysts have to rely on companies as their primary source of information. This information may be difficult to get and several months old by the time analysts get them.

1.5: Criticism of the Technical Analysis

1. The most evident criticism of technical analysis is the controversy over the efficacy of its strategies. Opponents of technical analysis claim that there is no real objective, substantial evidence that it works. Indeed the wealth of information supporting the weak form of efficient market hypothesis (EMH) rejects technical analysis.
2. Another criticism of the technical analysis is that it is self-fulfilling prophecy. Consider several hundreds of technicians using the same information and arriving of the some decision. For example if all the technicians follow ABC stock and the trend shows an upward move in the stock price. They will all call their brokers and place an order to buy. The increase in demand for ABC shares will force the price up, thus fulfilling the upward move.
3. Critics point out that technicians often act too late to be able to take advantage of the self-fulfilling prophecy situation. By the time they have processed the stock price information and acted, other technicians will have done same, the stock price will have reached its new equilibrium and it will be too late to make trading profit.
4. The final criticism deals with the difficulty of discerning chart patterns as they develop. A good instructional manual of charting will show nice examples of all the standard chart patterns, identifying the buy or sell points. However, actually identifying these patterns as they develop is not easy or simple.

Summary

In this session we discussed technical analysis. We said that technical analysis is essentially the search for recurrent and predictable patterns in the stock prices. In undertaking analysis the technical analyst uses chart as the basic tool. We also looked at the advantages that users derive from using technical analysis. We ended the session with the criticisms levied against technical analysis.

Self-Assessment Questions

Exercise 5.1

1. What is technical analysis?
2. Explain how technical analysis is undertaken.
3. Outline five assumptions underlying technical analysis.
4. State four advantages to be derived from using technical analysis Briefly explain four criticisms against the use

SESSION 2: FUNDAMENTAL ANALYSIS INCLUDING GLOBAL ECONOMY

Overview

You are once again welcome to session 2 of unit 5. In the previous session we discussed technical analysis. In this session and the sessions that follow, we are going to discuss the other aspect of security analysis namely fundamental analysis. We will start with the explanation of fundamental analysis and move on to an aspect of it; global economy.

Objectives

By the end of this session, you should be able to:

1. Explain fundamental analysis.
2. Explain how fundamental analysis is conducted.
3. How the global economy affects the operations of firms.

2.1: Fundamental Analysis

Before a security analyst can determine a proper price for a firm's stock, he/she has to forecast the earnings and dividend that can be expected from the firm. This is the heart of fundamental analysis, i.e. the analysis of the determinants of value of such as earnings prospects. Ultimately, the success of a business determines the dividend it will pay to shareholders and the price its shares will command in the security market. To determine the value of a firm, fundamental analysis relies on long-run forecast of the economy, the industry and the company's financial prospects. Short-run changes in business conditions are also important because they influence investors' required rate of return and expectation of corporate earnings and dividends.

Fundamental analysis uses earnings and dividend prospects of a firm, expectation of future interest rates and risk evaluation of the firm to determine stock prices. Fundamental analysis starts with a study of past earnings and the examination of a company's balance sheet, ordinarily including an evaluation of the quality of management of the firm, the firm's standing in the industry and prospect of the industry as a whole. Because the prospects of the firm are tied to those of the broader economy, fundamental analysis must consider the business environment in which the firm operates. For some firms macroeconomic and industry performance might have a greater influence within the industry. Therefore, the financial statement analysis of the firm should be combined with economic and industry analysis before a final judgement is made to purchase or sell specific security.

Fundamental analysis is therefore conducted on three levels: the economy, the industry and the firm. There are two types of fundamental analysis namely top-down analysis and bottom-up analysis. The top-down analysis starts with the broad economic environment, examining the state of the aggregate economy and even the international economy. Having analysed the broad economy, the analyst move the analysis to the industry level, looking at which industries are best positioned to be successful and profitable given the economic situation. Finally the firm's position within the industry is examined by looking at which firm(s) within the industry will maximise investment return.

The bottom-up analysis on the other hand, begins at the micro-level (firm's level) expand the analysis to the company's industry or sector and end with the analysis of the economic environment. The top-down analysis will be discussed in this book starting with the economic environment and ending with the analysis of the firm.

2.2: The Global Economy

A top-down analysis must start with the global economy. The international economy might affect a firm's export prospects, the price competition it faces from imported goods (competitors) or the profit it makes from investment abroad. In addition the global environment presents political risks of far greater magnitude than is typical in home-based investments. Political unrest can adversely affect an investment abroad. Similarly, the global environment may present a business opportunity for firms. For example the NEPAD initiative by the international community and AU can impact positively on firm's exports.

International events also have implication for firms' operations. The turmoil of several Asian countries (the Asian crisis), in 1997 and 1998 impacted negatively on international trade. Political developments offer significant opportunities to make or lose money. Other political issues that are less sensational but of extreme importance to economic growth and investment return is the issue of protectionism, trade policies, and free flow of capital.
Currently, the oil and food price increase being experience all over the world, have adversely affected the operations of all firms. This is because the oil crisis has increase the cost of energy. Since all firms use energy in one way other, their cost of operation has gone up. This has negatively affected the profits of firms since their expenditure has increased.

One major factor that affects international competitiveness of a country's industry is the exchange rate between the country's currency and other currencies. As the exchange rate fluctuates the value of goods price in foreign currency also fluctuates. An appreciation of the cedis will make our exports expensive in the international market

making our exports less competitive pricewise. On the other hand, firms which use foreign components and raw material will suffer when the cedis depreciates because their cost of production will go up. This will ultimately increase the price of their product(s). The end result will be either loss of market share or profit which invariably affects the value of the firm.

Summary

In this session we discussed fundamental analysis. We said that fundamental analysis uses earnings and dividend prospects of a firm, expectation of future interest rates and risk evaluation of the firm to determine stock prices. We mentioned that fundamental analysis can be conducted in two ways; top-down and bottom-up. The session ended with a discussion of the global economy. It was said that any event that happen on the global scene will impact on the operations of the firm. Exchange rates and global politics also affect the firm.

Self-Assessment Questions

Exercise 5.2

1. **Explain the term 'fundamental analysis' and how it is conducted.**
2. Explain how the global economy will impact on the operations of firms.

This is a blank sheet for your short notes on

- Issues that are not clear; and
- Difficult topics if any.

SESSION 3: FUNDAMENTAL ANALYSIS: DOMESTIC ECONOMY

Overview

You are once again welcome to session 3 of unit 5. We believe you have found your reading so far enjoyable. In this session, we are going to continue our discussions on fundamental analysis. We shall focus our attention in this session on domestic economy, an aspect of fundamental analysis. We shall look at the various variables within the domestic economy that can impact on the operations of the firm and hence the value of the firm's stock.

Objectives

By the end of this session you should be able to:

a) Explain the various variables that constitute the domestic economy
b) Explain how these variables impact on the firm's operation and hence its stock value.

Enjoy your reading.

3.1: The Domestic Economy

The economic environment has great influence over the performance of th groups and units that compose the economic system. There is a relations stock price and economic activity, increased earnings as a result of econom result in higher values for stocks.

In analysing the domestic economy, the following must be considered:

3.1.1: Gross Domestic Product (GDP)

GDP measures values of all goods and services produced domestically irrespective of who owns the resources used in production. Rapidly growing GDP indicates an expanding economy with ample opportunity for a firm to increase sales. Therefore GDP is more related to predicting sales of intermediate products (raw materials).

3.1.2: Gross National Product (GNP)

It is a measure of valve of all goods and services produced by resources owned by a nation. In a way it is a measure of income and can be used for forecasting sales. Firms using the prediction of GNP will be able to forecast the expected sales they will be making in the year relating to the forecast.

3.1.3: Employment

The unemployment rate of a country is the total labour force yet to find job. The unemployment rate measures the extent to which the economy is operating at capacity. In addition to considering the unemployment rate; analysts also look at the factory capacity utilization rate which is the ratio of actual output from factories to potential output. When people are employed, their purchasing power goes up. Therefore in a country where the rate of unemployment is low, all things being equal, demand for goods and services is expected to go up.

3.1.4: Inflation

Inflation is the rate at which the general level of price is rising. High rate of inflation is generally associated with 'overheat' economies, that is, economies where demand exceed supply. Inflation may impact negatively on the operations of a firm. This is because times of high inflation prices of goods and services go up. Thus, the operational cost of the firm is bound to rise and this will adversely affect the profit of the firm. Inflation may also be a sign that demand is higher than supply. In such a situation, firms can take advantage and increase their productivity to meet demand.

3.1.5: Interest Rates

High interest rate reduces the present value of future cash flows thereby reducing attractiveness of investment opportunities. It also makes borrowing very expensive to the firm. Thus high interest rates adversely affect firms.

3.1.6: Sentiments

Consumer and producer sentiments (optimism or pessimism) concerning the economy is an important determinant of economic performance. If consumers have confidence in their future income levels, they will be willing to spend more. Similarly businesses will increase their production and inventory levels if they anticipate high demand for their products. In this very sentiment influences how much consumption and investment will be pursued and affect the aggregate demand for goods and services.

3.1.6: Monetary Policy

Monetary policy is the government's control of money supply in the economy. Tools used in monetary policy are open market operations (buying and selling of T-bills) and discount rates (rates that are changed banks on short term loans). Monetary policy works largely through its impact on interest rates. Increasing money supply lower short-term interest rates, ultimately encouraging investment and consumption demand. In the long run however, most economists believe this can lead to inflation. There is a direct relationship between money supply and the price of stocks. Increase in money supply leads to increase in stock prices and vice versa. This is because an increase in money supply will increase the demand for stocks and the increase in demand will result in price increase of stocks.

3.1.7: Fiscal Policy

It refers to government spending and tap actions (policies). Fiscal policy is probably the most direct way to stimulate or show down the economy. Decrease in government spending directly reduces demand for goods and services. Similarly, increase in taxes siphon income from consumers and results in decrease in consumption. Increase in taxes also adversely affects the earnings of firms.

Different industries react differently to economic changes. For example consumer staples (food, drugs) industries are less sensitive to economic changes than luxury goods industries. Therefore the impact of the factors above will depend on the industry in which a particular firm finds itself. Also some factors will have adverse effect on the firms' operations whereas others will have a positive impact on the firms' operations and hence their stock price.

Summary

In this session, we continued our discussions on fundamental analysis. We focused our attention on domestic economy, an aspect of fundamental analysis. We looked at the various variables within the domestic economy that can impact on the operations of the firm and hence the value of the firm's stock.

Self-Assessment Questions

Exercise 5.3

1. Explain how fiscal policy and monetary policy will impact on stock prices of a firm.
2. Explain how inflation will have an adverse effect or a positive effect on the operations of a firm and hence its stock price.
3. Explain the following and how the affect firms' operations:
   a. Interest rates
   b. Sentiments
   c. Gross Domestic Product
   d. Gross National Product

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 4: FUNDAMENTAL ANALYSIS: INDUSTRY ANALYSIS

Overview

You are welcome to session 4 of unit 5. In this session we are going to continue our discussion on fundamental analysis. We will look at industry analysis. Industry analysis is made up of two main issues namely industry life cycle and industry structure. Under industry life cycle we will look at the various stages that industries go through from development to decline. Under industry structure we will discuss the economic structure of the industry, threat of entry, government regulations and competitive structure.

Objectives

By the end of the session you should be able to:

1. Explain industry life cycle and the various stages of an industry's life cycle.
2. Explain the various components of industry structure and their impact on the performance of the firm.

4.1: Industry Analysis

Industry analysis is the second step in the top-down analysis. Just as it is difficult for industries to perform well in an ailing economy, it is unusual for a firm in troubled industry to perform well. The discussion here will focus on industry life cycles, and industry structure.

4.2: Industry Life Cycle

Industry life cycle is created because of economic growth, competition, availability of resources and the resultant market situation by the particular goods and services offered. The particular phase in the life cycle of an industry determines the growth of earnings, dividends, capital expenditure and market demand for products. An industry analysis helps place an industry on the life cycle curve and in turn guides the analyst towards decision on industry growth, the duration of growth, profitability and potential rate of return. The analyst can determine where all companies in the industry are in the stage of the life cycle and translate company differences into various assumptions that will affect their valuation.The life cycle of an industry is divided into 5 stages:

4.2.1: Stage 1: Development

The development stage includes companies that are getting started in business with new product development or production technique that makes them unique. At this stage it is difficult to predict which firms will emerge as industry leaders. Some firms will turn

out to be successful and others will fail altogether. Therefore there is considerable risk in selecting one particular firm within the industry.

4.2.2: Stage II: Growth

4.2.3: Stage III: Expansion

At the expansion stage, sales and earnings expansion continues at a decreasing rate. Industrial leaders begin to emerge. The survivors in the industry are more stable and market share is easier to predict. The performance of the surviving firms more closely tracks the performance of the overall industry. At this stage, the dividend payout ratio increases from a low level to a moderate level. Some firm's in the industry continue to use stock dividend and stock splits.

4.2.4: Stage IV: Mature

At this stage the product has reached its full potential for use by consumers. The product has become standardised and producers compete on the basis of price. This leads to narrower profit margin and further pressure on profit.

Sales grow at a rate equal to the economy as measured by long-term trend in GDP. The automobile industry is a typical example of mature industry. By the time industries (firms) reach maturity stage, plant and equipment are in place, financing alternatives are available and cash flow from operations are more than enough to meet growth requirements of the firm. Such firms are characterised as cash cows, having reasonably stable cash flow but offering little opportunity for profitable expansion. The cash flows are best milked from rather than reinvested in the company. Therefore firms in mature industries normally pay high dividend to shareholders.

4.2.5: Stage V: Decline

At this stage, the industry might grow at a rate lower than the economy or it might even shrink. Industries suffer decline in sales as a result of competition from new products. It should be noted that it might not be the whole industry that goes into decline but rather weak firms within the industry.

The question to answer is at which stage in the industry life cycle is investment profitable or more attractive?

Conventional wisdom is that investors should seek firms in high-growth industries. This recipe for success is simplistic. If the security prices already reflect the likelihood for high growth, then it may be too late to make money from such stocks. Moreover the growth and high profit attracts competitors and new sources of supply that eventually

reduce prices. This is the dynamism behind the progressive from one stage of the industry life cycle to another.

4.3: Industry Structure

The structure of the industry is another area of importance for the analyst. Industry structure determines whether the companies in the industry are profitable, whether there are specific considerations such as government regulations that might positively or negatively affect the industry and whether cost advantages and product quality creates a dominant company within the industry. Factors to consider under the industry structure are the following:

4.3.1: Economic structure

Economic structure of industry determines how companies compete in the industry. The analyst has to find out whether the industry structure is monopoly, oligopoly or pure competition. If the industry is monopoly, it may be subject to government regulation with regards to rate of return on equity and assets and approval of consumer fees. This sets the limit for growth and profitability and creates a minimums and maximums for the analyst. Oligopolistic industries have few competitors but they face international competition, which affect profitability of such industries.

4.3.2: Government regulations

Most industries are affected by government regulations, for example the automobile industry where safety and exhaust emissions are regulated. In Ghana the food industry is regulated by the Food and Drugs Board. Most industries are affected by government expenditure, for example education, health, transportation and the agricultural industry to some extent are affected by government spending. Therefore it is very important for the analyst to understand each industry and how such industries are affected by government regulations.

4.3.3: Competitive structure

Industry consists of competing firms. Some industries have few firms whereas others have many firms. The firm in an industry compete with each another and employ different strategies for success. It should be noted that just that an industry is in a certain stage of life cycle does not mean that all firms will be in the same stage of a life cycle. A firm may have chosen a poor competitive strategy or an excellent competitive strategy which put them in different position.

Though the industry outlook is important, a firm may be able to create a competitive position that shapes the industry environment. There are profitable firms in poor industries and unprofitable firms in a good industry.

4.3.4: Threat of entry

New entrants to an industry put pressure on price and profits. Even if a firm has not entered an industry, the potential for it to do so put pressure on prices because high profit margins encourage new competitors to enter. Therefore barrier to enter can be a key determinant to industry profitability. Barriers can take many forms including brand loyalty of customers to existing producer (goodwill) and long term relationship with customers and suppliers which makes it difficult for new entrants to know whether the industry under consideration is easy to enter or not.

Summary

In this session we discussed fundamental analysis. Specifically, we looked at industry analysis. Industry analysis is made up of two main issues namely industry life cycle and industry structure. Under industry life cycle we discussed the various stages that industries go through from development to decline. Under industry structure we considered the economic structure of the industry, threat of entry, government regulations and competitive structure.

Self-Assessment Questions

Exercise 5.4

1 How will threat of entry impact on the operations of a firm?

2. Explain the first two stages of the industry life cycle and their importance to investors.

3. "Not all firms in the same industry will be in the same stage of life cycle" Explain.

SESSION 5: FUNDAMENTAL ANALYSIS: COMPANY ANALYSIS

Overview

You are once again welcome to session 5 of unit 5. This is going to be the last discussion on fundamental analysis using the top-down analysis. We started the analysis by looking at the global economy, the domestic economy, the industry and lastly the company; the main discussion of this session.

Objectives

By the end of this session you should be able to:

a) Explain the various components of company analysis.
b) Explain how the ownership structure impact on the operations of management.
c) Explain management should be monitored.

5.1: Company Analysis

The last leaf in the fundamental analysis is company analysis. Company analysis basically involves the analysis of the firm. Some aspects of the firm that will be analyzed are managerial monitoring (corporate governance), ownership structure and financial statements of the firm.

5.2: Managerial Monitoring

In a large organizations (such as public companies), ownership is separate from management. There is therefore the need to monitor the activities of management to ensure that management acts in the best interest of shareholders.

Managerial performance is monitored inside the firm by the board of directors who hire and oversee management. Board members represent the interest of shareholders in frequent meetings with top management. Institutional investors have some influence over management as large shareholders. Other people outside the organization may also attempt to monitor the activities of the firm. They include auditors who validate financial statements, financial analysts who advise investors, investment bankers who conduct diligence when issuing securities and credit-rating agencies who advise bondholders. The government also regulates the firm through agencies such as the Securities and Exchange Commission. How a firm is monitored will go a long way to affect the operations of the firm and this should to be analyzed.

5.3: Ownership Structure

Ownership structure is characterized by the amount of capital/equity owned by manager, board members, institutional investors and widely dispersed investors. Among

these, the percentage of insider equity financing receives the most attention. Insider equity is the amount of equity owned by the Chief executive Officer, top management, board members and employees of the organization. The balance of equity is obtained from large single-party outside shareholders, mutual funds, insurance companies, pension funds and the general public. When the share of insider holding is large, a similarly substantial insider self-interest in the ongoing performance of the firm can be presumed. Managers with a significant ownership interest have a strong incentive to run the firm in a value-maximizing manner. Similarly, when ownership is concentrated in the hands of few investors such as institutional investors, managers have the incentive to maximize corporate performance.

5.4: Financial Statements

The financial statements are normally analysed using ratios. Since the main aim of the analysis is to determine the value of the firm, analysts are interested in certain category of ratios, which include profitability ratios, efficiency ratios liquidity ratios and investment ratios.

Ratios by themselves are meaningless unless compared to a benchmark. Two such methods of comparison are the time series comparison and cross sectional comparisons. Time series comparison involves comparing current ratio with past ratios. Deviations from trend should be examined in line with policy changes to determine whether the change is intentional or unintentional. Cross sectional comparison involves comparing with peers. Deviations from industrial average should be examined for change in policy or performance reason. We review below some of the ratios mentioned above (which was full covered in unit three).

5.4.1: Profitability Ratios

These ratios are used to determine the profitability or other wise of a firm. It shows how earnings of a firm are growing. For example, a decline in return on equity (ROE) ratio is evidence that a firm's investments have offered a low ROE than its previous investments. Some ratios under profitability are:

1. $\text{ROE} = \dfrac{\text{Earning before tax and interest (EBIT)}}{\text{Net Assets}}$

2. $\text{Profit to turnover Ratio} = \dfrac{\text{EBIT}}{\text{Turnover}}$

3. Gross Profit ratio = $\frac{\text{Gross Profit}}{\text{Turnover}}$

4. Return on shareholders fund = $\frac{\text{Attributable Profit}}{\text{Shareholders fund}}$

Where attributable profit is earnings after tax, interest and preference share dividend and shareholders fund consist of equity plus retained earnings.

5. Earning per share (EPS) = $\frac{\text{Profit after tax}}{\text{Number of common shares outstanding}}$

6. Dividend yield = $\frac{\text{Gross Dividend}}{\text{Total market value of equity}}$

7. Dividend Cover = $\frac{\text{Profit available for dividend}}{\text{Dividend paid}}$

5.4.2: Efficiency Ratios

These ratios show how management is effectively and efficiently utilizing the assets (resources) of the firm. For example, the fixed asset ratio measures sales per cedi of the firm's money tied up in fixed assets. The average collection period, which shows the number of days it takes a firm to collect credit sales from customers, indicates the total credit extended to customers per cedi of daily sales. It also indicates the number of days' worth of sale tied up in accounts receivable. Efficiency ratios include:

1. Fixed Asset Turnover = $\frac{\text{Sales}}{\text{Fixed Assets}}$

2. Capital turnover = $\frac{\text{Turnover}}{\text{Net Assets}}$

3. Stock Turnover = $\frac{\text{Average Stock}}{\text{Cost of Sales}}$

4. Average Collection Period = $\frac{\text{Average Accounts Receivable x 365 days}}{\text{Sales}}$

5. Average Payment Period = $\frac{\text{Average Accounts Payable x 356 days}}{\text{Purchases}}$

5.4.3: Liquidity/ Leverage Ratios

These ratios measure the financial strength of the firm. The liquidity ratios measure the firm's ability to meet its current obligations. Leverage ratios, on the other hand, are

used to measure the long-term financial position of the firm. In a way, these ratios indicate the riskiness of the firm as far as bankruptcy is concerned. Some of the ratios in this category are:

1. Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$

2. Quick (Acid Test) Ratio = $\frac{\text{Current Asset – Stock}}{\text{Current Liabilities}}$

3. Gearing Ratio = $\frac{\text{Long-term Debt}}{\text{Long-term debt + Equity}}$

4. Debt to Equity Ratio = $\frac{\text{Long-term Debt}}{\text{Equity}}$

5.4.4: Investment Ratios

These ratios indicate the potential and actual growth rate of a business. These ratios help equity shareholders and other investors to assess the value and quality of an investment in the ordinary shares of the company. Therefore, it is of great importance to potential investors who are thinking of investing in a company. Investment ratios include:

a) Earnings Per Share (EPS)= $\frac{\text{Earnings available to Ordinary Shareholders}}{\text{Number of Ordinary Shares}}$

b) Dividend per Share = $\frac{\text{Total Dividend}}{\text{Number of Ordinary Share}}$

c) Dividend Cover = $\frac{\text{Total Equity Earnings}}{\text{Total Equity Dividend}}$

or

$\frac{\text{Earnings per Share}}{\text{Dividend per Share}}$

d) Payout Ratio = $\frac{\text{Equity Dividend}}{\text{Equity Earnings}}$

e) Price/Earning Ratio= $\frac{\text{Market Price per Ordinary Share}}{}$

Earnings per Share

f) Earnings Yield = $\frac{\text{Earnings per Share}}{\text{Market Price per Share}}$

g) Dividend Yield = $\frac{\text{Dividend per Share}}{\text{Market Price per Share}}$

h) Net Book Value per Share = $\frac{\text{Net Book Value of Ordinary Shares}}{\text{Number of Ordinary Shares}}$

Summary

In this session we discussed company analysis. Some aspects of company analysis such as management monitoring, ownership structure and financial statement analysis were discussed. Management monitoring looked at how the activities of management could be monitored to ensure the best performance from management.

Self-Assessment Questions

Exercise 5.5

1. Explain why ownership structure of a firm is significant too the operation of a firm.
2. Explain how management performance can be monitored by stakeholders.
3. How will financial statement analysis benefit a potential investor?

This is a blank sheet for your short notes on

- Issues that are not clear; and
- Difficult topics if any.

SESSION 6: EFFICIENT MARKET HYPOTHESIS

Overview

You are warmly welcome to the last session of unit 5. We believe you have so far enjoyed reading this unit. In this session we are going to discuss efficient market hypothesis. We will explain what efficient market hypothesis and the various forms of market hypothesis.

Objectives

By the end of the session, you should be able to:

a) Explain efficient market hypothesis.
b) Explain what an efficient market is.
c) Outline the various forms of efficient market hypothesis.

6.1: Efficient Market Hypothesis (EMH)

The interplay between investors in financial security and company managers responsible for issuing securities takes place in the financial market. These markets exist to enable investors and issuers of securities to make contact and transact business. When investors buy equity, they undertake perpetual investment since equity has no maturity date. They encourage investors make such commitment, they should have the assurance that a market exists where they could sell their shares at a fair price some time in the foreseeable future. This can only happen if there is an efficient market which prices shares efficiently by incorporating into the price all information currently available about the company and the economy as a whole. If this is the case, investors will be encouraged to buy shares through the security market because they know they are paying a fair price for the shares and would be able to sell at a fair price. Financial managers would be able to obtain funds at the financial market at a cost that reflect the riskiness of the business activities being undertaken. The notion of efficiency is therefore not only important to the investor but also company managers. The efficient allocation of resources is an important determinant of economic growth. In many countries, it is believed that the market provides the best means of allocating resources. Funds are efficiently allocated from individuals/organizations with few productive opportunities and greater wealth to those with many opportunities but insufficient wealth.

6.2: The Efficient Market

An efficient capital market is the one in which prices of traded securities always fully reflect all available information concerning those securities. Thus, by implication, security prices adjust instantaneously and in an unbiased manner to any piece of information released to the market. A capital market is, therefore, efficient if it does not neglect any information relevant to the determination of security prices and also has

rational expectation, i.e. it uses the information correctly. This means that the market uses all available information and it understands the true implication of this information to the joint distribution of payoffs. Thus, any change in price can only occur in the event of new information. In an efficient market therefore, investors would not be able to use available information to make above-normal profits because as soon as any information becomes known, it will quickly be implanted into the price of the security. The basic intuition of efficient market is that individual traders process the information that is available to them and take positions in assets in response to the information as well as their personal situations.

6.3: Forms of Efficient Market Hypothesis

6.3.1: The Weak Form

The weak form of efficient market hypothesis (EMH) states that security prices fully reflect information regarding the historical events (past prices, return and trade volumes) and investors knowing the historical sequence of prices can neither abnormally enhance their investment return nor improve their ability to select shares. In other words, the market is weak form efficient if the information in past prices is fully incorporated in current stock prices so that no investor can earn excess return by the use of trading rules based on this information. The weak form of EMH suggests that there is no relationship between past and future prices of securities. They are presumed to be independent. In other words, stock prices follow a random walk.

6.3.2: Semi-Strong Form

The semi-strong form of EMH maintains that all public information is impounded into the value of a security and therefore one cannot use fundamental analysis to determine whether a stock is undervalued or overvalued. Therefore, a market is semi-strong efficient if all public information (company annual reports, newspapers, dividend declaration etc.) is fully incorporated in the current price of the security so that no investor can make excess return based on this information. Therefore, the semi-strong form is concerned with whether current prices of securities reflect public knowledge about the underlying company and whether the speed of price adjustment to the public announcement of the information is fast enough to eliminate the possibility of abnormal gain.

A number of researches have shown consistency with the semi-strong form of EMH. These include Fama et al (1969) on stock splits in the U. S. A, Firth (1977) on stock split in U. K and Marsh (1977) on right issue.

6.3.3: Strong Form

The strong from of EMH goes beyond the semi-strong form by stating that stock prices reflect not only all public information but all information. Thus, a market is strong from

efficient if insider information is also fully incorporated in current prices so that no investor can earn excess return by the use of trading rules. Unlike the weak and semi-strong forms, major research results are not supportive of the strong form. It has been found out that professional analysts and certain people within companies have access to privileged information or significant information, which the general public (investors) don't have and have earned superior return on invested capital. Another research has shown evidence that 'insiders' do have what is called information advantage (Finnerty 1976). This means that they are able to use this price-sensitive information to earn abnormal return from their investment. Using insider information is known as insider trading and is a crime in many countries.

Summary

This is the last session of unit 5. In this session we discussed the efficient market hypothesis. We looked at the various forms of efficient; strong form, semi-strong form and the weak form of efficient market hypothesis.

Self-Assessment Questions

Exercise 5.6

1. What is efficient market hypothesis?
2. Explain the three forms of efficient market hypothesis and their implication on stock prices.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

Unit 6: Dividend Policy

Session 1: Types of Cash Dividend
Session 2: Stock Dividends, Split and Repurchase
Session 3: Dividend Irrelevant Argument
Session 4: Dividend Relevant Argument
Session 5: Dividend Relevant Argument (Continued**)**
Session 6: Other Dividend Policy Issues

Unit Overview

Have you ever wondered the effects of dividends on the fortunes of a firm? In other words, does paying dividend affect the value of a firm's stock? Dividends constitute a major cash outlay for firms and so the debate has been whether profits made by businesses should be retained in the business or distributed to stockholders as dividends. In fact, this issue of paying profit to stockholders as dividend or re-investing it in the business for the stockholders is the heart of dividend policy. In other words, should firms pay out a huge portion of profit now as dividend, or just a small (or even nothing) proportion to stockholders? There are two schools of thought on this contentious issue and we shall look at the argument for both schools in this Unit. We shall also look at how dividends are paid and the factors that affect dividend payment.

Unit Objective

After studying this unit you should be able to:

1. list and explain the types of dividend
2. Explain how dividends are paid
3. Outline how companies decide on dividend payment
4. Explain the dividend relevant argument
5. Discuss the dividend irrelevant argument
6. Explain dividend payment in practice and factors that affect dividend payment.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics, if any.

SESSION 1: TYPES OF CASH DIVIDEND

Overview

Dividend refers to cash paid out of earnings to business owners (stockholders or shareholders). It constitutes a significant cash outlay for firms. However, if a payment is made by a firm to its owners from sources other than current or accumulated retained earnings, the term distribution is used rather than dividend. Dividends may come in several forms but we shall consider on cash dividends in this session.

Objectives

After studying this session, you should be able to:

a) Explain the various types of cash dividends.
b) List and explain the dividend payment process.

Read on now …

The returns that investors expect on their investment are in the form of dividend and capital gain. If management decides to pay cash dividend to shareholders, that decision should be informed by the firm's major objective of maximizing shareholder value. In the light of this, the percentage of net income to be paid out as cash dividend (***target payout ratio***) must be based on investors' preference for dividend. We can use the constant growth stock valuation model to explain the effect of either a high or low dividend on stock price. The constant growth stock valuation model is given as: $P_0 = \frac{D1}{r-g}$

From the above, if management decides to raise up the payout ratio, then D_1 would go up, resulting in stock price going up too. But you should be aware that as D_1 goes up, less money would be available for reinvestment. This will lead to a decline in the expect growth rate, resulting in a fall in the stock's price. It is clear now that any change in payout ratio would have two opposing consequences and hence the optimal dividend policy must ensure a balance between current dividend and future growth prospects in order to maximize share price.

We shall consider theories of investor preference later on in this issue, but for now let us turn attention to types of cash dividend that are paid to stockholders and the procedures involved in the payment.

Types of Cash Dividend

Cash dividend is the most common type of dividend paid by firms to its owners. The basic types of cash dividends and discussed next.

i. Regular Cash Dividends

These are cash payments made directly to shareholders and are made in the regular course of business. In the United States of America for instance, regular cash dividends are paid four times in a year. In fact, as the name implies, this type of dividend would be paid regularly since management sees no reason why it should not be continued. Shareholders are given dividend warrant to claim their monies at specified locations.

ii. Extra Cash Dividends

Sometimes after paying regular cash dividend, management may decide to pay extra cash dividend to shareholders. By referring to this payment as extra, management is notifying shareholders that it may or may not repeat such payment in future.

iii. Special Dividends

This dividend is special in the sense that it is a one-off event that would not be repeated. They are unusual.

iv. Liquidating Dividends

This dividend is paid when the business or company is being liquidated or sold off.

Dividend Payment Procedures

In theory, dividends may be paid quarterly, semi annually or annually, but in Ghana, companies pay dividend annually. The decision as to whether to pay or not to pay dividend is made by the Board, and once the decision has been made to pay dividend. That is once dividend has been declared, it becomes binding or a liability on the firm and as such that decision cannot be changed or rescinded easily. When paying dividend, the following procedure must be followed:

(i) Declaration Date

This is the date on which the board of directors pass a resolution to pay dividend and publicly announces their decision. As mentioned earlier, this decision becomes a liability and cannot be rescinded easily.

(ii) Ex-Dividend Date

The ex-dividend date is two (2) business days before the date of record. This date is fixed in order to ensure that dividends are paid to the right people. Brokerage firms and stock exchanges establish an ex-dividend date to avoid the confusion as to who should

receive dividend and who should not. Investors who buy stock before this date are entitled to receive dividend. If an investor buys share on or after this date, the previous owner gets the dividend. The ex-dividend date convention removes any ambiguity about who is entitled to dividends.

(iii) Date of Record
On this date, the company closes its stock transfer books and prepares a list of all individuals believed to be shareholders as of that date. All investors whose name appears on the list on the date of record are designated to receive dividend. Declared dividends are distributable to shareholders of record on a specified date.

(iv) Payment Date
This is the date that dividend cheques (warrant) are mailed to shareholders of record.

Summary

We have seen that there are various forms of cash dividend that companies distribute to stockholders. They include regular cash dividend, extra cash dividend, special dividends and liquidating dividends. We also outlined the procedures that companies follow when dividends are to be paid. We hope you are in a position to explain the issues in your own words and would be able to answer questions when called to do so. We wish you success in all your endeavours.

This is a blank sheet for your short notes on:

- Issues that are not clear; and
- Difficult topics if any.

SESSION 2: STOCK DIVIDENDS, SPLITS AND REPURCHASES

Overview

As we noted in the first session, there are various types of cash dividends that companies distribute to stockholders, but that is not the only form of dividend. Companies also give stocks or shares as dividend to stockholders. This session would deal with stock dividends and other related issues – stock splits and stock repurchases.

Objectives

After studying this session, you should be able to:

a) Explain stock dividends, stock split and stock repurchases.
b) State the conditions under which stock dividend and stock split are used.
c) List some merits and demerits of stock repurchases

Stock Dividends

These are additional shares distributed to ordinary shareholders rather than cash. In the opinion of Brigham et al (1999) stock split "divides the pie into smaller slices" without affecting the fundamental position of the current stockholders. In fact stock dividends are not actual or true dividend because it is not paid in cash. Usually, stock dividends are expressed as a percentage, for instance a 10 percent stock dividend entitle each shareholder to receive one new share for every ten shares currently held. Because each shareholder received 10 percent more shares, total shares outstanding increases by 10 percent, and as a result the value of each share of the company reduces by 10 percent. We would conclude that stock dividend leads to an increase in total number of shares, but earnings, dividend and share price all decline.

Stock Split

Stock splits are similar to stock dividends except that stock splits are expressed in ratios rather than percentages. In fact, splits are accounting decision to alter the total number of stock outstanding without the sale of additional shares to the general public. They do not affect the position of the current shareholder. For instance a two-for-one split would convert each old share into two new shares. Stock split can be either a forward split or revere split. A forward split, which is also referred to as direct split results in shareholders receiving a greater number of shares than before the split. For instance if a shareholder has 5,000 in a company, a two-for-one split would increase the shareholders total shares to 10,000 after the split.

A reverse split on the other hand reduces the number of shares outstanding. They are stated as *one-for- two, one-for-five* and the like.

As an investment consultant, if a company intends to reduce its share price, which of the two concepts above (stock dividend and stock split) would you recommend? You should realise that when used on regular annual basis, stock dividends has the effect of holding down price movements. Stock splits on the other hand are usually used to produce a large reduction in stock prices. Small stock dividends, according to Brigham et al (1999) create bookkeeping problems and unnecessary expenses, so firms today use stock splits far more often than stock dividends.

Stock Repurchases

This involves a firm buying back shares of its own stock and thereby decreasing the number of shares outstanding and increasing the earning per share and price per share. There are two basic types of repurchases.

- Situation where the company has excess cash and inadequate investment opportunities may buy shares from shareholders rather than paying cash dividends.
- When the company's capital structure is weighted with equity, it sells its debt and uses the proceeds to by back its shares.

Stock that has been repurchased by a firm is called Treasury Stock. But we need to ask ourselves why stock repurchase programmes are becoming so popular these days. The simple answer is that stockholder value is enhanced. This is because fewer stocks remain outstanding after a repurchase programme, and it follows logically that if the repurchase does not affect the future earnings of the company, then earnings per share on the remaining shares will go up, culminating in a higher market price per share.

Stock repurchase can be made in one of these 3 ways:

1. A publicly owned company can buy back its stock through a broker on the open market.
2. It can make a tender offer where shareholders send in tenders within a particular time and period.
3. It can buy a block of shares from a shareholder on negotiated basis.

Advantages of Repurchase

1. They are viewed as positive signals by investors. Investors believe the repurchase is motivated by management belief that the firm's shares are undervalued.
2. Shareholders have a choice when the company distributes cash by repurchasing stock. They can decide to sell or not. With cash dividend however, they must accept and pay tax. Those who need cash can sell some of the shares.
3. Repurchase can remove a large block of stock that is overhanging in the market and keeping the price of the stock down.

4. They can be used to produce large-scale changes in capital structure (Debt: equity).
5. Dividend is "sticky" in the short run, because management is reluctant to raise dividend if they cannot be raised. Cutting cash dividends sends negative signals.

Disadvantages of Repurchase

1. Stockholders may be indifferent between dividends and capital gains. Cash dividends are generally dependable than repurchase agreements.
2. The selling shareholder may not be fully aware of all implication of a repurchase or may not have all information they need to make a decision based on the company's present and future prospects.
3. The corporation may pay too high a price for the repurchased stock to the disadvantage of other shareholders.

Summary

Stock dividends and stock splits are very similar. However the two are treated differently in accounting. Also, the conditions under which they are used to achieve maximum effects are different. Stock repurchases have the advantage of enhancing stockholder value. Note that a stock that has been repurchased by a firm is called Treasury Stock.

This is a blank sheet for your short notes

- Issues that are not clear; and
- Difficult topic if any.

SESSION 3: DIVIDEND IRRELEVANT ARGUMENT

Overview

Dividend decision of a firm is very crucial area of financial management. There are two schools of thought with regards to dividend policy. One school of thought believes that dividend policy is relevant whereas the other believes that dividend policy is irrelevant when it comes to the value of a firm. This session will attempt to explain the reasoning behind the dividend irrelevant argument.

Objective

After studying this session, you should be able to:

a) Explain dividend irrelevant argument put forward by Franco Modigliani and Merton Miller.

Read on now …

The importance of dividend policy is to determine the amount of earnings to distribute to shareholders and the amount to retain in the firm. Such a decision is crucial because the amount of earnings retained is the main internal source of fund for taking advantage of investing in projects that will add value to the firm. On the other hand, dividends may be desirable to shareholders and for that matter, they will like that major part of earnings be paid to them in a form of dividend to increase their current income. Therefore, management should try and strike a balance so that the shareholders' investment would be maximized.

Modigliani and Miller Model

This theory was developed by Franco Modigliani and Merton Miller (M&M) in 1961. They argued that under certain set of condition dividend is irrelevant in that share price is independent of a particular dividend policy followed. They argued that the value of a firm depends on the firm's earnings which result from the firm's investment policies. Thus, when investment decision of the firm is given, the dividend decision – the split of earnings between dividends and retained earnings is of no significance in altering the value of the firm. In other words, how a firm generates earnings is more significant than how such earnings are distributed.

Under the MM assumptions, if divided policy is to be irrelevant then the price of each share must be such that the rate of return on every share will be the same throughout the market over any given interval. This discount rate is defined as the discount rate that equates the present value of dividends received during the period and the end-of period selling price to the current share price.

$$P_0 = \frac{Pi + Di}{1+k} \longrightarrow \text{Equation 1}$$

If we assume an all-equity firm and multiply the above equation by the current number of shares outstanding (N_0), the total market value of the firm (V_0) can be written as:

$$V_0 = P_0N_0 = \frac{DiNo + PiNo}{1+k} \longrightarrow \text{Equation 2}$$

However, as dividend policy affects the amount of new equity financing required, P_1 in the above equation depends on D_1. To overcome this apparent dependency, it is necessary to equate the sources of corporate cash to uses during period 1.

$$Y_1 + P_1(N_1\text{-}N_0) = D_1N_0 + I_1 \longrightarrow \text{Equation 3}$$

Where Y_1 and I_1 are earnings and investment in period 1 and N_1 is total number of shares outstanding at period 1

Equation (3) states that for an all-equity firm, earnings plus receipt from sale of new shares must equal dividends plus investment. From equation (3), we can observe that:

$$D_1N_0 = Y_1 - I_1 + P_1(N_1 - N_0)$$

Substituting this into equation (7) gives:

$$V_0 = \frac{Y_1 - I_1 + P_1(N_1 - N_0) + P_1N_0}{1+k}$$

$$V_0 = \frac{Y_1 - I_1 + V_1}{1+k} \longrightarrow \text{Equation 4}$$

Where V_1 is P_1N_0

Since D_1 does not appear in equation (4) directly, the only way current dividends can influence share price is if they affect one of the variables in the equation. Y_1 depends on the profitability of past investments and hence it is independent of dividends, in any period. Similarly, k depends on the risk borne by equity shareholders and not on dividend policy. V_1 might depend on dividends but if so

the dependency will be on future not current dividends. It can therefore, be concluded that given a firm's investment policies, the dividend payout it chooses to follow will affect neither the current price nor the total return to its shareholders.

The model also illustrates a number of interesting facts about ordinary share values. The market value of an all-equity firm equal the difference between the present value (PV) of its earnings and its investments discounted at the cost of equity capital. Secondly, the model stresses the fact that the value of a share is not simply the PV of future earnings per share. This would only be true in the extreme case where the firm made no new investments and issued no new shares. Because of the need for sources and uses of funds to be equal (equation 3), any change in dividend payment will lead to an equal and opposite change in the amount of money raised from new issues. In other words, dividend increases will only be possible if new shares are issued to raise equivalent amount.

Suppose a company is valued at GH¢100,000 and has issued one million shares valued at 10GP each. Assume that the firm's current income exactly covers its investment requirement. In this situation any dividend to be paid will have to be met by issuing new equity. If the company is paying a dividend of 2GP a share this will be the implication.

	Company		**Share (equity)**
	GH ¢		GP ¢
Current Value	100,000	0.10	
Less dividend payment	20,000		0.02
	80,000		**0.08**
New issue	20,000		
	100,000		

To raise GH¢20,000 at GH¢0.08 a share (the value of share after paying dividend), the company will issue 250,000 shares (GH¢20,000/GH¢0.08). The value of company after dividend payment and the new issue remain unchanged, GH¢100,000. However, there are now 1,250,000 shares valued at GH¢0.08 a share. Old shareholders have received a dividend of GH¢0.02 and this has reduced their share value to GH¢0.08, but their total wealth is unchanged. New shareholders have purchased shares at a price, which reflects the true value for the company. If the company did not pay dividend the existing (old) shareholders could manufacture their own dividend by selling some of their shares in the market to new shareholders. A holder of 100 shares would need to sell 20 shares at GH¢0.10 to receive cash equivalent of dividend of GH¢0.02 a share. The shareholder would be

left with 80 shares at GH¢0.10 which is equal to 100 shares at GH¢0.08. However, this result is based entirely on the assumption that perfect capital market exists.

Summary

The thrust of dividend irrelevant argument is that the value of a firm depends on the firm's earnings which result from the firm's investment policies. Therefore, when investment decision of the firm is given, the dividend decision – the split of earnings between dividends and retained earnings is of no significance in altering the value of the firm. In other words, the argument is that how a firm generates earnings is more significant than how such earnings are distributed.

SESSION 4: DIVIDEND RELEVANT ARGUMENT

Overview

In the preceding session, we looked at the dividend irrelevant argument put forward by Franco Modigliani and Merton Miller. In this session, we shall consider the dividend relevant argument. There are a number of contributors to this argument and hence, we shall use this session and the one after this to exhaust the argument.

Objectives

By the time you have finished studying this session, you should be able to:

a) Use Walter's Model to explain the dividend relevant argument.
b) Use Gordon's Model to explain the dividend relevant argument.

Read on now …

Walter's Model

Prof. James W. Walters argues that dividend policies almost always affect the value of a firm. His model shows the importance of the relationship between the firm's internal rate of return (r) and its cost of capital (k). Walter's model is based on the following assumptions:

1. The firm finances all investment through retained earnings
2. The firm's internal rate of return (r) and its cost of capital (k) are constant
3. All earnings are distributed as dividend or reinvested immediately
4. Beginning earnings and dividends never change
5. The firm has a very large or infinite life

Walter's formula to determine the value of shares is:

$$P_0 = \frac{DIV}{k} + \frac{r(EPS - DIV)/k}{k}$$

$$P_0 = \frac{DIV + r/k(EPS - DIV)}{k} \longrightarrow \text{Equation 5}$$

The implications of the above model on firms are as presented below:

Growth firms, r > k

- Rapidly expanding firms
- Investment opportunities yielding returns (r) higher than cost of capital (k)
- Able to reinvest retained earnings at a rate higher than the rate expected by investors/shareholders
- Maximise share value by following policy of retaining all earnings for reinvestment.

Normal Firms, r = k

Such firms do no have unlimited surplus-generating investment opportunities generating returns higher than the opportunity cost of capital.

Thus, there is no unique optimum payment ratio for a normal firm. One dividend policy is as good as the other. The market value per share is not affected by the payout ratio when r = k.

Declining firms (Mature firms) r < k

- Some of the firms do not have any profitable investment opportunities to invest the earnings.
- Return on investment is less than the minimum rate required by investors.
- Investors of such firms want the earnings distributed so that they can either spread it or invest it somewhere else.
- The market value per share increases as the payment ratio increases.

Gordon's Model

Gordon's model is a very popular model, explicitly relating the value of the firm to dividend policy. It was developed by Myron Gordon (1962).

Assumptions

- The firm is all-equity firm.
- No external financing is available.
- The internal rate of return (r) is constant so is the appropriate discount rate (k).
- The firm and its stream of earnings are perpetual.
- Corporate taxes do not exist.
- The retention ratio (b), once decided upon is constant.
- The discount rate is greater than the growth rate, k > g

$$P_0 = \frac{DIV(1+g)}{1+k} + \frac{DIV(1+g)2}{(1+g)2} + \frac{DIV(1+g)\infty}{(1+k)\infty}$$

$$P_0 = \frac{DIV}{k-g} \longrightarrow \text{Equation 6}$$

Substituting *EPS (1-b)* for dividend and *rb* for growth

$$P_0 = \frac{EPS(1-b)}{k-rb}$$

Where EPS is expected earnings

b is payout ratio

r is internal profitability and

k is cost of capital.

EPS = rA where A is total asset per share

$$P_0 = \frac{rA(1-b)}{k-rb}$$

If r = k then,

$$P_0 = \frac{EPS(1-b)}{k(1-b)} = \frac{rA(1-b)}{r(1-b)}$$

$$P_0 = \frac{EPS}{k} = = A \longrightarrow \text{Equation 7}$$

The equation above shows that regardless of the firm's earnings (EPS) or riskiness (k), the firm's value is not affected by dividend policy. In other words, **where r = k (the case of normal firms)** dividend policy is irrelevant since b which

represents the firm's dividend policy is cancelled out. This means that the opportunity cost of capital (k) must be equal to the rate of return (r) generally available to investors in comparable shares. Any fund distributed as dividend may be invested in the market at a rate equal to the firm's internal rate of return. As a result, shareholders would neither gain nor lose as a result of a change in dividend policy and the market value of their shares must remain unchanged.

Where r< k (declining firms)

$P_0 = \frac{rA}{k}$, if b = 0 ⟶ Equation 8

If r < k, then r/k is less than 1, and from equation 4 it follows that P_0 is smaller than the firm's investment per share in assets (A). If r is less than k, then profit retention becomes undesirable from the shareholders perspective. This is because any earnings retained and reinvested yields a return lower than the amount shareholders could earn elsewhere if the earnings were paid to them as dividend for them to undertake investment in more profitable firms. Under such circumstances, the company should adopt a policy of contraction and disinvestments, which would allow investors to transfer profit and paid in capital (dividend) to more remunerative firms.

Where r > k (growth firms)

If r > k, then r/k is greater than one and therefore, the value of a share (P_0) will be greater than the firm's investments per share in assets (A). The value of share will increase as the retention ratio increases under the condition r > k. This is because any earnings retained and reinvested yields a return (r) greater than the prevailing opportunity cost of capital. Investors would therefore, like earnings to be retained and reinvested.

SUMMARY

Summary

We have looked at the contributions of James W. Walters and Myron Gordon to the dividend relevant argument. Waiters showed the importance of the relationship between the firm's internal rate of return (r) and its cost of capital (k), where as Gordon explicitly related the value of a firm to dividend policy adopted by a firm

SESSION 5: DIVIDEND RELEVANT ARGUMENT (CONTINUED)

Overview

In the preceding session we looked at the contributions of James W. Walters and Myron Gordon to the dividend relevant argument. This session is a continuation of the dividend relevant argument. We shall look at the Bird-In-Hand Argument and other theoretical issues that can question or affect our views on dividend policy.

Objectives

By the end of this session, you should be able to:

a) Explain the thrust of the bird-in-hand argument.
b) Explain the issues of information asymmetry, information content, signalling, tax preference, and clientele effect.

Read on now …

The Bird-In-Hand Argument

The support for the bird-in-hand argument came from Myron Gordon. According to the Gordon growth model, dividend policy is irrelevant if r = k and when all other assumptions are held valid. But when these assumptions are modified to confirm more closely with reality, Gordon concludes that dividend policy affects the value of a share even when r = k.

The view is based on the assumption that under conditions of uncertainty, investors tend to discount distant dividends at a higher rate than they discount near dividend. This is so because the longer the period, the more the uncertainty associated with dividends.

Krishman put the bird-in-hand argument forward first when he said, ***"****of two stocks with identical earning record and prospects, the one paying a larger dividend will undoubtedly command a higher price merely because stockholders prefer present values to future values. Myopic vision plays a part in the price-making process. Stockholders often act upon the principle that a bird-in-hand is worth two in the bush and for this reason are willing to pay a premium for the stock with the higher dividend rate, just as they discount the one with the lower rate".*

Graham and Dodd also hold a similar view when they stated, "***the typical*** *investor would most certainly prefer to have his dividend today and let tomorrow take care of*

itself. No instances are on record in which the withholding of dividends for the sake of future profits has been hailed with such enthusiasm as to advance the price of the stock. The direct opposite has invariably been true".

Given two companies in the same general position and with the same earning power, the one paying the larger dividend will always sell at a higher price.

Gordon has expressed the bird-in-hand argument more convincingly and in formal terms. According to him, uncertainty increases with futurity. Accordingly, when dividend policy is considered in the context of uncertainty, the appropriate discount rate (k) cannot be assumed to be constant. In fact it increases with uncertainty. Investor prefers to avoid uncertainty and would be willing to pay higher price for shares than pay higher dividend. If investors discount distant dividend at a higher rate than near dividends, increasing the retention ratio has the effect of raising the discount rate (k) or equivalently lowering share prices. Therefore, dividend policy affects the value of a share. Gordon's argument justifies the behaviour of investors who value dividend more than capital gains.

The bird-in-hand theory suggests that investors feel that they can invest in cash flow of the firm as well as the firm's managers. It suggests that a cedi in an investor's hand is valuable because managers cannot be trusted to spend it wisely. Other scholars, in particular Myron Gordon and John Lintner, argued that the required rate of return on equity (k) decreases as dividend paid out is increased because investors are less certain of receiving the capital gains which should result from retained earnings than they are receiving dividend payment.

Gordon and Lintner said investors value a cedi of expected dividend more highly than a cedi of expected capital gain because the dividend yield D_1/P_0 is less risky than the *g* component in the total expected return equation.

$$k = \frac{Di}{Po} + g$$

MM disagreed saying k was independent of dividend policy which implies that investors are indifferent between D_1/P_0 and *g*.

They also called Gordon-Linter argument the bird in hand fallacy because investors plan to reinvest their dividends in the stock of the same company and in any event the value of the firm is based on income generated by the assets and business risk and not on dividend policy.

Information Asymmetry

The ownership and management of a company are separated. Management of a company will have information about the firm. It is rare, in practice, to find management sharing information about the business with shareholders. Most

shareholders do not have insight about the day-to-day operations and management of their companies. This gap between information available to management and what is actually shared with shareholders is known as information asymmetry. Information asymmetry leads to conflicts arising between the management and the shareholders. Dividend policy helps to reduce this conflict.

It is believed that companies which pay dividend regularly will be raising funds externally from the capital market and financial institutions to finance other investment activities. Therefore, the lenders or providers of funds in the capital market and financial institutions will be monitoring the performance of these companies. If such monitoring is continuously done by the professionals in capital markets and financial institutions, shareholders need not incur monitoring cost. Shareholders are assured that the activities of management will be monitored and can therefore, be assured that their money is being put in a good use.

Dividend payment also allocates financial resources in favour of shareholders as against the lenders. Lenders have prior claims over a company's cash flows generated internally. The payment of dividend changes this as shareholders receive cash flow before the loans are redeemed.

Information Content

It is contended that dividends are relevant because they have information of value. A company can make statement about its expected earnings growth to inform shareholders in order to create a favourable impression on them.

Miller and Rock (1985) argued that holding investment policy fixed, dividend conveys information about earnings. Higher dividends signal increased future earnings. Accordingly, the price of a share is affected by the changes in dividend policy. Solomon (1963) contends that dividends may offer a tangible evidence of the firm's ability to raise earnings and as a result, dividend policy affects the share price. He states "*in an uncertain world in which verbal statements can be ignored or misinterpreted, dividend action does provide a clear-cut means or making a statement that speaks louder than a thousand words*".

It is argued that the announcement of changes in dividend policy influence share prices and that managers use the dividend changes to convey information about the future earnings of their companies. If a company follows a stable dividend policy the management of companies may influence the perception of investors that the riskiness of the company remains unchanged.

DIVIDEND RELEVANT ARGUMENT (CONTINUED)

Signalling

The primary goal of management is to maximise the company's stock price. The stock price is the present value of expected future cash flow and the present cash is the dividend stream. The present cash flow also depends on earnings. Management should therefore take actions to produce high, stable future earning and convince investors and shareholders that future earnings will be high. Some managers are not to be trusted because they provide biased or overly optimistic reports so as to sell out their stocks.

Other managers are reluctant to make explicit their earning for fear of being wrong and having investors file suit claiming they have lost money.

This uncertainty causes investors to look at signals such as management actions, historical accounting data when deciding how much a given stock is worth.

An increase in dividend signals management confidence that future earnings will be strong enough to support the new and higher dividend, a cut on the other hand signals that management is uncertain about the future earnings.

Dividend signals are therefore honest because they require cash payment, unlike reported earnings. Companies thus recognise the importance of dividend signalling and so issue statements regarding dividend policy.

Tax Preference

The tax preference theory has been expanded to recognise that different investors have different tax situations. Investors in high tax brackets prefer low dividend paying stock and vice versa.

In some situations capital gains are taxed at a maximum rate of say 28% whereas that of dividends is 39.6%. Wealthy investors will prefer to have companies retain and plough back money into business. The earning growth would lead to increase in stock price. This is because such investors are normally in the high tax bracket and will therefore like to reduce their tax liability by investing in growth stocks. Thus the lower taxed capital gains will be substituted for higher fixed dividends. Taxes are not paid on gains until they are sold. Due to time value of money a cedi of taxes paid in future has a lower effect cost than a cedi paid today. Because of these tax advantages investors prefer to have their companies retain most of their earning.

The conclusion based on this theory is that investors will be willing to pay more for low pay out companies than high pay out companies. On the other hand investors in lower tax bracket will prefer high dividend paying stocks (high yield stocks) since they will not be affected much with the tax they pay on the dividends.

Clientele Effect

The shareholders of a company have a variety of tax positions, variety of expectations and a variety of preferences. A company cannot expect to satisfy all its shareholders when it takes a financial decision. However, it has been suggested that the characteristics of a company including its dividend policy attracts a particular class of shareholders. This is the tendency for a firm to attract the type of investors who like its dividend policy. This is known as the clientele effect. A dividend policy followed for a long period in the past tends to create the clientele effect. That is it attracts those investors who consider the dividend policy to be in accord with their investment requirements. If a company suddenly change its dividend policy, it may work to the detriment of the shareholders, as they may have to switch from another investment to fulfil their investment needs. However investment switching is costly because of brokerage cost (transaction cost), likelihood that selling shareholders will have to pay taxes on their capital gain and a possible shortage of investors who like the firm's newly stated policy.

M&M also alluded to the fact that a clientele effect might exist when a firm sets out a dividend policy in such a way that it attracts a category of clientele. Some shareholders prefer current income to future gains (retirees). Others want future capital gain. Retired and old people generally invest in shares to get regular income. They use their savings or pension funds to purchase shares. These people may therefore select high yielding stocks with history of paying regular and liberal cash dividends. The same thing goes for institutional investors. Some institutional investors do not invest in firms that do not pay cash dividend. Wealthy investors are very much concerned with the dividend policy followed by a firm. They have a definite investment policy of increasing their wealth and minimizing taxes. These people are in high tax brackets and therefore cash dividends they receive will be taxed at a high rate. In view of this, they generally prefer dividend policy of retaining earnings and distributing bonus shares.

Therefore, those investors in favour of cash dividend will want a high pay out ratio. If the firm retains and invests earning rather than pay dividends, those clients who prefer current income will be disadvantaged. They might have to sell some of the shares so as to get some cash. They will however have to pay some brokerage costs. This will be expensive and inefficient. Thus those clients who desire current income would purchase shares in high dividend pay out firms and those who want future income in low payout firms. This establishes the fact that management should establish a policy that it deems appropriate.

Summary:

We have looked at the bird-in-hand argument along side theoretical issues of information asymmetry, information content, signalling, tax preference, and clientele effect. We believe that you were able to follow along with the reasoning behind the theories. Constant reading would reinforce the issues in your memory, so we encourage you to do more reading.

SESSION 6: OTHER DIVIDEND POLICY ISSUES

Overview

It is said that those who practice before learning the theory behind the practice are like sailors who go to sea without a rudder. Therefore, having reviewed the various arguments for and against dividend payment, it is now time to look at the real world situation. That is to say in practice, how do firms handle the dividend policy? We shall conclude the session with factors that affect dividend policy.

Objectives

By the end of this session, you should be able to:

a) Explain dividend policy in practice.
b) Outline the factors that effect dividend policy

Now read on …

Dividend Policy in Practice

i, Residual Dividend Policy

When management decides to pay dividend to shareholders, two major issues must inform them as to how much to distribute to shareholders. The first is the objective to maximize shareholder value. The second is refraining from retaining earnings unless management can reinvest such funds to earn higher returns than shareholders could earn if they invested the funds by themselves in investments of equal risk. A policy in which the dividend paid is set equal to the actual earning less the amount of retained earnings needed to finance the firm's capital budget. In practice dividend policy is influenced by:

a. investment opportunities
b. availability of funds to finance those new investments.

The steps involved are as outlined below, but firms must realised that establishing dividend policy is not a straight jacket decision. In other words, one size does not fit all when it comes to dividend policy decisions.

1. determine the optimal budget
2. determine amount of capital needed to finance the budget
3. use retained earnings to supply equity component to the extent possible
4. pay dividend only if more earnings are available than needed to support optimal capital budget.

The basis of residual policy is the fact that investors prefer to have the firm retain and reinvest earning rather than pay them out as dividend if the rate of return the

firm can earn on reinvested earning exceeds the rate of investors' investments of comparable risk.

The residual dividend policy minimises the cost to the company of raising outside fund. The disadvantage is that it does not offer a stable cash flow to investors.

ii, Constant or Steadily Increasing Dividends

Firms do not want to reduce the dividends paid to shareholders. When dividends are reduced, the value of stocks usually falls substantially. To guard against this, firms tend to push earning so that their policy of stable dividend payout can be maintained. In inflationary periods this requirement is more reasonably met. The reasons are not far fetched as shown below:

1. Stable dividend payment increases stock price.
2. Fluctuating payment policy would lead to greater uncertainty leading to higher k and lower stock prices.
3. Many shareholders may use dividends for current consumption and would be troubled if they had to sell part of the shares.
4. A long-range constant payout ratio may be a desirable dividend policy.

iii, Low Regular Dividend plus Extra

A policy of low regular dividend plus a year-end extra in good year is a compromise between stable dividend and constant payout rate. Such a policy gives the following advantages:

1. It is flexible
2. The directors can set a relatively low regular dividend and supplement it with an extra dividend in years when excess funds are available.

This policy attempts to maintain a form of stability in dividend payment.

Factors Affecting Dividend Policy

i, Managerial Considerations

Transaction costs, floatation cost and institutional restrictions are a few specific factors that must be considered in setting a particular dividend policy. All these factors affect the finances of the organization and should therefore be considered in dividend policy decisions.

ii, The Firm's Financial Reserves

The amount of dividend paid by a company is influenced by the firm's financial reserves. If a firm has an unexpected temporary reduction in earnings, rather than lower the amount of dividend paid, it could rely on its financial reserves to keep

dividends at their originally planned levels. If they are high, it can increase the dividends temporarily.

iii, Flexibility

If the firm has ability to postpone some new investments, this flexibility may help the manager to keep a stable dividend policy. Temporary decreases in earnings need not lead to decreases in dividends because it can postpone some new investments to a later date. The funds marked for specific investment can be used to pay the dividends.

iv, Access to Equity Capital

Access to equity capital plays a major role in managers' decisions to pursue a residual or non-residual dividend policy. Dividends are paid only after an amount adequate for capital budgeting has been done (investment). Following a residual policy may spare the company the problem of fresh equity financing. If equity capital is available at a reasonable cost than the company can deviate from its residual policy – if a shareholder wanted a large dividend. If however a fresh equity will be expensive and not readily available it will stick to its old policy.

v, Control Considerations

A liberal dividend policy increases probability of raising fresh capital at some future date. If present shareholders do not or cannot subscribe for new shares, new shareholders can dilute their control. Therefore shareholders who are sensitive to potential loss of control will prefer a low dividend pay out policy.

vi, Ability to Borrow

Companies finance their activities through debt and equity. Short term borrowing capacity through established lines of credit with bank increases flexibility land consequently dividend payout. Ability to float a new bond issue increases the company's ability to pay dividends. A company may be unwilling to increase its debt (if it is too expensive – debt to equity) it will not be generous with its dividend payment.

vii, Inflation

During periods of inflation, it is possible that funds generated from depreciation are insufficient to replace outdated machines and plants. Replacement costs may be so high that companies may have to acquire new machines, with fresh influx of capita. In a situation like this the company may want to reduce pay out ratio to ensure that it

will have enough funds internally generated to finance necessary replacements or new expansion.

Summary

We have seen that the dividend policy in practice comprise residual dividend policy, constant or steadily increasing dividends, and low regular dividend plus extra. We also outlined factors that affect dividend policy. These factors are managerial considerations, the firm's financial reserves, flexibility, access to equity capital, control considerations, ability to borrow, and inflation. It is our firm believe that you have found the material presented very useful. We wish you success in all your endeavours, even as you prepare for your exams.